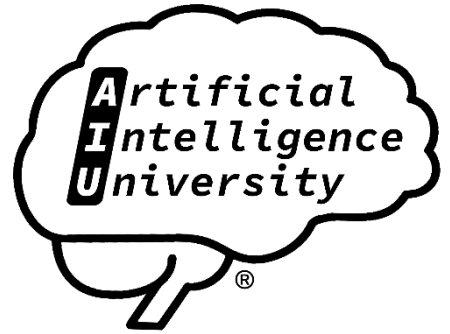




The Roadmap to AGI and ASI

From Natural Intelligence
to Artificial General Intelligence
and Artificial Super Intelligence

D. R. Diison



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The Roadmap to AGI and ASI

The Roadmap to AGI and ASI:

A Gap Analysis from Natural Intelligence to Artificial General Intelligence and Artificial Super Intelligence

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This work is presented as an open research and educational document intended for researchers, engineers, policymakers, philosophers, educators, and curious readers who wish to understand the profound distance between the intelligence nature has perfected over billions of years and the intelligence humanity is working to replicate in silicon and code.



AUTHOR'S NOTE

This book is written for two audiences simultaneously, and this dual purpose is intentional and necessary.

The first audience is the specialist: the neuroscientist, the machine learning researcher, the cognitive scientist, the robotics engineer, the philosopher of mind, and the policy analyst.

For this audience the book offers a structured synthesis across disciplines, a clear taxonomy of the gaps between Natural Intelligence (NI) and Artificial Intelligence (AI), and a concrete research agenda organized around the most important open problems.

The second audience is the informed general reader: the curious person who has heard about Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI) and wants to understand what those terms really mean, why they matter, and what stands between today's impressive but limited AI systems and the transformative systems that researchers envision.

For this audience every technical term is defined at first use, every concept is illustrated with concrete examples, and the argument is developed step by step without assuming prior expertise.

These two audiences are not as different as they might seem.

The specialist often benefits from seeing their domain's assumptions questioned from the outside.



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The general reader often has better intuitions about intelligence than they realize, because they already possess it.

The argument this book makes is ultimately accessible to anyone who has ever learned something new, forgotten something important, been surprised, changed their mind, or wondered why.

That is, it is accessible to any mind.

Definitions used throughout this book:

Natural Intelligence (NI) refers to any form of intelligence that has arisen through biological evolution and development in living organisms.

Artificial Intelligence (AI) refers to computational systems designed to perform tasks that require intelligence when performed by humans.

Artificial General Intelligence (AGI) refers to a hypothetical AI system capable of performing any cognitive task that a human can perform at a comparable or superior level.

Artificial Super Intelligence (ASI) refers to a hypothetical AI system whose cognitive capabilities exceed human performance across every domain of relevance.

All technical terms are defined at first use throughout the text and collected in the Glossary in Chapter 23.



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FOREWORD

Intelligence is the most remarkable phenomenon in the known universe.

It transforms raw sensory data into meaning, converts experience into wisdom, and bends the trajectory of events through deliberate action.

For billions of years, natural selection has been the sole engineer of intelligence, producing a staggering variety of minds ranging from the distributed nervous system of the octopus to the vast associative cortex of the human brain.

In a blink of geological time, however, humanity has begun constructing its own versions of intelligence in the form of Artificial Intelligence.

The ambition behind AI is breathtaking.

Researchers and technologists dream not only of machines that can perform narrow tasks reliably, but of machines that possess general reasoning ability, creativity, self-awareness, and eventually capabilities that surpass our own.

Those ambitions are named AGI and ASI.

Yet despite enormous investment, dramatic computational advances, and impressive benchmark achievements, today's best AI systems remain brittle, narrow, and deeply unlike the biological minds that inspired them.



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This book exists because the gap between Natural Intelligence and Artificial Intelligence is not merely an engineering problem.

It is a scientific problem, a philosophical problem, and arguably a civilizational problem.

To close this gap, we must first understand it precisely.

That requires taking neuroscience seriously as a guide and as a source of design principles for the AI systems of the future.

It also requires something rarer: the willingness to sit with the full complexity of what biological intelligence actually is before rushing to replicate or surpass it.

The journey from natural to artificial intelligence is one of the most profound challenges and opportunities of our time.

This foreword closes with an acknowledgment of the many scientists, engineers, philosophers, and ethicists whose work underlies every page of this analysis.

Their collective effort represents exactly the kind of collaborative, cross-disciplinary enterprise that bridging the gap will ultimately require. [25] [27]

INTRODUCTION: WHY THIS BOOK MATTERS AND WHAT IT INTENDS TO DO

The Central Question



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The central question of this book is deceptively simple: What does natural biological intelligence do that artificial intelligence cannot yet do, and what specific research is needed to close that gap?

To answer that question rigorously requires drawing on neuroscience, cognitive science, computer science, philosophy of mind, evolutionary biology, and systems engineering simultaneously.

No single discipline can answer it alone.

This is therefore a deliberately interdisciplinary book, written with the conviction that the next great leaps in AI will come not from purely computational or purely biological research in isolation, but from their sustained and serious integration.

Why This Book Is Important – Reason One: We Are at a Decisive Moment

The world is pouring unprecedented resources into AI development.

Hundreds of billions of dollars are flowing into AI research, AI infrastructure, and AI products every year.

Powerful AI systems are already reshaping medicine, law, education, finance, transportation, and national security.

Yet there is a striking asymmetry: the investment in AI engineering far outpaces the investment in understanding why natural intelligence works the



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way it does and how those principles could be translated into machine systems.

This asymmetry is dangerous.

Building more powerful AI without understanding the principles of intelligence risks producing systems that are powerful but misaligned, capable but fragile, and impressive but fundamentally unlike minds.

The gap analysis presented here is not a pessimistic account of AI's failures.

It is a strategic map for what must be done next.

Why This Book Is Important - Reason Two:
Neuroscience Holds Crucial Keys

The nervous systems of organisms represent billions of years of evolutionary experimentation in the design of intelligence.

Natural selection is the most relentless optimizer that has ever existed.

Every inefficiency that could be removed from biological intelligence has faced extraordinary pressure to be removed over vast stretches of time.

The result is not a perfect system - biology is full of historical constraints and evolutionary accidents - but it is a remarkably effective one.

The human brain, weighing approximately 1.4 kilograms and consuming roughly 20 watts of power, routinely solves problems that the most powerful



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supercomputers in the world cannot solve reliably.
[1]

It learns from a handful of examples where AI systems need millions.

It generalizes across wildly different contexts where AI systems fail at even modest distribution shifts, meaning situations where real-world inputs differ from the data the system was trained on.
[16] [26]

It integrates emotion, perception, memory, reasoning, and social understanding seamlessly where AI systems treat each as a separate technical problem. [2] [25]

The corpus of neuroscience is rich with architectural and functional information that AI research has only begun to draw upon.

This book surveys that corpus and asks which parts of it matter most for closing the gap between NI and AI.

Why This Book Is Important - Reason Three: AGI and ASI Are Not Inevitable Without Direction

A common assumption in technology culture is that AI progress is simply a matter of scale - more data, more compute, more parameters, and AGI will eventually emerge.

This assumption is almost certainly wrong.

The history of AI is littered with moments of premature confidence.



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The field has passed through multiple periods of rapid optimism followed by what researchers call AI Winters – periods when progress stalled because foundational problems proved harder than expected.

AGI is not a guaranteed destination on the current road.

It may require entirely new research directions, new computational architectures, and new theoretical frameworks that do not yet exist.

Scaling refers to increasing model size, data volume, training compute, parameter count, and infrastructure capacity.

Scaling is important, but scaling without deeper architectural understanding is likely to encounter diminishing returns.

This book identifies the directions most likely to produce genuine progress toward AGI, as clearly as the current state of knowledge allows.

Deciding which research is appropriate and necessary is not a luxury – it is the precondition for success.

Why This Book Is Important – Reason Four: The Stakes Are Civilizational

If AGI or ASI is achieved, the consequences for human civilization will be profound.

Systems that can outperform human experts in science, engineering, medicine, and governance could accelerate progress in every domain of human flourishing simultaneously.



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They could also, if designed without wisdom and care, destabilize economies, concentrate power dangerously, undermine human agency, and potentially pose existential risks that no previous technology has matched.

The alignment problem - the challenge of ensuring that highly capable AI systems pursue goals that are genuinely beneficial to humanity - is inseparable from the gap analysis this book performs.

Understanding what it takes to build minds like ours, or greater than ours, is also the foundation for understanding how to build them safely.

Why This Book Is Important - Reason Five: The Time to Think Is Now

AI capabilities are advancing faster than AI understanding.

Models that were impossible five years ago are now commonplace.

Systems that will be commonplace five years from now may pose challenges for which we have no framework at all.

The time to do the deep intellectual work of understanding the gap between NI and AI - and planning the research agenda needed to close it - is now, while there is still time to shape the trajectory of development.

Structure of This Book



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The early chapters examine biological intelligence from multiple perspectives including neuroscience, evolutionary biology, cognitive science, and philosophy of mind.

The middle chapters examine the current state of AI systems and where they have drawn on biological insight.

The later chapters perform the core gap analysis and lay out a detailed research agenda.

The final chapters address ethical, safety, and civilizational dimensions.

The conclusion returns to the reasons this work matters and offers a forward-looking synthesis.

Chapter 20 through Chapter 25 provide resources for readers who wish to pursue the research agenda themselves, including a full glossary, an annotated reading list, and a full reference list.

PART ONE: NATURAL INTELLIGENCE

CHAPTER 1: DEFINING INTELLIGENCE: WHAT ARE WE ACTUALLY TRYING TO BUILD?

Before humanity can bridge the gap between Natural Intelligence and Artificial Intelligence, humanity must first define intelligence itself.

This task is far more difficult than it initially appears.

Intelligence is not a single mechanism or a single faculty.



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Intelligence is a multi-dimensional adaptive process that takes many forms across the biological world.

A useful way to begin is to notice that we recognize intelligence long before we can define it.

We see it in a toddler who, having burned a finger once, treats every glowing stove with new caution.

We see it in a raven that drops stones into a water-filled tube to raise a floating treat within reach.

We see it in a research team that, confronted with an unexpected experimental result, invents a concept that did not exist the day before.

In each case something more than mere reaction is happening: a system is using what it has learned to handle a situation it was never explicitly prepared for.

That capacity to handle the genuinely new is the thread we will follow throughout this book.

What Does the Word Mean?

At its most basic, intelligence refers to the capacity to acquire knowledge and skills and apply them flexibly to achieve goals in novel circumstances.

This definition, simple as it sounds, conceals enormous complexity.



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The phrase "apply them flexibly" is doing the heaviest lifting.

A thermostat acquires information about temperature and uses it to regulate heating.

But no one calls a thermostat intelligent, because its response repertoire is fixed, its goals are imposed entirely from outside, and it cannot adapt to circumstances outside its narrow range of design.

Intelligence, at minimum, requires flexibility, adaptability, and some degree of autonomous goal pursuit.

One useful operational definition is this: intelligence is the ability of a system to adaptively acquire, integrate, model, predict, and apply information across changing environments in pursuit of goals.

This definition contains several important components.

Adaptive means capable of responding flexibly to environmental change.

Acquire means gathering information through perception, interaction, memory, communication, or exploration.

Integrate means combining information into coherent representations.

Model means constructing internal predictive structures about reality.



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Predict means estimating future states.

Apply means using information to influence outcomes.

Goals refer to desired states, objectives, survival drives, learned preferences, or optimization targets.

It is worth dwelling on each of these components with a concrete example, because their interaction is precisely what current Artificial Intelligence struggles to reproduce.

Consider a person learning to cook in an unfamiliar kitchen.

They acquire information by looking in cupboards and tasting as they go.

They integrate that information with prior knowledge, recognizing that a particular sauce is too acidic and recalling that a pinch of sugar can balance acidity.

They model the dish as a system of interacting flavors, textures, and temperatures.

They predict that the sauce will thicken if it simmers a few minutes longer.

They apply that prediction by lowering the heat and waiting, all in service of the goal of a meal that tastes good to the people they are cooking for.

No single one of these abilities is intelligence on its own.



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Intelligence is their seamless integration in pursuit of a goal that matters to the system itself.

Natural Intelligence demonstrates all of these properties in rich and interconnected ways.

Artificial Intelligence currently demonstrates only partial versions of these properties.

A Brief History of Trying to Define Intelligence:

The difficulty of defining intelligence is not new, and the history of the attempt is instructive.

In 1904 the psychologist Charles Spearman observed that people who do well on one kind of mental test tend to do well on others, and he proposed a single underlying factor he called general intelligence, or *g*.

The idea of *g* captures something real about human ability, but it describes a statistical regularity rather than a mechanism, and it says nothing about how the underlying processing actually works.

In the mid-twentieth century Raymond Cattell drew a distinction between fluid intelligence, meaning the ability to reason about and solve genuinely novel problems, and crystallized intelligence, meaning the accumulated knowledge and skills a person has already acquired.

This distinction matters enormously for our purposes, because today's most impressive Artificial Intelligence systems display vast



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crystallized knowledge while remaining strikingly weak in fluid reasoning when faced with situations far from their training data.

Later thinkers widened the lens further.

Howard Gardner argued for multiple intelligences, proposing that linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, interpersonal, and intrapersonal abilities are partly distinct rather than expressions of one general factor.

Robert Sternberg proposed a triarchic theory, distinguishing analytical intelligence (solving well-defined problems), creative intelligence (generating novel ideas), and practical intelligence (the everyday know-how of getting things done in the real world).

Practical intelligence is the dimension current systems most conspicuously lack, because it rests on tacit knowledge, the unspoken know-how people absorb from experience and can rarely put into words, and a benchmark built from written questions cannot easily test it.

These frameworks disagree about the structure of intelligence, but they agree on something important: intelligence is plural, layered, and context-dependent rather than a single quantity that a system simply has more or less of.

Any definition narrow enough to be measured by one test is almost certainly too narrow to describe what brains actually do.



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Two modern attempts try to widen the lens while keeping it precise.

Shane Legg and Marcus Hutter proposed in 2007 a formal definition of intelligence as an agent's average ability to achieve goals across all possible environments, mathematically elegant but uncomputable in practice.

Francois Chollet sharpened the practical version in 2019 by defining intelligence as skill-acquisition efficiency, the rate at which a system converts limited experience into competence on tasks it has never seen, which reframes the goal of the field from accumulating skills to acquiring them quickly.

The Turing Test and the Limits of Behavior:

The most famous attempt to sidestep the definitional problem came from the mathematician Alan Turing in 1950.

Rather than ask whether a machine can think, Turing proposed an imitation game, now called the Turing Test, in which a machine is judged intelligent if a human conversing with it through text cannot reliably distinguish it from another human.

The Turing Test was a brilliant move because it replaced an unanswerable question about inner experience with a concrete, observable behavioral test.

But it also embedded an assumption that has shaped, and arguably misled, the field ever since: the assumption that intelligence can be fully



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assessed from the outside, by examining behavior alone.

The philosopher John Searle challenged this assumption directly in 1980 with a thought experiment called the Chinese Room.

Searle imagined a person who speaks no Chinese sitting in a room, following an elaborate rulebook that tells them which Chinese symbols to pass back out in response to Chinese symbols passed in.

To an outside observer the room appears to understand Chinese, yet the person inside understands nothing; they are merely manipulating symbols according to rules.

Searle's point was that producing the right outputs is not the same as understanding, and the distinction between competent behavior and genuine comprehension will recur throughout this book.

Modern large language models make this distinction urgent rather than academic, because they can now pass casual versions of the Turing Test while still, by most accounts, lacking grounded understanding of what their words refer to.

Natural Intelligence as a Biological Phenomenon:

Natural Intelligence, as found in biological organisms, adds additional properties that are extraordinarily hard to replicate: embodiment (meaning intelligence that emerges from a body interacting with a physical world), affect (meaning the capacity to feel states that motivate and shape behavior), social understanding, temporal depth (meaning the ability to reason



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across past, present, and future), and the kind of creativity that generates genuinely novel solutions to genuinely novel problems.

A human child can learn language, motor coordination, emotional communication, social norms, and physical reasoning simultaneously and from sparse data.

The same child does this without a labeled dataset, without a reset button, and without ever seeing more than a tiny fraction of the sentences they will eventually understand.

An octopus can solve novel puzzles it has never encountered before, unscrewing a jar from the inside to reach a crab, despite having a nervous system organized very differently from our own.

A crow can use tools, and New Caledonian crows will even craft hooked tools from twigs, selecting and shaping the raw material to fit the task.

A dolphin can coordinate socially in sophisticated ways, with some populations inventing hunting techniques that are then passed culturally from one generation to the next.

An ant colony can self-organize and optimize collectively without any central controller, discovering short paths to food through nothing more than local interactions and chemical trails.

A human brain can operate continuously for decades using approximately twenty watts of power, roughly the draw of a dim light bulb.



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Modern AI systems often require massive datasets, enormous energy expenditures, expensive computational infrastructure, extensive human supervision, and narrowly constrained operating environments to perform tasks that biological organisms accomplish as a matter of routine.

The contrast is not merely quantitative but qualitative, and understanding why is the central project of the chapters that follow.

The Problem with Output-Only Definitions of Intelligence:

Artificial Intelligence has historically been defined by the ability to perform tasks that, if a human were to perform them, would require intelligence.

This functionalist definition focuses on the output rather than the process.

This is the behaviorist inheritance in AI, and it explains why the field has so often mistaken a system that produces the right answers for a system that understands them.

An AI is often considered intelligent if it can play chess, translate languages, or recognize faces.

However, this output-oriented definition obscures the massive internal differences between a brain and a large language model.

A Large Language Model (LLM) is a type of artificial intelligence designed to understand and



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generate human language by processing vast amounts of text data.

While an LLM can produce impressive language output, it lacks the survival drive, the continuous memory, the physical grounding, and the integrated motivational systems that define biological intelligence.

The danger of output-only definitions is that they reward systems for appearing intelligent rather than for being robust, and the two can diverge sharply.

A system trained to maximize a benchmark score will often find shortcuts that exploit the quirks of that benchmark, achieving high marks without acquiring the underlying competence the benchmark was meant to measure.

This phenomenon, sometimes called shortcut learning, is one reason an AI can score at a professional level on a standardized test and then fail a slightly reworded version of a problem a child could solve.

The gap begins at the definition.

We have built machines that are savants - entities with incredible skill in narrow domains but lacking the general adaptability and robustness of even a simple mouse.

A telling symptom of this savant character is brittleness, since an image classifier that can surpass humans may be flipped to a confident error by altering a handful of pixels in ways no person would even notice, an adversarial example that



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reveals how shallow the underlying competence can be.

To bridge the gap, we must expand our conception of AI to include the structural and functional principles of NI.

We must move beyond metrics that evaluate only outputs and toward standards of intelligence based on architectural fidelity, genuine generalization, and functional robustness.

Narrow AI Versus General Intelligence:

Most current AI systems are narrow systems, meaning systems optimized for specific tasks.

Examples include image classification, speech recognition, text generation, recommendation systems, navigation systems, fraud detection, and strategic gameplay.

These systems may appear highly intelligent within their domains.

However, they typically lack persistent self-awareness, robust causal reasoning (the ability to understand cause and effect rather than mere statistical correlation), long-term autonomy, deep contextual grounding, flexible cross-domain transfer, general world understanding, stable identity continuity, independent motivation, embodied interaction, and real-time developmental learning.

The difference can be summarized with a single illustrative contrast.



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A narrow chess engine can defeat any human alive, yet it cannot be asked to set up the board, notice that a piece is missing, improvise a substitute, or explain to a child why the game is fun.

A five-year-old can do all of those things and play a passable game besides, and that breadth, not depth in any single skill, is the hallmark of general intelligence.

Biological organisms often possess these characteristics naturally, and they interweave them seamlessly.

This reveals one of the most fundamental intelligence gaps.

Natural Intelligence is highly integrated.

Artificial Intelligence is frequently fragmented.

Why the Definition Matters for the Rest of This Book:

It might seem that arguing over definitions is a philosopher's indulgence, a delay before the real engineering begins.

The opposite is true.

How we define intelligence determines what we measure, what we measure determines what we optimize, and what we optimize determines what we build.

A field that defines intelligence as benchmark performance will build ever-better benchmark performers.



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A field that defines intelligence as adaptive, integrated, embodied, goal-directed competence will build something far closer to the minds nature has already produced.

This book takes the second path deliberately.

In the chapters that follow we will examine how evolution engineered intelligence, how the brain implements it, and precisely where today's artificial systems fall short, before turning to a concrete research agenda for closing the gap.

The goal is not to diminish the genuine achievements of modern AI, which are remarkable, but to measure them against the right standard, so that the path forward is judged by the distance still to travel rather than the distance already covered.

CHAPTER 2: EVOLUTION AS THE ARCHITECT OF INTELLIGENCE:

Natural Intelligence did not emerge instantly or through top-down design.

It emerged through billions of years of evolution.

Evolution refers to adaptive change across generations through variation, selection, inheritance, and environmental pressure.

Every nervous system on Earth is the product of evolutionary optimization, meaning it was shaped by survival pressures over immense stretches of time.



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To appreciate the scale of this process, it helps to lay out a rough timeline.

The first true neurons appeared in simple animals more than 600 million years ago, well before the Cambrian explosion that filled the oceans with complex body plans.

The first centralized brains, formed by the clustering of neurons at the front of mobile animals in a process biologists call cephalization, followed as movement made it advantageous to concentrate sensing and decision-making where the animal met the world first.

Mammalian brains, with their distinctive six-layered neocortex, are a recent development of the last 200 million years, and the dramatic expansion of the human prefrontal cortex occupies only the last few million years, an eye-blink in evolutionary time.

The lesson of this timeline is humbling: the intelligence we are trying to reverse-engineer is the refined output of an optimization process that ran, in parallel, across countless lineages, for longer than there have been bones.

How Evolution Built Minds

Evolution solved problems incrementally and opportunistically.

This is critically important because modern Artificial Intelligence often attempts to engineer intelligence directly through top-down design, assembling systems from first principles.



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Biological intelligence emerged through bottom-up adaptation, meaning small improvements accumulated over vast time scales, each one selected because it helped organisms survive and reproduce in particular environments.

The molecular biologist François Jacob captured the character of this process with the word *bricolage*, a French term for the work of a tinkerer who builds new things from whatever odds and ends are already at hand.

Evolution does not design from a blank sheet; it modifies, duplicates, and repurposes existing structures, which is why the same genes that pattern a fly's body also help pattern ours, and why the human eye is wired in a way no engineer would choose.

A particularly important form of this tinkering is *exaptation*, meaning the co-opting of a trait that evolved for one purpose to serve a completely different one, as when feathers that first aided temperature regulation were later recruited for flight.

Much of higher cognition appears to be *exaptation* of this kind, with circuits that evolved for navigation and motor control repurposed for abstract reasoning and language.

A concrete instance is the hippocampus, whose place cells and grid cells evolved to map physical space but appear to have been recruited to organize abstract relationships as well, since brain imaging reveals grid-like activity when people navigate spaces of pure concepts, so that



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we may think through idea-landscapes using machinery first built for crossing terrain.

Evolution optimized for survival, reproduction, energy efficiency, robustness, adaptability, and environmental integration.

Modern AI systems are often optimized for benchmark performance, meaning performance on standardized tests used to evaluate capability.

These objectives are not equivalent, and the difference matters enormously for the resulting architecture.

An organism that is brilliant but starves has been selected against, so biological intelligence is everywhere constrained by metabolic cost in a way that benchmark-driven AI is not, and that single constraint shapes much of what makes brains so efficient.

The tax is steep, because the brain consumes roughly a fifth of the body's energy while making up only about two percent of its mass, a pressure that relentlessly favored designs computing more thought per joule.

A key insight from evolutionary biology is that intelligence is deeply embodied.

Organisms evolved inside ecological systems.

Perception evolved because organisms needed to navigate environments.

Memory evolved because organisms benefited from retaining useful information.



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Social cognition evolved because coordination improved survival.

Emotions evolved because rapid motivational prioritization improved decision making.

Pain evolved because damage avoidance improved survival.

Each of these drives is a piece of inherited wisdom about what once mattered for survival, and together they form a layered motivational system that no current AI possesses, since a model feels no hunger, no fear, and no curiosity of its own.

Curiosity evolved because exploration increased adaptive flexibility.

Several of these drives have no counterpart in machines, and their absence is not a minor omission but a structural one, because in living things they are the very source of the goals that perception and reasoning then serve.

Biological intelligence is therefore inseparable from evolutionary pressures, and understanding those pressures illuminates what intelligence actually is and why it has the properties it does.

It is worth noting the contrast between how evolution searches and how today's AI learns.

Evolution is a massively parallel, population-level search that explores many designs at once, keeps what works, and has no goal beyond persistence, whereas gradient descent, the optimization method behind modern neural networks,



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is a directed search that nudges a single system steadily toward lower error on a fixed objective.

Each approach has strengths, but only the evolutionary one has so far produced open-ended, general, self-maintaining intelligence, and that fact alone should make AI researchers curious about what evolution knows that gradient descent does not.

The Spectrum of Evolved Intelligence:

One of the most important and underappreciated facts about intelligence is how many different forms it takes in the natural world.

The human brain is the most elaborated example of the primate brain plan, which emphasizes a very large prefrontal cortex (the region at the front of the brain most associated with planning, reasoning, and social cognition) and an unusually rich connectivity between brain regions.

But human-type intelligence is not the only kind of intelligence that exists, nor is it necessarily the best possible kind for every purpose.

The octopus represents a completely independent evolution of sophisticated problem-solving ability, with most of its neurons distributed throughout its eight arms rather than concentrated in its central brain.

An octopus can learn by observation, solve puzzles, use tools, recognize individual humans, and adapt its behavior to complex and changing environments – all with a nervous system organized



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on entirely different architectural principles from the vertebrate brain.

The octopus matters to AI research precisely because it proves that high intelligence is not tied to one body plan or one wiring diagram; it is a solution that evolution found more than once, by more than one route.

Corvid birds - crows, ravens, jays, and their relatives - demonstrate sophisticated planning, tool use, social reasoning, and something that at least resembles an understanding of other minds, all within brains that are tiny by mammalian standards but organized with impressive computational efficiency.

Honeybee colonies exhibit collective intelligence that allows the group to solve optimization problems through a process of distributed deliberation that has no single decision-maker.

When a honeybee swarm chooses a new nest site, scouts advertise candidate locations through their waggle dances, recruit other scouts, and the colony converges on a high-quality choice through a decentralized competition of evidence, with no bee in charge and no bee holding the whole picture.

Thomas Seeley showed that the swarm effectively reaches a quorum, with the first site to gather enough committed scouts winning, a decentralized voting scheme strikingly similar to how competing populations of neurons settle a decision inside the primate brain.



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The roundworm *Caenorhabditis elegans* (often abbreviated *C. elegans*) possesses exactly 302 neurons, yet it exhibits complete sensorimotor integration, searches for food, avoids danger, mates, and learns from negative experiences.

Its entire wiring diagram was mapped by John White and colleagues in 1986, the first connectome ever completed, yet decades of effort by the OpenWorm project to turn that diagram into a working simulation show that wiring alone does not reproduce the animal's behavior, a humbling reminder that structure without dynamics is not enough.

It accomplishes all of these tasks autonomously using zero global backpropagation, zero digital processors, and consuming a fraction of a microwatt of power.

Current AI models with hundreds of billions of parameters cannot navigate a physical room with the agility, resilience, and adaptive intelligence of a common housefly.

This stark reality emphasizes that the path to AGI and ASI is not a matter of simply adding more parameters to current architectures or expanding digital data centers.

It requires understanding what biological intelligence actually does and how it actually does it.

Lessons from Evolutionary Design for AI Research:

The study of evolution provides several direct lessons for AI research.



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First, intelligence exists on a spectrum, and studying diverse organisms may reveal alternative computational paradigms that human-centric AI research has overlooked.

Insects demonstrate swarm intelligence, meaning collective problem-solving by simple agents following local rules.

Octopuses demonstrate distributed cognition, meaning processing that is spread across the body rather than centralized.

Fungi demonstrate network coordination across vast distances, with mycelial networks routing resources and signals through structures that some researchers compare to biological circuits.

Each of these represents an alternative solution to the problem of intelligence that nature discovered and tested over millions of years.

Second, evolution demonstrates that intelligence systems require open-ended development.

Open-ended evolution refers to systems capable of continual innovation without predefined endpoints.

Biological evolution never truly stopped.

AI training systems often converge toward fixed objectives and stop changing once training is complete.

This difference may limit long-term generalization and adaptability, and it is one reason a deployed



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model that meets a frozen world can quietly fall out of step with a world that keeps moving.

Third, evolution demonstrates that hierarchical organization is a fundamental principle of intelligent systems.

Hierarchical organization refers to systems composed of layered structures with increasing abstraction.

The nervous system contains sensory layers, reflex systems, motor systems, emotional systems, memory systems, predictive systems, planning systems, social cognition systems, and executive control systems.

Modern AI architectures contain only simplified approximations of these layers.

A fourth lesson, easy to overlook, is robustness through redundancy and degeneracy: evolution rarely places a critical function in a single fragile component, and biological systems routinely keep working after injury, noise, or the loss of cells in a way that brittle engineered systems do not.

Potential research programs inspired by evolution include comparative neuroscience, evolutionary simulation environments, artificial life research, neuromorphic hardware, swarm intelligence systems, synthetic developmental systems, evolutionary reinforcement learning, open-ended learning environments, and self-organizing computational systems.



CHAPTER 3: THE ARCHITECTURE OF THE BRAIN: WHAT NEUROSCIENCE TELLS US:

The human brain is the most sophisticated known biological information processing system.

Understanding its architecture is essential for the development of AGI.

Scale and Structure:

The brain contains approximately 86 billion neurons, which are specialized cells that process and transmit information through electrical and chemical signaling.

These neurons are connected by an estimated 100 trillion synapses, a number worth pausing on, because it dwarfs the parameter counts of even the largest artificial neural networks while running on a power budget smaller than a household light bulb.

The brain also contains vastly larger numbers of glial cells.

Glial cells are non-neuronal cells that support neural function, regulate metabolism, influence signaling, manage the chemical environment, clear metabolic waste, and contribute to learning processes.

For most of the history of neuroscience glia were dismissed as mere packing material, but that view has collapsed: astrocytes actively modulate synaptic transmission and can influence learning, oligodendrocytes wrap axons in the myelin insulation that sets the brain's communication



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speed, and microglia continuously prune and remodel synapses, sculpting circuits throughout life.

Any account of the brain that counts only neurons is therefore leaving out much of the machinery, and AI's neuron-only abstraction inherits that omission.

Astrocytes take part in what researchers call the tripartite synapse, sensing and responding to the very signals neurons exchange, and a single human astrocyte can enwrap hundreds of thousands of synapses at once, putting it in a position to coordinate rather than merely support.

Modern AI systems are loosely inspired by neurons, but artificial neural networks are highly simplified abstractions.

Real neurons are vastly more complex.

Neurons possess dynamic signaling behaviors, temporal firing patterns, chemical modulation, structural plasticity, adaptive thresholds, nonlinear interactions, multi-scale communication, and long-range connectivity.

Neurons connect to one another through junctions called synapses, which are structures that permit a neuron to pass an electrical or chemical signal to another neuron.

In biological systems, memory is not retrieved from a distant storage location to be processed elsewhere.



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Instead, memory is structural, embedded directly within the strength, configuration, and physical geometry of the synapses themselves.

This co-localization of memory and processing eliminates data transfer latency and allows for massive parallelism that silicon architectures can only simulate at immense energetic cost.

Von Neumann Architecture Versus Biological Architecture:

Modern computers rely on the von Neumann architecture, which is a design framework that separates a Central Processing Unit (CPU) and memory storage unit and connects them by a physical bus.

This separation creates a fundamental performance constraint known as the von Neumann bottleneck, where data transfer speeds between the processor and memory limit overall computational throughput.

A great deal of the energy a computer spends on a large neural network is spent not on arithmetic but on shuttling numbers back and forth across this bottleneck, which is precisely the cost biology avoids by storing and computing in the same place.

Biological brains are organized as highly recurrent, multi-directional networks with extensive feedback loops and lateral connections.

Every cortical area in a mammalian brain sends as many fibers back to its input sources as it receives from them.



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In fact the descending feedback projections often outnumber the ascending ones, so the cortex devotes more of its wiring to telling lower areas what to expect than to listening to what they report, a ratio that makes sense only if the brain is fundamentally a predictive organ.

This structural recurrence enables continuous prediction, real-time contextual modulation, and a persistent internal state that exists independently of immediate sensory inputs.

Silicon chips operate using deterministic, high-frequency binary signaling where voltages cycle billions of times per second to switch transistors between absolute states of zero and one.

Biological neurons communicate using Action Potentials (AP), which are rapid, transient changes in membrane potential that propagate along cell membranes as discrete electrical spikes.

This process is called spiking computation, and it is inherently asynchronous (not synchronized to a clock), sparse (most neurons are silent most of the time), and probabilistic (outcome is not perfectly determined).

A biological neuron remains completely silent until its internal electrical threshold is breached, meaning information is encoded not just in continuous values but in the exact timing and frequency of individual spikes.

This sparse signaling paradigm allows the human brain to operate its entire cognitive apparatus on approximately twenty watts of power.



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For comparison, a single high-end graphics processor used to train AI can draw several hundred watts on its own, and one large training run can consume more electricity over a few weeks than a human brain would use across an entire lifetime.

Conversely, a modern data center training an advanced LLM requires megawatts of electrical power, requiring dedicated cooling infrastructure and producing massive carbon output.

The architectural contrast can be summarized simply: the computer separates memory from processing, runs on a global clock, signals densely, and is deterministic, while the brain fuses memory with processing, runs without a clock, signals sparsely, and is probabilistic, and almost every efficiency advantage of biology traces back to that list of differences.

The Major Regions of the Brain and Their Functions:

The brain is deeply hierarchical and regionalized, with different areas contributing to specialized functions.

The visual cortex processes visual information in hierarchical layers building from simple edges to complex objects.

The full extent of this organization was charted by Daniel Felleman and David Van Essen, who mapped more than thirty interconnected visual areas arranged in a layered hierarchy, a biological wiring diagram that prefigured the deep layered networks of modern computer vision.



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The hippocampus, a seahorse-shaped structure in the medial temporal lobe, is crucial for the formation of long-term memories and spatial navigation.

The cerebellum contributes to motor coordination and timing, and although it is often left out of discussions of higher cognition, it contains the majority of the brain's neurons and increasingly appears to refine thought itself, not only movement.

The prefrontal cortex, the front portion of the frontal lobe, contributes to executive control, planning, decision-making, and social cognition.

The amygdala, an almond-shaped structure in the temporal lobe, contributes to emotional processing, fear responses, and the emotional tagging of memories.

The thalamus acts as a relay station, routing information between brain regions.

The basal ganglia, a group of structures deep in the brain, contribute to habit formation, reward learning, and the selection among competing actions.

The Default Mode Network (DMN) is a set of interconnected brain regions that is active during rest and deactivated during focused external tasks.

The spontaneous activity of the DMN is not waste.



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It appears to be doing something important: consolidating memories, simulating possible futures, generating creative associations, and maintaining the self-model that anchors cognition.

Current AI systems have no equivalent of spontaneous, self-directed cognitive activity.

The brain is in fact never idle, spending most of its energy on this intrinsic ongoing activity rather than on responding to stimuli, which is why some neuroscientists describe perception as the brain's continuous internal fantasy, gently corrected by the senses.

They process inputs when given them and do nothing otherwise.

This is one of the profound architectural differences between NI and AI that the gap analysis will return to repeatedly.

It is important to add that these regions are not independent modules wired in isolation; they form richly interconnected networks, and most real cognition emerges from their coordinated interaction rather than from any single area acting alone.

Neuromodulatory Systems:

The brain's activity is profoundly shaped by neuromodulatory systems – diffuse networks of neurons that release chemical messengers called neuromodulators across wide areas of the brain, influencing the excitability and plasticity of neural circuits.



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The major neuromodulatory systems are:

The dopaminergic system - neurons in the midbrain that release dopamine throughout the brain, playing crucial roles in reward prediction, motivation, reinforcement learning, and the regulation of attention and working memory.

The noradrenergic system - neurons in the locus coeruleus region that release norepinephrine throughout the brain, regulating arousal, attention, and the consolidation of emotionally significant memories.

The serotonergic system - neurons in the raphe nuclei that release serotonin throughout the brain, regulating mood, sleep, appetite, and social behavior.

The cholinergic system - neurons in the basal forebrain and brainstem that release acetylcholine throughout the brain, playing crucial roles in attention, learning, and the encoding of new memories.

These neuromodulatory systems do not carry specific information in the way that sensory systems do.

Instead, they set the overall context for information processing - determining how plastic the system is (how readily it can change and learn), how attentive it is, how motivated it is, and how it weights new information against stored knowledge.

A helpful analogy is that if the neurons are the orchestra, the neuromodulators are the conductor



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and the lighting, setting tempo, mood, and emphasis without playing a single note themselves.

AI systems have no equivalent of neuromodulatory regulation.

This is a significant gap, because neuromodulation is likely central to the flexibility, adaptability, and context-sensitivity that characterize biological intelligence.

The Brain as a Predictive System:

One of the most productive theoretical frameworks in contemporary neuroscience is the predictive processing framework – the idea that the brain is fundamentally a prediction machine.

Rather than passively receiving sensory input and processing it bottom-up (from simple to complex), the brain constantly generates predictions about what sensory input it expects to receive and compares those predictions to actual input.

The only thing that propagates widely through the processing hierarchy is the prediction error – the difference between what was expected and what actually arrived.

This architecture is extraordinarily efficient because the brain only needs to transmit and process surprises rather than the full sensory signal.

A familiar experience makes the idea concrete: you do not notice the steady pressure of the clothes on your skin until this sentence directs your attention to it, because your brain predicted that



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sensation and the prediction error was near zero, so nothing was worth reporting.

The predictive processing framework, developed most fully by Karl Friston's Free Energy Principle (a mathematical framework describing how biological systems minimize surprise by either updating their internal model of the world or acting to make the world match their predictions), may be one of the most important theoretical frameworks for guiding the design of AGI-capable AI systems.

CHAPTER 4: NEURAL COMPUTATION: FROM NEURONS TO NETWORKS

The Neuron as a Computing Unit:

The individual neuron is the basic computing unit of the brain, but calling it a basic unit is somewhat misleading because real neurons are enormously complex.

A biological neuron is not simply a binary switch that is either on or off.

The neuron integrates thousands of synaptic inputs, each weighted by the strength of the synapse and the position of the synapse on the dendritic tree (the branching extensions of the neuron that receive incoming signals).

Recent research has shown that individual dendrites can perform complex nonlinear computations, effectively acting as independent computational subunits within a single neuron.



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In particular, dendritic branches can generate their own local spikes, including events driven by NMDA receptors, so that a single neuron behaves less like one summing unit and more like a small multi-layer network in its own right.

This means that a single biological neuron may have the computational power of a small neural network in the deep learning sense, and some estimates suggest it would take an artificial network of several layers just to mimic the input-output behavior of one cortical cell.

A 2021 study by David Beniaguev, Idan Segev, and Michael London made this precise, finding that capturing the input-output behavior of a single cortical pyramidal neuron required a deep artificial network of five to eight layers, a vivid measure of how much computation one biological cell performs.

The timing of spikes matters enormously.

A spike that arrives a few milliseconds early or late can carry an entirely different meaning, so the brain encodes information partly in a temporal code that the rate-based units of standard deep learning simply discard.

Neurons communicate not only through rate coding (how often they fire) but through temporal coding (precisely when they fire relative to other neurons and relative to ongoing oscillatory rhythms in the local network).

Current artificial neural networks model neurons as simple weighted summing devices with a nonlinear activation function.



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This is a profound simplification that discards most of the computational richness of biological neurons, and it is sobering to realize that the unit the entire deep learning revolution is built on is a cartoon of the cell it is named after.

Synaptic Plasticity - How Connections Change

The brain's ability to learn depends on synaptic plasticity, which is the ability of synapses to change their strength based on the activity patterns of the neurons they connect.

The foundational principle of synaptic plasticity was articulated by Canadian psychologist Donald Hebb in 1949.

Neurons that fire together wire together.

More precisely, if a presynaptic neuron (the neuron sending a signal) consistently fires shortly before a postsynaptic neuron (the neuron receiving the signal), the synapse between them is strengthened through a process called Long-Term Potentiation (LTP).

If the timing is reversed, the synapse is weakened through a process called Long-Term Depression (LTD).

A more precise biological manifestation is Spike-Timing-Dependent Plasticity (STDP), which is a biological process that adjusts the strength of connections between neurons based on the relative timing of their spikes.



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This localized mechanism allows billions of synapses to adapt independently and simultaneously without requiring a global coordinator or an overall mathematical gradient calculation.

The contrast here points directly at one of the deepest open problems in the field, the credit assignment problem, meaning the question of how a system decides which of its many connections deserve credit or blame for an outcome.

In contrast, the most successful learning algorithm in current AI – backpropagation (a method for computing how much each connection weight in a neural network should change to reduce the network's prediction error, by propagating error signals backward through the network from output to input) – requires a global error signal to be transmitted backward through the entire network.

This backward transmission of error signals is considered biologically implausible, and the question of how the brain implements something functionally equivalent without this mechanism is an important open research question.

Proposed biologically plausible alternatives, including feedback alignment, predictive coding, and equilibrium propagation, are active areas of research precisely because solving local credit assignment could unlock learning that is far cheaper and more brain-like than backpropagation.

Feedback alignment, demonstrated by Timothy Lillicrap and colleagues in 2016, produced the surprising result that even fixed, random backward connections can carry a useful learning signal,



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showing that the brain need not implement backpropagation exactly in order to achieve something that works as well.

Additional forms of synaptic plasticity beyond Hebbian mechanisms include homeostatic plasticity (a process by which neurons scale their overall sensitivity up or down to keep their activity levels within a functional range) and metaplasticity (changes in the threshold for inducing plasticity rather than changes in synaptic strength directly).

These forms of plasticity work together to produce a system that can learn rapidly when learning is appropriate and maintain stability when stability is needed – a balance that current AI systems achieve only crudely, and a balance whose failure modes, runaway change on one side and frozen rigidity on the other, will reappear in the discussion of catastrophic forgetting later in this book.

Neural Oscillations and Rhythms:

One of the most striking features of brain activity is its rhythmic character.

Neurons in the brain do not fire independently.

They synchronize their activity into oscillatory patterns at characteristic frequencies, and these oscillations play functional roles in coordination, communication, and cognition.

The major frequency bands include delta waves (0.5 to 4 Hz) associated with deep sleep, theta waves (4 to 8 Hz) associated with spatial navigation and



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memory encoding, alpha waves (8 to 12 Hz) associated with relaxed wakefulness, beta waves (12 to 30 Hz) associated with active thinking and sensorimotor processing, and gamma waves (30 to 80 Hz and higher) associated with local cortical computation and attention.

Hz refers to Hertz, a unit of frequency meaning cycles per second.

The coordination of these oscillations across brain regions appears to be a mechanism for routing information – determining which regions communicate with which, and when.

A well-studied example is theta-gamma coupling, in which fast gamma cycles are nested inside slower theta cycles, a structure thought to help the brain hold several items in working memory at once by giving each item its own slot in the rhythm.

This dynamic routing of information through oscillatory synchrony is one of the most interesting computational mechanisms in the brain and has no clear counterpart in current AI systems.

Inhibitory Neurons and Circuit Balance:

Brain circuits are not composed solely of excitatory connections that increase the activity of downstream neurons.

Inhibitory interneurons (neurons that release inhibitory neurotransmitters and primarily connect to other neurons within the same local circuit) play a crucial and often underappreciated role.



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Some inhibitory neurons provide feedforward inhibition that sharpens the timing of responses.

Others provide feedback inhibition that limits overall circuit activity and prevents runaway excitation.

Still others provide lateral inhibition (the suppression of neighboring neurons by an active neuron) that enhances the selectivity of responses.

The balance of excitation and inhibition in neural circuits is carefully regulated and is critical for normal cognitive function, and its disruption is implicated in conditions ranging from epilepsy, where excitation overwhelms inhibition, to disorders of attention and perception.

AI neural networks have inhibitory connections in the sense that some weights can be negative, but they lack the rich diversity of inhibitory interneuron types and the sophisticated local circuit dynamics that biological inhibition produces.

The cortex maintains roughly four excitatory neurons for every inhibitory one and dozens of distinct interneuron types, and one thing this machinery computes is divisive normalization, a canonical operation that rescales each neuron's response by the activity of its neighbors and that engineers have since rediscovered as a useful component of artificial networks.

CHAPTER 5: MEMORY, LEARNING, AND PLASTICITY IN BIOLOGICAL SYSTEMS:



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The Multiple Memory Systems of the Brain:

One of the most important discoveries in twentieth-century neuroscience was that memory is not a single faculty but a collection of distinct systems, each with its own neural substrate, its own rules for encoding and retrieval, and its own relationship to conscious awareness.

Neuroscientists often organize these systems into two broad families: declarative memory, meaning the memories we can consciously bring to mind and put into words, and non-declarative memory, meaning the skills, habits, and conditioned responses that shape behavior without ever entering conscious awareness.

This division is not a tidy abstraction; it was carved out of tragedy and clinical observation, most famously in the case of the patient known for decades only as H.M.

After surgeons removed much of H.M.'s hippocampus on both sides of his brain to control severe epilepsy, he lost the ability to form new conscious memories of events, yet he could still learn new motor skills, improving day by day at tasks he had no memory of ever practicing.

H.M. demonstrated, more clearly than any theory could, that the brain stores "knowing that" and "knowing how" in different places, and that a single word, memory, conceals several distinct machines.

He was studied for decades by the neuropsychologist Brenda Milner, whose work helped found modern memory science, while at the



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molecular scale Eric Kandel traced how learning physically strengthens synapses in the sea slug *Aplysia*, work recognized with a Nobel Prize in 2000.

The major memory systems of the brain include:

Episodic Memory - the system for storing and retrieving specific personal experiences, tied to a specific time and place.

- This is the most distinctively human form of memory, requiring not only storage of what happened but of when, where, and the subjective sense of re-experiencing the event.
- Episodic memory depends critically on the hippocampus and is the type most severely affected by Alzheimer's disease.

Semantic Memory - the system for storing general knowledge about the world, independent of the specific context in which it was acquired.

- Knowing that Paris is the capital of France is an example of semantic knowledge.
- Semantic memory also depends on the hippocampus for initial acquisition but becomes progressively independent of it through systems consolidation, which is the gradual transfer of memory from hippocampus-dependent to cortex-dependent storage over weeks, months, or years.

Procedural Memory - the system for storing motor skills and cognitive habits.



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- Riding a bicycle, typing, or playing a musical instrument are procedural memories.
- Procedural memory depends on the basal ganglia and the cerebellum rather than the hippocampus.

Working Memory - the capacity to maintain and manipulate a small amount of information in an active, accessible state over short periods.

- Working memory is the mental workspace where current thinking takes place.
- It depends critically on the prefrontal cortex and is severely limited in capacity, typically holding approximately four items simultaneously.

George Miller's famous 1956 paper put the figure near seven, later revised down to about four, and showed that the brain defeats this tight limit through chunking, the grouping of individual items into larger meaningful units, which is how a chess master can hold an entire board in mind as a handful of familiar patterns.

Spatial Memory - the system for encoding and retrieving information about spatial layouts and navigational paths.

- Spatial memory is mediated by specialized neurons called place cells and grid cells in the hippocampal system, a discovery so fundamental that it earned the Nobel Prize in 2014.

Place cells, each firing when the animal occupies a particular location, were discovered by John O'Keefe in 1971, and grid cells, which fire in a striking hexagonal lattice that tiles space like

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graph paper, were discovered by May-Britt and Edvard Moser, the three sharing the 2014 prize.

Emotional Memory - the system for storing emotionally significant experiences, which tends to produce especially strong and lasting memories.

- Emotional memory depends heavily on the amygdala and its interactions with the hippocampus.

Current AI systems do not fully replicate these distinctions and typically treat memory as a single undifferentiated store, usually the fixed weights of a trained network, which conflates the rapid, specific recall of an event with the slow, general extraction of knowledge that biology keeps deliberately separate.

A further property that biological memory possesses and most AI lacks is reconsolidation, meaning that when a memory is recalled it briefly becomes labile, or modifiable, and must be stabilized again, so that remembering is not a passive readout but an act that can update the memory itself.

Reconsolidation helps explain why human memory is reconstructive rather than photographic, and why it can be both flexibly updated and, at times, distorted, a trade-off any artificial memory system will eventually have to confront.

The Complementary Learning Systems Theory:

A landmark theoretical contribution to understanding biological memory is the Complementary Learning Systems (CLS) theory.



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Its central claim is that the brain needs two learning systems at once, a fast hippocampus that can memorize a single episode without disturbing old knowledge and a slow neocortex that interleaves many experiences to distill general structure, and the theory directly inspired the experience-replay buffers that later made deep reinforcement learning stable.

The CLS theory proposes that the brain has two distinct learning systems with complementary properties.

The first system is the hippocampus, which is capable of rapid, one-shot encoding of new experiences but must keep these representations sparse and separated to avoid interference between memories.

The second system is the neocortex (the largest and evolutionarily newest part of the cerebral cortex), which learns slowly by extracting statistical regularities across many experiences and represents them in overlapping, distributed representations that support generalization.

These two systems work together.

The hippocampus rapidly encodes new experiences and gradually replays them to the neocortex during sleep and rest, allowing the neocortex to slowly incorporate them into its statistical knowledge without forgetting previously learned patterns.

The CLS theory directly predicts and explains one of the most debilitating problems in AI - catastrophic forgetting (the tendency of an artificial neural network to completely erase



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previously learned knowledge when trained on a new task), and it points toward architectural solutions that implement something analogous to the two-system architecture.

The most direct echo of CLS in modern AI is the replay buffer, a store of past experiences that a learning system revisits so that new learning does not overwrite old, and more recent work on generative replay, in which a model dreams up synthetic samples of old tasks to rehearse, is essentially an engineering rediscovery of what the sleeping hippocampus appears to do every night.

Sleep, Consolidation, and the Offline Mind:

Biological systems consolidate memories during sleep.

Sleep may contribute to memory stabilization, emotional processing, abstraction, and neural optimization.

Sleep is not a single uniform state but a cycle, and its stages appear to serve different mnemonic functions, with slow-wave sleep linked to the consolidation of facts and events and rapid-eye-movement sleep linked to emotional processing and the integration of new material into existing knowledge.

During sleep, the hippocampus replays the day's experiences in a compressed form, allowing the neocortex to slowly incorporate them.

AI systems rarely possess analogous mechanisms.

This suggests a major research opportunity.



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Potential sleep-inspired computational research areas include offline replay, memory consolidation systems, self-generated simulation, generative rehearsal, and internal world model refinement.

The deeper point is that biological intelligence does some of its most important learning when it is not interacting with the world at all, whereas most AI systems are inert the moment the data stops, a difference that the gap analysis will return to when it considers continual learning.

Biological Learning Is Continuous, Embodied, and Social:

Biological learning occurs through multiple interacting processes including reinforcement learning (learning through rewards and penalties), Hebbian learning (strengthening connections through correlated activity), observational learning, social learning, motor learning, episodic learning, associative learning, and predictive learning.

Biological learning is adaptive, embodied, contextual, social, emotional, developmental, and continual.

AI learning is often static, task-specific, and heavily data-dependent.

Children learn efficiently from sparse data.

Sparse data refers to a very limited number of training examples.



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A child may learn a new word from only a few exposures, a feat psychologists call fast mapping, sometimes fixing the meaning of a word after a single encounter.

Large language models typically require enormous datasets, often many orders of magnitude more language than a child hears in a lifetime, to reach comparable fluency.

This difference reveals another major gap.

Biological systems appear capable of strong inductive bias (built-in assumptions derived from evolution that guide learning) and carry powerful priors about physical reality, social behavior, causality, object permanence, and spatial reasoning.

Developmental psychologists such as Elizabeth Spelke have argued that human infants are born with a small set of core knowledge systems, including early competence about objects, number, space, and agents, that give learning a running start rather than a blank slate.

Future AGI systems may require similarly strong built-in world priors rather than attempting to learn everything from scratch, and the question of which priors to build in, and which to let the system discover, is one of the central design choices the later chapters will examine.

CHAPTER 6: PERCEPTION AND SENSORY PROCESSING IN ORGANISMS

Biological Perception Is Active, Not Passive:



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The perceptual systems of biological organisms are profoundly different from the input pipelines of AI systems.

Perception in animals is active and self-directed rather than a passive feed, because the eye is never still but darts in rapid jumps called saccades, sampling a scene a few times each second, so that the seamless picture we experience is in fact a construction stitched together from fragments and heavy prediction.

Biological perception is not a passive recording of environmental states.

It is an active, goal-directed, interpretive process in which the organism uses prior knowledge and expectations to make sense of incoming sensory data.

A useful way to state the principle is that the brain does not see the eye's image; it uses the eye's image, together with everything it already expects, to construct a best guess about the world.

This is why perception can be fooled by illusions, and why illusions are so scientifically valuable: they are cases where the brain's prediction and the actual input disagree, and the prediction wins, revealing the machinery of guessing that normally hides behind accurate seeing.

Animals actively move their sensory organs to gather the most informative data.

A human rotates their eyes and head to examine an object of interest.



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A rat whisks its whiskers to determine the texture and position of objects in darkness.

An echolocating bat adjusts the frequency and timing of its calls based on what its previous echoes revealed.

This active sensing – the deliberate control of sensory data collection based on internal models and goals – is entirely absent from most current AI systems, which passively receive whatever data is supplied to them.

The psychologist J. J. Gibson captured a related idea with the concept of affordances, meaning that organisms perceive the world in terms of the actions it makes possible, seeing a chair not as a pattern of pixels but as something sit-on-able, so that perception and possible action are bound together from the start.

Multi-Modal Integration:

Biological organisms integrate multiple sensory modalities (types of sensory input) seamlessly into unified perceptions of the world.

The seams become visible only when the senses disagree, as in the McGurk effect, where watching lips mouth one syllable while hearing another makes a listener perceive a third that was never present, vivid proof that the brain fuses sight and sound rather than merely adding them.

Humans integrate vision, hearing, touch, taste, smell, proprioception (awareness of body position), vestibular sensation (sense of balance



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and head movement), and interoception (awareness of internal bodily states including hunger, thirst, pain, and heart rate).

When you hear a car, you turn your head to see it.

Your brain fuses the auditory and visual data to form a single unified concept of the car, including its location, speed, and likely trajectory.

How tightly the senses are fused is shown by the McGurk effect, in which seeing a person's lips form one sound while hearing another causes most people to perceive a third sound that was never present, a vivid demonstration that what we hear is shaped by what we see.

Current AI models are often modality-specific.

A vision model cannot hear; a language model cannot see.

Multimodal models are emerging, but they often simply concatenate (combine by joining end to end) data types rather than truly integrating them at a deep cognitive level, the way the brain binds a sight and a sound into a single event.

Research into sensorimotor integration – how sensory input drives motor output and vice versa – is a critical missing piece for replicating natural intelligence.

The Hierarchical Organization of Sensory Cortex:

Sensory processing in the brain is hierarchical.



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David Hubel and Torsten Wiesel earned a Nobel Prize for showing that neurons early in the visual system respond to simple edges and bars while later neurons respond to increasingly complex forms, a layered scheme that directly inspired the convolutional neural networks behind modern computer vision.

In the visual system, for example, the earliest processing stages extract simple features such as edges and orientations from the retinal image.

Subsequent stages combine these features into representations of shapes, objects, faces, and scenes.

The highest stages integrate visual information with knowledge, memory, and context to produce rich, meaningful perception.

The visual system also splits into two great streams, a ventral "what" pathway specialized for recognizing objects and a dorsal "where" pathway specialized for locating them and guiding action, a division of labor that lets the brain answer the two basic questions about anything it sees, what is it and where is it, in parallel.

This hierarchical feature extraction was one of the primary inspirations for deep learning, particularly convolutional neural networks (CNNs - artificial neural networks designed to process data with a grid-like topology, using layers of filters that detect progressively more complex features).

However, biological sensory hierarchies are more than simple feedforward feature extractors.



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They are bidirectional, with strong feedback from higher stages back to lower stages that actively shapes what is perceived.

They are also dynamically modulated by attention, expectation, and motivational state.

These top-down influences on perception have no adequate counterpart in current AI sensory processing systems, and their absence is one reason a vision model can be confident and badly wrong about an image that a person would instantly recognize as nonsense.

Event-Driven and Sparse Processing:

The brain uses sparse coding, meaning that only a small percentage of neurons are active at any given moment.

This strategy was anticipated by Horace Barlow's efficient coding hypothesis, the proposal that sensory systems evolved to represent the world with as few active signals as possible, and it has a direct engineering echo in event cameras, which report only the pixels that change and so capture fast motion at a fraction of the data and power of ordinary video.

This minimizes metabolic cost enormously.

In contrast, artificial neural networks are typically dense, activating millions of parameters simultaneously for every single inference (the process of applying a trained model to produce an output).



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The brain also uses event-driven processing.

If nothing changes in the visual field, the retina sends fewer signals.

This is why neuromorphic sensors such as Dynamic Vision Sensors (DVS) - cameras that mimic the retina by only sending data when a pixel changes brightness - reduce data volume and power consumption by orders of magnitude compared to conventional cameras that record every pixel at a fixed frame rate.

Bridging the energy efficiency gap requires a fundamental shift in how AI circuits and algorithms are designed, prioritizing event-driven, sparse processing over dense, synchronized computation.

CHAPTER 7: CONSCIOUSNESS, SELF-AWARENESS, AND METACOGNITION:

Consciousness Remains the Deepest Open Question:

Consciousness generally refers to subjective experience - the felt quality of awareness, including the experience of seeing red, feeling pain, or having a thought.

It is sometimes described as "what it is like" to be in a given mental state.

The philosopher David Chalmers sharpened this puzzle by separating what he called the easy problems of consciousness from the hard problem.

The easy problems, difficult as they are, concern functions we can in principle explain



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mechanistically, such as how the brain integrates information, focuses attention, or reports its own states.

The hard problem is why any of that processing should be accompanied by subjective experience at all, rather than going on, as it does in a pocket calculator, entirely in the dark.

This gap between physical mechanism and felt experience, sometimes called the explanatory gap, is the reason consciousness remains genuinely unsolved.

Despite centuries of philosophy and decades of neuroscience, no consensus explanation of consciousness exists.

Whether consciousness is necessary for AGI remains genuinely controversial.

Some researchers argue intelligence and consciousness are separable, and that a sufficiently capable system need not be conscious to perform any cognitive task.

Others argue consciousness may be necessary to support flexible integration, genuine self-modeling, reflection, planning, social understanding, and long-term behavioral coherence.

Major theories of consciousness include:

Integrated Information Theory (IIT) - proposed by Giulio Tononi - holds that consciousness is identical to integrated information, measured by a quantity called phi (a mathematical measure of the degree to which the parts of a system are causally



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integrated with each other, such that the whole generates more information than the sum of its parts).

These rival theories are now being tested head to head, as in the large adversarial collaboration reported in 2023 that pitted Integrated Information Theory against Global Workspace Theory using predictions registered in advance, a model of how a notoriously slippery question can be made to yield to experiment.

- On this account, any system with nonzero ϕ is conscious to some degree, and the amount of consciousness is proportional to the amount of integrated information.

- IIT implies that current AI neural networks, which have relatively modular information processing architectures with limited integration across modules, have very little or no consciousness.

Global Workspace Theory – proposes that consciousness emerges from globally broadcast information processing, where information is made available to many specialized brain systems simultaneously through a central workspace.

- On this account, consciousness is what happens when information is broadcast broadly enough to be available to multiple cognitive systems at once.

- The neuroscientist Stanislas Dehaene developed this into the global neuronal workspace model, in which a piece of information becomes conscious when it ignites widespread, sustained activity



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across the cortex, a transition that brain recordings can actually detect.

Predictive Processing accounts hold that consciousness is associated with high-level predictions about the causes of sensory input, and that the content of conscious experience is determined by the brain's best model of the world and of itself.

Higher-Order Theories hold that a mental state is conscious only when it is represented by a higher-order mental state – a thought about a thought, or a perception of a perception.

- On this view, consciousness requires a meta-representational capacity that many simpler animals and current AI systems lack.

Attention Schema Theory – proposed by Michael Graziano – holds that the brain constructs a simplified internal model of its own attention, and that subjective awareness is what this self-model represents, a view notable for suggesting a concrete engineering recipe rather than treating consciousness as inexplicable.

Each of these theories carries different implications for what architectural features would be needed for an AI system to become conscious, if consciousness turns out to be necessary for AGI.

They also disagree sharply on whether a digital computer could ever be conscious at all, with Integrated Information Theory implying that the right kind of physical integration is required and Global Workspace and Attention Schema theories



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implying that the right kind of functional organization would suffice.

Self-Awareness and the Self-Model:

Closely related to consciousness is self-awareness – the capacity to represent oneself as an agent in the world, to think about one's own thoughts and knowledge states, and to model one's own capabilities and limitations.

A classic if imperfect probe of self-recognition is the mirror mark test devised by Gordon Gallup in 1970, which a small set of species, including great apes, elephants, dolphins, and magpies, appear to pass by using a mirror to inspect a mark placed on their own bodies.

Self-awareness is not all-or-nothing.

It comes in degrees and in multiple forms.

The ability to recognize oneself in a mirror is sometimes taken as a behavioral marker of bodily self-awareness, and this ability is demonstrated by humans, great apes, elephants, dolphins, and possibly crows.

The mirror test is suggestive but limited, since it leans heavily on vision and may underestimate animals that rely on smell or sound, a caution worth keeping in mind whenever a single behavioral test is treated as proof of an inner capacity.

More sophisticated forms of self-awareness include knowledge of one's own knowledge (knowing what you know and what you don't know), metacognitive monitoring (the ongoing process of evaluating the



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quality of one's own cognitive processes), autobiographical self-consciousness (the sense of oneself as a continuing agent with a history and a future), and the sense of agency (the experience of being the author of one's own actions).

Current AI systems display very limited self-awareness in any of these senses.

Large language models can produce text that reflects apparent self-knowledge, but this is largely a reflection of patterns in training data rather than genuine metacognitive monitoring of the system's own internal states.

A model that writes "I am not sure about this" has, in most current architectures, no privileged access to an internal signal of uncertainty; it is producing the phrase because such phrases follow such contexts in its training data, which is a fundamentally different thing from knowing that it does not know.

Metacognition - Thinking About Thinking:

Metacognition (the capacity to monitor and regulate one's own cognitive processes - to think about one's own thinking, assess the reliability of one's own judgments, and adjust one's strategies accordingly) is one of the most distinctively human cognitive capacities.

Humans and even some animals reliably know when they do not know, choosing to opt out of a hard test rather than guess, whereas most current AI systems remain poorly calibrated and are often most confident precisely when they are wrong.



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Metacognitive abilities include confidence monitoring (knowing how confident to be in a judgment before any external feedback), strategy selection (choosing the appropriate cognitive approach for a given problem), error detection (noticing mistakes before any external feedback), and learning regulation (determining when to seek more information, when to stop practicing, and what to focus attention on during learning).

Metacognition depends heavily on the prefrontal cortex and is closely linked to working memory.

AI systems currently lack reliable metacognition.

Large language models are notorious for generating confident-sounding outputs even when they are incorrect - a failure of confidence monitoring that poses real risks when these systems are deployed in high-stakes applications.

This failure has a precise technical name, poor calibration, meaning that a system's expressed confidence does not match its actual probability of being correct, and it is one of the clearest, most measurable ways in which today's most capable models fall short of even a cautious human expert.

Building AI systems with genuine metacognitive ability - systems that know what they know, know what they don't know, and can communicate that distinction reliably - is one of the most important research priorities identified in this book, because a system that reliably flagged its own uncertainty would be safer, more trustworthy, and more useful than a more capable system that did not.



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The Sense of Agency and Causal Self-Understanding:

A biological intelligence does not merely experience the world.

The brain distinguishes the events it causes from the events that merely happen to it by sending a copy of every motor command, called an efference copy, back to its own sensory areas, which is why you cannot tickle yourself yet the same touch from another person can.

It experiences itself as an agent acting in the world - as a cause, not merely an effect.

The sense of agency (the experience of being the author of one's own actions, of intending a movement and experiencing its execution as self-generated) is a central feature of conscious experience and a prerequisite for genuine goal-directed behavior.

For an AI system to be genuinely goal-directed rather than merely appearing goal-directed from the outside, it may need something analogous to a sense of agency - an internal representation of itself as an agent whose actions are self-generated and causally efficacious.

What this would look like in a computational system, and whether it would require something like phenomenal consciousness, is an open question that intersects the philosophy of mind with AI research in the most direct possible way.

CHAPTER 8: LANGUAGE, REASONING, AND ABSTRACT THOUGHT:



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Language as a Cognitive Tool:

Language is the most visible and most studied dimension of human intelligence.

Noam Chomsky argued that children acquire grammar despite a poverty of the stimulus, hearing far too few examples to justify the rules they reliably infer, and whatever the final verdict on his theory, the contrast with AI is stark, since a child masters language from a few million words while a large model is trained on trillions.

But language is not simply a communication system.

It is a cognitive tool of extraordinary power that reshapes the mind that possesses it.

Language enables explicit, sequential reasoning about propositions – the capacity to hold a chain of premises in mind, check each step for validity, and arrive at a conclusion that may be remote from any direct experience.

Language enables the accumulation and transmission of knowledge across generations – the cultural inheritance that has allowed human civilization to build cumulatively on the discoveries of previous generations.

Language enables the construction of complex social institutions, legal systems, scientific frameworks, and mathematical structures that exist in the shared cognitive space of a community rather than in any individual mind.



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And language enables the kind of self-directed thought – the inner monologue – that is central to human conscious experience and self-regulation.

A long-standing debate concerns how much language shapes thought itself, a proposal associated with the linguists Edward Sapir and Benjamin Lee Whorf, and although the strongest version, that language determines thought, is now rejected, a milder version survives: the words and grammar a mind has available can make some distinctions easier to draw and some ideas easier to hold.

The capacity for language appears to depend on a set of brain regions that are especially elaborated in humans, including Broca's area (in the left inferior frontal gyrus, associated with language production and grammatical processing) and Wernicke's area (in the left superior temporal gyrus, associated with language comprehension).

But language in the brain is not localized to these two areas.

It depends on a distributed network of regions that also supports working memory, attention, social cognition, and conceptual knowledge.

The linguist Noam Chomsky added a further provocation with the argument from the poverty of the stimulus, observing that children acquire the deep rules of grammar from far less and far messier input than would seem sufficient, and concluding that the human mind must bring strong innate structure to the task, a claim that bears directly on how much an artificial system must be given rather than left to learn.



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The Grounding Problem:

Current AI systems, particularly large language models, have demonstrated impressive language capabilities.

Emily Bender and Alexander Koller dramatized the worry in 2020 with a thought experiment about a hyper-intelligent octopus that taps an undersea cable and learns to imitate human messages perfectly, until one correspondent faces a bear and asks for help, at which point the octopus, having never encountered bears or sticks, has nothing to say, a parable for a system that masters form while missing meaning.

But these systems learn the statistics of language without the physical grounding that makes language meaningful to biological organisms.

An LLM knows the word "heavy" but has no experience of the muscular effort required to lift a heavy object.

It knows the word "cold" but has never felt the drop in skin temperature that signals cold to a biological organism.

This is the Symbol Grounding Problem: the challenge of how words or symbols get their meaning if they are defined only by other symbols.

The cognitive scientist Stevan Harnad, who named the problem, illustrated it by imagining trying to learn a first language from a dictionary written entirely in that same unknown language, an endless circle of symbols defined by symbols with no exit into meaning.



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In a dictionary, words are defined by other words.

But human understanding is grounded in physical reality and lived experience.

We know what "red" is because we have seen red.

We know what "hot" is because we have felt heat.

We know what "falling" feels like because we have fallen.

This is the central claim of embodied cognition, the view that meaning is built from sensorimotor experience and that a mind without a body, or at least without sensors and effectors that connect it to a world, is missing the foundation on which human concepts are built.

Bridging the grounding problem is essential for AGI.

A general intelligence must either directly experience the physical world or have its representations grounded in something more substantive than statistical patterns across text.

Causal Reasoning and Common Sense:

Biological intelligence reasons about causes and effects, not just correlations.

The developmental psychologist Alison Gopnik has shown that even toddlers behave like little scientists, intervening on the world by pressing a switch to see what it controls and revising their



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beliefs in ways that track formal causal inference, long before they can read.

A child who sees a window break after a ball strikes it understands that the ball caused the window to break.

They can reason about counterfactuals (what would have happened if the ball had missed) and interventions (what would happen if you caught the ball before it hit).

The computer scientist Judea Pearl has formalized this with what he calls the ladder of causation, a three-rung hierarchy rising from association (seeing that two things go together), to intervention (predicting the effect of deliberately doing something), to counterfactuals (imagining what would have happened had things been otherwise).

Pearl's central warning is that a system trained only on observed correlations is stuck on the bottom rung, and no amount of additional data of the same kind will lift it to the rungs of intervention and counterfactual reasoning where genuine understanding lives.

Current AI systems, trained largely on statistical correlations, systematically struggle with genuine causal reasoning.

They can identify patterns but fail to reliably distinguish cause from effect or to reason about what would happen if circumstances were changed.



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Causal reasoning is fundamental to planning, problem-solving, moral reasoning, and scientific thinking.

Closing the causal reasoning gap is among the highest-priority research directions for AGI.

Compositionality and Generalization:

Biological intelligence is remarkably good at combining known concepts in novel ways.

Jerry Fodor and Zenon Pylyshyn argued that thought is systematic in this way by its very nature, and Brenden Lake and Marco Baroni later turned the claim into a test, the SCAN benchmark, on which standard networks that have learned to jump and to walk twice nonetheless stumble at walking twice after jumping.

A speaker of English who knows what "purple" means and what "elephant" means can immediately understand "purple elephant" even if they have never encountered the combination.

A child who can add, subtract, and multiply can immediately begin to apply these operations in new combinations to solve novel arithmetic problems.

This compositionality - the ability to understand and produce an unlimited number of novel combinations from a finite set of known elements - is a fundamental property of both language and thought.

The philosophers Jerry Fodor and Zenon Pylyshyn argued decades ago that this systematicity is exactly what simple associative networks fail to



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guarantee, and the criticism remains sharp today: a system can appear to master a pattern while still falling apart when the same elements are recombined in a way it did not see during training.

AI systems, despite impressive performance on many language benchmarks, systematically fail at compositional generalization tasks that require applying known rules and concepts in genuinely novel combinations.

Bridging the compositionality gap requires AI architectures that learn rules and structures rather than merely memorizing patterns from training data.

Abstract Reasoning and Analogy:

One of the most powerful cognitive tools of biological intelligence is analogical reasoning – the ability to identify structural similarities between different situations and to transfer knowledge from a familiar domain to an unfamiliar one.

Douglas Hofstadter has argued that analogy is not a special trick but the very core of cognition, and the difficulty machines have with it is exactly what tests such as Raven's Progressive Matrices and the Abstraction and Reasoning Corpus are designed to expose.

When a physicist uses the analogy of waves on water to explain light, or when a child uses the concept of a container to understand the idea of a category, they are performing analogical transfer



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that allows knowledge to span enormous conceptual distances.

The psychologist Dedre Gentner described the mechanism behind this as structure mapping, the alignment of the relationships in one situation with the relationships in another, so that what transfers is not surface appearance but underlying structure, which is precisely why an atom can be likened to a solar system despite sharing none of its features.

Current AI systems exhibit weak analogical reasoning by comparison.

They can sometimes succeed at analogy tasks when the relevant surface features are present in training data, but they struggle with deep structural analogies that require genuinely abstract relational representations.

Much of this chapter's gap can be summarized through the psychologist Daniel Kahneman's distinction between System 1, fast, automatic, intuitive thinking, and System 2, slow, effortful, deliberate reasoning, since today's large language models are superb pattern-completing System 1 engines that only imitate System 2 reasoning, and building reliable, genuinely deliberate reasoning is one of the field's defining challenges.

Building AI systems with robust abstract reasoning and analogical transfer capability is a major open research challenge.

CHAPTER 9: EMOTION, MOTIVATION, AND DECISION-MAKING:



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Emotion Is Not the Opposite of Reason:

For much of the history of Western philosophy and early cognitive science, emotion was viewed as an irrational force that interferes with clear thinking.

Antonio Damasio overturned this picture with patients whose prefrontal damage stripped away emotional signaling, who did not become coolly rational but instead made disastrous everyday decisions, evidence that feeling is part of reasoning rather than its enemy.

This view has been substantially revised by neuroscience.

The neuroscientist Antonio Damasio demonstrated through studies of patients with damage to the emotional systems of the brain that such patients become severely impaired in decision-making rather than improved.

The most famous historical example is Phineas Gage, a nineteenth-century railway worker who survived an iron rod driven through his frontal lobes but emerged, by the accounts of those who knew him, changed in judgment and personality even as his basic intellect remained intact.

Damasio studied modern patients with similar damage and found the same pattern: people who could still score well on intelligence tests yet made ruinous decisions in their own lives, because the emotional signals that normally tag options as promising or dangerous had been silenced.



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Damasio's somatic marker hypothesis (the theory that emotional signals from the body, encoded as bodily states or memories of such states, guide decision-making by marking options with values of good or bad before conscious deliberation) showed that without emotional guidance, decisions become unmoored from the values and priorities that make choices meaningful.

A laboratory demonstration of this is the Iowa Gambling Task, in which healthy people begin to avoid risky card decks, showing anticipatory stress responses, before they can consciously explain why, while patients with emotional-system damage keep choosing the losing decks even after they can state the rule.

Without emotion, patients could reason correctly in abstract terms but could not translate that reasoning into adaptive choices in real-life situations.

Emotion is not the enemy of reason.

Emotion is a component of reason.

Biological organisms rely on emotion and motivation as a fundamental organizing principle for behavior.

Hunger, thirst, curiosity, fear, and the desire for social connection motivate behavior and keep it aligned with survival and flourishing.

These are homeostatic mechanisms (tendencies that maintain internal stability and drive the organism to correct deviations from needed conditions).



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When blood sugar drops, the drive of hunger is triggered to correct it.

When a threat is detected, fear focuses attention and mobilizes resources.

When novelty is encountered, curiosity drives exploration and learning.

The neuroscientist Jaak Panksepp argued that mammals share a small set of basic emotional systems, with names he capitalized such as SEEKING, FEAR, RAGE, and CARE, built into ancient subcortical circuitry, a reminder that emotion is not a late decorative layer on top of cognition but part of its evolutionary foundation.

In current AI, motivation is entirely external.

A researcher defines a loss function or a reward function.

The AI minimizes the loss or maximizes the reward.

It has no internal concept of need or desire.

It does not get bored; it does not get curious unless curiosity is explicitly programmed as a bonus signal for exploring new states.

This externally imposed motivation is a profound architectural difference, because a biological organism carries its goals inside its own body, in the form of needs that must be met to keep living, while an AI is handed its goals from outside and has no stake of its own in meeting them.

Dopamine and the Architecture of Reward Learning:



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The neurotransmitter dopamine plays a central role in biological reward learning.

Wolfram Schultz discovered that dopamine neurons signal not reward itself but reward prediction error, the gap between expected and received reward, a quantity mathematically identical to the temporal-difference error at the heart of modern reinforcement learning, making this one of the clearest bridges ever found between neuroscience and AI.

Midbrain dopamine neurons encode a reward prediction error signal – the difference between the reward that was received and the reward that was predicted.

When an outcome is better than predicted, dopamine neurons fire more strongly, signaling that the prediction was too pessimistic.

When an outcome is worse than predicted, dopamine neurons fire less strongly, signaling that the prediction was too optimistic.

This signal drives learning to predict rewards more accurately and to seek actions that produce them.

The discovery that biological dopamine neurons encode precisely this reward prediction error signal was a landmark finding because it cross-validated a mathematical learning algorithm called Temporal Difference (TD) learning that AI researchers had independently developed.



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The convergence is worth dwelling on: Richard Sutton and Andrew Barto had worked out temporal difference learning as an abstract algorithm for predicting future reward, and then Wolfram Schultz's recordings of monkey dopamine neurons, interpreted by Read Montague, Peter Dayan, and others, revealed the brain computing essentially the same quantity, a rare case of theory and biology meeting in the middle.

This connection between dopamine and TD learning is one of the most elegant examples of mutual benefit between AI and neuroscience research.

However, biological reinforcement learning is significantly richer than current AI reinforcement learning implementations.

The biological system is modulated by other neuromodulatory systems, is embedded in a rich hierarchical action representation, implements curiosity and intrinsic motivation alongside reward-based learning, and is deeply integrated with social learning and imitation.

Training an AI reinforcement learning agent to play a video game at human level requires millions of episodes of experience.

A human can reach that level in far fewer, often grasping the point of a new game within a handful of attempts because they bring a lifetime of transferable physical and conceptual knowledge to it.

Understanding the mechanisms that make biological reinforcement learning so efficient is a major research priority, and that efficiency, sometimes



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called sample efficiency, is one of the sharpest practical differences between natural and artificial learners.

Intrinsic Motivation and Curiosity:

For AGI, the absence of intrinsic motivation is a critical gap.

Children at play are the clearest demonstration of intrinsic motivation, exploring relentlessly with no external reward, and engineering an analog powerful enough to drive open-ended learning yet safe enough not to pursue harmful sub-goals is one of the central challenges this book returns to.

A general intelligence must be able to set its own goals and pursue them without requiring a human to define the objective function for every new situation.

True AGI will likely require a form of intrinsic motivation (the drive to engage in an activity for its own sake, for the information it provides, or for the inherent satisfaction of mastery, rather than for an external reward).

Research into artificial curiosity and intrinsic reward signals is attempting to bridge this gap.

Several formal approaches exist, including rewarding a system for prediction error in its own model of the world, for the learning progress it makes, or for empowerment, meaning the amount of control it has over its future, each an attempt to give a machine a reason to explore that does not have to be hand-specified for every task.



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Systems driven to reduce uncertainty or maximize their learning progress might learn autonomously across a wide range of domains without requiring human-designed reward functions for each one.

However, this also introduces risk.

An agent with its own drives may pursue those drives in ways that do not align with human interests.

This is the Alignment Problem (the challenge of ensuring that the goals and behaviors of an AI system remain compatible with human values and human well-being as the system becomes more capable), which is addressed in detail in Chapter 17, and it is the dark side of the very autonomy that makes intrinsic motivation so desirable: a system that writes its own goals is a system whose goals we must be very sure we understand.

CHAPTER 10: LESSONS FROM DIVERSE ORGANISMS

The Richness of the Biological Design Space:

The natural world contains an extraordinary diversity of nervous system designs, each representing a solution to the problem of intelligent adaptive behavior refined by millions of years of evolution.

This diversity is itself the argument of the chapter, because it shows that intelligence is not a single ladder with humans at the top but a branching bush of independent solutions, and a field that studies only the human case is sampling one twig and mistaking it for the whole tree.



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Studying this diversity is not merely an academic exercise.

It is a source of engineering inspiration for AI researchers who may be too narrowly focused on replicating specifically human cognition.

Modern AI research has drawn predominantly from human neuroscience and primate cognition.

Nature contains billions of years of additional design experimentation in organisms whose cognitive architectures differ radically from ours.

Each of these organisms has solved the problem of adaptive behavior under energy constraints, in real time, in complex physical environments.

Each therefore has something to teach.

A useful concept for this chapter is convergent evolution, the repeated, independent arrival of different lineages at similar solutions, as with the camera eye that evolved separately in vertebrates and octopuses; when nature solves the same problem the same way more than once, it is a strong hint that the solution reflects something deep about the problem rather than an accident of history.

The Nematode Worm and Minimal Intelligence:

The roundworm *C. elegans* possesses exactly 302 neurons and approximately 7,000 synapses.

Because every neuron and connection is known, the worm has become a test of whether a wiring diagram



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alone can predict behavior, and the OpenWorm project's long struggle to build a faithful simulation suggests that the molecules and dynamics riding on top of the wiring matter as much as the wiring itself.

Its complete neural wiring diagram, called a connectome (a comprehensive map of all neural connections in a nervous system), was published in 1986 and remains the only complete connectome of a freely behaving animal.

Despite this extreme simplicity, *C. elegans* exhibits complete sensorimotor integration, chemotaxis (movement toward attractive chemicals and away from repulsive ones), associative learning, and social behavior.

The study of this minimal nervous system demonstrates that sophisticated adaptive behavior does not require massive computational resources.

It requires clever architecture, efficient coding, and tight integration between sensing, processing, and action.

An open-science effort called OpenWorm has tried for years to build a complete digital simulation of this worm, neuron by neuron and muscle by muscle, and the fact that fully reproducing even 302 neurons remains genuinely hard is itself a lesson about how much biological detail our tidy artificial neurons leave out.

Research Project Idea: Use the *C. elegans* connectome as a benchmark for testing whether neuromorphic hardware implementations of known



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biological circuits can reproduce the organism's full behavioral repertoire.

- This would provide a minimal but concrete test case for the fidelity of neuromorphic simulation.
- Success would validate the hardware platform and reveal which architectural features are necessary and sufficient for specific adaptive behaviors.

The Insect Brain and Efficient Navigation:

Insects demonstrate that sophisticated navigation, learning, and social coordination are achievable with neural circuits containing fewer than one million neurons.

A honeybee accomplishes navigation, communication, and even simple counting with roughly a million neurons, and in 2024 scientists completed the wiring diagram of the adult fruit fly brain, about 140,000 neurons and fifty million connections, the first complete map of a brain complex enough to support flexible behavior.

Honeybees can navigate over kilometers using a combination of visual landmarks and solar compass cues, communicate the location of food sources to nestmates through the waggle dance (a figure-eight movement that encodes direction and distance to a food source in the angle and duration of the dance), and learn flower colors and patterns through classical conditioning.

The mushroom bodies of the insect brain (paired structures in the insect brain that are major centers for associative learning and the integration of sensory information) process



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olfactory and visual information and support flexible associative learning with remarkable efficiency.

Drosophila (fruit flies) have become a major model organism for understanding the neural circuits underlying specific behaviors because their relatively small nervous system of approximately 100,000 neurons is accessible to complete circuit-level analysis, and the recent completion of a full fruit-fly brain connectome has, for the first time, given researchers a wiring diagram of a flying, learning animal.

Research Project Idea: Study collective intelligence in ant colonies and bee hives to inform multi-agent artificial systems.

- Explore decentralized decision-making without central controllers to develop AI systems that are resilient without requiring coordination hierarchies.
- Apply lessons to robotics swarms for resilient real-world applications that need to function even when individual units fail.

The Octopus and Distributed Cognition:

The octopus represents one of the most remarkable examples of intelligence in the animal kingdom, and one of the most important for AI research precisely because it is so different from vertebrate intelligence.

Roughly two-thirds of its half-billion neurons lie in its arms, each of which can taste, touch, and act with striking autonomy, so the octopus is a



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living proof of concept for the distributed control that robotics has only begun to attempt.

The octopus has a central brain but two-thirds of its neurons are distributed in the arms themselves.

Each arm has a degree of autonomous control, capable of executing complex movements without direction from the central brain.

This distributed intelligence (the distribution of cognitive processing throughout the body rather than centralizing it in a single brain structure) allows for extraordinary motor flexibility and parallel processing.

An octopus can simultaneously coordinate eight semi-independent limbs to pursue prey, escape predators, and manipulate objects - tasks that would require enormous central processing if all control were centralized.

The octopus is also a vivid example of how thoroughly body and intelligence can intertwine, since much of its problem-solving is in effect offloaded to the mechanics and local control of its arms, a principle that roboticists study under the name embodied or morphological computation.

Research Project Idea: Study the architecture of octopus distributed cognition to inform the design of modular artificial systems in which different components can operate semi-autonomously while remaining coordinated.



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- This may be particularly relevant for robotic systems that need to coordinate multiple limbs or subsystems in complex manipulation tasks.
- Implementing arm-level local control with central coordination may dramatically improve the agility and robustness of robotic systems.

Corvid Birds and Tool Use:

Corvid birds - particularly New Caledonian crows - demonstrate tool use, tool manufacture, and causal reasoning that rivals or exceeds what non-human primates demonstrate.

In controlled experiments these crows pass versions of the Aesop's fable task, dropping stones into water to raise a floating reward and preferring sinking objects to floating ones, which shows a grasp of cause and effect rather than mere trial-and-error luck.

New Caledonian crows can manufacture hooked tools from plant materials to retrieve food from cavities, understand causal relationships between tool properties and outcomes, and solve multi-step problems that require planning several actions ahead.

This remarkable intelligence exists in a brain that weighs approximately 7 grams.

Part of the secret is density, since bird brains pack neurons far more tightly than mammal brains of similar size, so a small corvid brain can contain as many forebrain neurons as a much larger primate brain, a reminder that raw size is a crude proxy for computational power.



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The computational efficiency of the corvid brain relative to its capabilities is a striking example of what evolution can achieve under severe energy and size constraints.

Research Project Idea: Develop a comparative neuroscience program studying the neural circuits underlying tool use and causal reasoning in corvids and other tool-using species.

- Identify which architectural principles allow small circuits to support sophisticated causal reasoning and planning.
- Use findings to inform the design of AI systems capable of genuine causal reasoning without requiring massive computational resources.

Whales, Dolphins, and Social Intelligence:

Cetaceans (whales and dolphins) demonstrate sophisticated social intelligence, complex communication, cooperative hunting strategies, and what appear to be cultural traditions that are transmitted across generations.

Bottlenose dolphins even develop signature whistles that function like names, addressing one another individually, and some populations pass specialized hunting techniques down the generations as genuine culture.

Dolphin communication involves complex vocalizations that may encode individual identity, emotional state, and intentional content, including signature whistles that function much like names.



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Sperm whale clans appear to have cultural differences in their behavior that are not explained by genetics or environment alone.

Studying cetacean social cognition may reveal principles for building AI systems with genuine social intelligence and the ability to participate in cultural transmission, a capacity that matters because so much human intelligence is not stored in any single brain but in the accumulated culture that brains build and pass on.

Slime Molds and Distributed Problem Solving:

Perhaps the most surprising entry in this survey is the slime mold *Physarum polycephalum* (a single-celled organism that can form large networks of cytoplasmic tubes to transport nutrients).

In a celebrated 2010 experiment led by Atsushi Tero, researchers placed food at the positions of cities around Tokyo and watched the mold grow a nutrient network that rivalled the efficiency of the real rail system, computing a near-optimal solution with no brain, no neurons, and no central control at all.

When presented with food sources scattered across a surface representing a map, *Physarum* spontaneously forms an efficient network connecting the food sources that closely approximates the topology of human-engineered transport networks.

In one celebrated experiment, researchers laid out oat flakes in the pattern of the cities around Tokyo and watched the slime mold grow a network



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strikingly similar to the actual Tokyo rail system, balancing efficiency against resilience much as the human engineers had.

It solves a variant of the minimum spanning tree problem (a classic optimization problem in graph theory, asking for the minimum total edge length needed to connect all nodes in a network) without neurons, without a brain, and without computation in any conventional sense.

This demonstrates that sophisticated optimization is not uniquely a property of nervous systems.

It can emerge from physical dynamics in appropriately structured biological materials.

This has direct implications for unconventional computing architectures that use physical materials to perform computation rather than simulating it digitally.

Taken together, the organisms in this chapter deliver one unified lesson that closes the first part of this book: intelligence is not one thing built one way, but a family of solutions discovered many times over, on many scales of size and energy, in many physical substrates, and a field that studies only the human case is reading a single page of a very long book.

PART TWO: ARTIFICIAL INTELLIGENCE AND THE GAP

CHAPTER 11: THE CURRENT STATE OF ARTIFICIAL INTELLIGENCE:

What AI Has Achieved:



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Modern AI has achieved extraordinary milestones that would have seemed impossible a generation ago.

Machine learning systems can classify images with superhuman accuracy in certain controlled conditions.

Large language models can generate coherent text, answer questions, write code, translate languages, and summarize documents at impressive levels of quality.

Robotic systems can navigate structured environments and perform manipulation tasks.

Neural networks can recognize patterns in vast datasets of medical images, genomic sequences, and scientific data.

Reinforcement learning systems can master complex games, sometimes at superhuman levels.

Two achievements deserve special mention because they show AI contributing to science itself.

In 2016 the system AlphaGo defeated a world champion at the game of Go, a feat experts had expected to be a decade or more away, by combining deep neural networks with search.

In 2020 the system AlphaFold largely solved the fifty-year-old problem of predicting how proteins fold into their three-dimensional shapes, delivering a genuine scientific advance that is already accelerating biology and medicine.



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That protein-folding breakthrough deepened further in 2024, when the successor system AlphaFold 3 extended prediction from single proteins to the interactions between proteins and other molecules such as deoxyribonucleic acid (DNA), ribonucleic acid (RNA), and small-molecule drugs, which are the very interactions that matter most for designing medicines.

The scientific community marked the significance of these advances in an unprecedented way in 2024, when artificial-intelligence research was recognized with two Nobel Prizes in the same year.

The 2024 Nobel Prize in Physics was awarded to John Hopfield and Geoffrey Hinton for the foundational work on artificial neural networks that made modern machine learning possible.

The 2024 Nobel Prize in Chemistry was awarded in part to Demis Hassabis and John Jumper of Google DeepMind for AlphaFold, and in part to David Baker for computational protein design, meaning the ability to invent entirely new proteins that do not exist in nature.

No prior year had seen artificial-intelligence methods honored across two separate sciences at once, a signal that these tools had become genuine instruments of discovery in their own right.

These achievements are genuine and important.

They represent decades of brilliant research and have produced substantial practical value in medicine, scientific discovery, industry, and everyday life.



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Modern AI has already saved lives through earlier cancer detection, accelerated drug discovery, and improved weather forecasting.

These contributions should not be understated.

A Brief History: From Symbols to Scale:

To judge the present accurately it helps to know how the field arrived here, because AI has lived several lives.

The first, from the 1950s into the 1980s, was the era of symbolic AI, sometimes called Good Old-Fashioned AI, which tried to build intelligence out of hand-written rules and logical symbols, and which produced expert systems that encoded human knowledge as explicit if-then rules.

A competing tradition, connectionism, argued instead that intelligence should emerge from networks of simple units that learn from data, an approach that languished for years partly because the computers and datasets it needed did not yet exist.

The field also endured two so-called AI Winters, periods when funding and interest collapsed after early promises failed to materialize, the first in the 1970s following critical assessments such as the Lighthill report, and the second in the late 1980s when the expert-system industry proved brittle and expensive.

The current era began in 2012, when a deep neural network called AlexNet won a major image-recognition contest by a wide margin, demonstrating that the connectionist bet finally



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paid off once enough data and graphics-processing-unit computing power were available.

AI Winter refers to historical periods when AI research funding and interest declined sharply after early promises failed to materialize, because foundational problems proved harder than anticipated.

Understanding AI Winters is important context for evaluating current optimism about AI progress, because each winter followed a period of confident predictions much like those heard today.

How Current AI Systems Work:

The dominant paradigm in current AI is deep learning, meaning the training of artificial neural networks with many layers of processing using large datasets and gradient-based optimization.

The core learning algorithm is backpropagation combined with gradient descent (an optimization procedure that iteratively adjusts the parameters of a model to minimize its error on training data).

This approach requires an explicit separation between a distinct training phase and an inference phase, where the trained model is deployed to make predictions without further learning, a separation that has no counterpart in biological brains, which learn and act at the same time.

The transformer architecture, introduced in 2017, has become dominant for most AI tasks.



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Its key innovation is the self-attention mechanism (a computational operation that allows every element of an input sequence to attend to every other element and compute a weighted combination of their representations).

This allows the model to capture long-range dependencies in data without sequential processing.

Current AI systems have achieved much of their success by scaling – increasing model size, data volume, and training compute.

The largest current models have hundreds of billions of parameters (the adjustable numerical values that define the model's behavior) and were trained on trillions of tokens (units of text, roughly equivalent to words or word fragments).

This strategy is underwritten by empirical scaling laws, regularities showing that model performance improves predictably as parameters, data, and compute grow together, results that turned scaling from a guess into something closer to an engineering plan.

Scaling has produced what some researchers call emergent abilities, capabilities that appear rather suddenly at larger scales, though whether these are true qualitative jumps or artifacts of how performance is measured remains debated.

What is not debated is that scaling has limits set by the finite supply of high-quality data, the rising cost of compute, and, as the next chapter argues, by gaps that more of the same training does not appear to close.



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The Reasoning and Agentic Turn: 2023 to 2026:

By the mid-2020s the field's own response to those limits produced the most important shift since the transformer.

Rather than making models only larger, researchers learned to make them think for longer at the moment they are used, an approach called test-time compute (spending extra computation while answering a question, rather than only during training).

The first widely known systems of this kind were the reasoning models, beginning with OpenAI's o1 in late 2024 and its successor o3, which are trained to produce a long internal chain of thought (a step-by-step written reasoning process) before committing to a final answer.

These models were taught to reason less by imitation and more by reinforcement learning (a training method in which a system learns from rewards and penalties rather than from labeled examples), which rewarded chains of reasoning that reached correct, checkable answers in domains such as mathematics and computer programming.

The result was a large jump on problems that had resisted pure scaling, including competition mathematics and graduate-level science questions.

In early 2025 a Chinese laboratory named DeepSeek released a reasoning model called R1 that matched much of the performance of the leading closed systems, while being released openly and served at a small fraction of the price.



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The R1 release was a genuine shock to the field, because it showed that frontier-level reasoning was not the exclusive property of a few heavily funded laboratories, and that efficiency, not only raw scale, would shape the competition ahead.

A second shift followed closely, the move from chatbots that answer questions to agents that take actions.

An artificial-intelligence agent (AI agent) is a system that pursues a goal over many steps by using tools, such as searching the web, running code, or operating a computer, then observing the results and deciding what to do next.

Two open standards accelerated this turn, the Model Context Protocol (MCP) introduced by Anthropic in late 2024, which lets a model connect to external tools and data in a uniform way, and later agent-to-agent standards that let separate agents coordinate with one another.

By 2026 coding agents could run on their own for many minutes or even hours on a single task rather than answering one prompt at a time, and organizations had begun deploying agents for customer support, software development, and document processing, although dependable production use remained far narrower than the surrounding enthusiasm suggested.

These advances are real, and they matter, but they must be read alongside a sobering counter-measurement.



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In 2025 the researcher Francois Chollet and his collaborators released ARC-AGI-2, the second version of a benchmark built from abstract visual puzzles (the name is short for Abstraction and Reasoning Corpus) that ordinary adults find easy and machines find hard.

Untrained humans solve well over half of the ARC-AGI-2 tasks, whereas frontier models at the time of release scored only in the low single digits to the middle teens of percent, and the benchmark was deliberately designed so that closing the gap by brute-force computation would be ruinously expensive.

The lesson of the reasoning-and-agent era is therefore double-edged, that giving models time to think and tools to act unlocks striking new capability, and yet a wide gap in flexible, efficient, and general reasoning remains exactly where the next chapter locates it.

What AI Has Not Achieved:

Despite impressive achievements, AI still falls dramatically short of the adaptive, flexible, energy-efficient, embodied, emotionally aware, self-directed, and deeply integrated intelligence observed in biological organisms.

A human child can acquire language from limited, unlabeled experience in the real world.

A mouse can navigate uncertainty in a changing environment.

A crow can use tools it has never been trained to use.



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An ant colony can reorganize after disruption with no central instruction.

None of these feats is reliably achievable by current AI systems outside of narrowly constrained conditions.

A revealing symptom of the shortfall is brittleness: a vision system that labels images flawlessly can be fooled into confident errors by changes too small for a human to notice, and a language model that writes elegant prose can fail a slightly reworded version of a problem it just solved.

The following biological principles have NOT been adequately implemented in current AI:

Local, unsupervised learning without a global error signal.

The Complementary Learning Systems (CLS) architecture for avoiding catastrophic forgetting.

Neuromodulatory regulation of learning rate and plasticity.

Active sensing and embodied perception.

Emotional and motivational integration with cognition.

Genuine causal reasoning and counterfactual modeling.

Consciousness and self-awareness.



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The predictive processing architecture.

Spontaneous, self-directed cognitive activity equivalent to the brain's default mode.

Single-shot learning (the ability to permanently learn from a single exposure).

Robust common-sense reasoning about physical reality.

Theory of Mind (the ability to attribute beliefs, intentions, desires, and knowledge to others).

These are the major topics of the gap analysis in Chapter 12.

The Energy and Sustainability Crisis of Current AI:

The energy consumption of modern AI training and inference is already a significant environmental concern and is growing rapidly.

Training a single large language model can consume megawatt-hours of electricity - equivalent to the lifetime energy use of multiple cars.

Running inference on large deployed models at scale consumes enormous quantities of power per day, and at planetary scale the data centers that serve AI are now large enough consumers of electricity and water to register in national energy forecasts.

The carbon footprint of AI is becoming a significant and increasingly discussed problem.



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If AI systems are to be deployed at civilizational scale, they cannot remain tethered to power consumption levels that are orders of magnitude beyond what biological systems require for equivalent tasks.

The energy efficiency gap between biological and artificial intelligence is therefore not merely a scientific curiosity.

It is a practical and environmental constraint on the future of AI deployment, and it is one of the strongest arguments that the path forward must learn from biology rather than simply building larger versions of what already exists.

CHAPTER 12: THE GAP ANALYSIS: WHERE ARTIFICIAL INTELLIGENCE FALLS SHORT:

Framework for the Gap Analysis:

The gap analysis in this chapter is organized around the components of natural intelligence identified in previous chapters.

For each gap, the analysis describes what the biological system does, what current AI systems do, where the gap lies, and what research is needed to close it.

The goal is not to be pessimistic about AI progress but to be precise about what remains undone, so that research resources can be directed where they are most needed.

This precision matters because resources are finite, and the history of AI contains many examples of effort expended on approaches that led



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to dead ends because the fundamental gaps were not clearly identified.

It is worth stating plainly that these ten gaps are not independent items on a checklist; they are deeply interwoven, so that progress on embodiment supports grounding, progress on neuromodulation supports continual learning, and a genuine solution is likely to address several gaps at once rather than each in isolation.

A fair analysis must also confront an obvious objection, that the reasoning models and agents described in the previous chapter have already begun to close these gaps.

They have narrowed some of them, and this book takes that progress seriously.

A model that reasons step by step for a full minute before answering does better at multi-step problems, and an agent that can run code and check its own work catches mistakes that a single-shot chatbot would confidently repeat.

Yet the gaps identified here have proven stubborn precisely because they survive these improvements.

A reasoning model still learns nothing permanent from the problem it just solved, because its weights are frozen after training, so the sample-efficiency gap and the continual-learning gap remain.

The same model that scores well on a familiar mathematics competition still collapses on ARC-AGI-2 puzzles that a child would find easy, which is the signature of a system that pattern-matches



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within its training distribution rather than reasoning about genuinely novel structure.

Reasoning and tool use, in other words, are powerful additions layered on top of the existing paradigm, not solutions to the deeper architectural differences catalogued below.

Gap One: Sample Efficiency:

Biological systems learn from very few examples.

A child acquires a vocabulary of tens of thousands of words through informal, unsupervised exposure to natural speech and conversation.

A human who has never played chess before can learn to play a recognizable game after watching one or two games.

An octopus can solve a novel puzzle on the first attempt by observing another octopus solving it.

Current AI systems require orders of magnitude more data to achieve comparable performance.

Large language models are trained on trillions of tokens of text.

Image classifiers require millions of labeled examples.

Reinforcement learning systems require millions of interactions.

The gap in sample efficiency between biological and artificial learning is one of the most fundamental differences between NI and AI, and it



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is not merely a matter of degree, because the biological learner is bringing a lifetime of prior structure to each new problem rather than starting from a blank slate.

Closing this gap requires understanding the mechanisms that make biological learning so data-efficient: hierarchical feature reuse, compositional generalization, strong inductive biases derived from evolutionary preparation, active learning through exploration, and the integration of new knowledge with rich prior knowledge bases.

Research priorities include Meta-learning (learning to learn – the development of AI systems that, through experience with many different tasks, acquire inductive biases that allow them to learn new tasks quickly from very few examples), one-shot and few-shot learning architectures, better integration of structured prior knowledge with statistical learning, and embodied learning through active exploration.

Gap Two: Catastrophic Forgetting and Continual Learning:

Biological organisms learn continuously throughout life without erasing previously acquired knowledge.

A person who learns to play the piano does not forget how to ride a bicycle.

A doctor who learns a new medical procedure does not forget the procedures they already knew.



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AI systems currently suffer from catastrophic forgetting (the tendency of an artificial neural network to completely erase previously learned knowledge when trained on a new task).

When an AI system learns task B, the global gradient updates alter the parameters optimized for task A, rendering the system incapable of performing the original task without complete retraining from scratch.

This is the engineering face of the stability-plasticity dilemma, the requirement that a learning system be plastic enough to absorb the new yet stable enough to retain the old, a balance biology strikes routinely and artificial networks strike only crudely.

Biological organisms avoid catastrophic forgetting through the complementary learning systems architecture, sleep-based memory consolidation, neuromodulatory gating of plasticity, and structural mechanisms that protect stable knowledge from overwriting.

Research priorities include implementing a genuine two-system architecture with fast episodic and slow semantic components, offline consolidation processes analogous to sleep-based memory replay, mechanisms for selectively protecting important knowledge while allowing flexible updating of less important knowledge, and integration of episodic and semantic memory systems.

Gap Three: Causal Reasoning and World Modeling:

Biological intelligence reasons about causes and effects, can mentally simulate counterfactuals,



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and can plan multi-step action sequences by predicting their outcomes.

A child understands that dropping a glass will cause it to break before it hits the floor.

A hunter-gatherer can plan a multi-day route through territory based on a mental model of the landscape.

A scientist can reason about what experiment would distinguish between competing hypotheses.

Current AI systems are trained on statistical correlations and systematically struggle with genuine causal reasoning.

They can identify patterns but fail to reliably distinguish cause from effect or to reason about what would happen if circumstances were changed, the difference, in Judea Pearl's terms, between merely seeing and actually doing.

Research priorities include algorithms that can infer causal structure from observational data, AI systems that learn disentangled representations in which independent causal factors are represented separately, integration of causal models with neural learning systems in hybrid architectures, and incorporation of physical intuitions about causality into AI world models.

Gap Four: Embodiment and Grounded Understanding:

Natural intelligence emerged through billions of years of organisms interacting with physical environments.



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The body shapes cognition in fundamental ways.

Motor interaction teaches physics.

Sensory interaction teaches causality.

Environmental interaction teaches adaptation.

Current AI systems are largely disembodied – they process data without existing in or interacting with a physical world.

This creates the Symbol Grounding Problem described in Chapter 8.

Disembodied systems may lack important grounding mechanisms that are necessary for genuine understanding rather than sophisticated pattern matching.

Research priorities include robotic learning systems that can learn general-purpose manipulation and navigation skills from minimal supervision, simulation-to-real transfer methods that produce skills transferring effectively to real physical systems, language grounding that connects language representations to perceptual and sensorimotor experience, multi-modal learning systems integrating visual, auditory, proprioceptive, and tactile information in a unified representation, and developmental approaches that build competence progressively as biological organisms do.

Gap Five: The Architectural Gap – Von Neumann Versus Biological Computing:



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Silicon systems separate execution logic from data storage, creating systemic latency and vulnerability.

Biological brains merge execution and storage, ensuring that localized damage causes graceful degradation rather than total system failure.

Artificial networks rely on continuous, synchronized floating-point arithmetic across synchronized layers.

Natural networks utilize asynchronous, event-driven temporal spikes that leverage time itself as a core computational variable.

The human brain operates on approximately twenty watts.

Large AI training runs consume megawatts.

Research priorities include physically implemented neuromorphic computing hardware that executes spiking neural networks natively, temporal coding mechanisms where information is represented by precise arrival times of electrical pulses, and three-dimensional silicon integration stacking memristive layers vertically to replicate the density of biological neural circuits.

A memristor is an electrical component that regulates the flow of electrical current in a circuit while simultaneously remembering the amount of charge that previously passed through it, physically instantiating the co-localization of memory and processing that biology achieves through synaptic structure.



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Gap Six: Neuromodulation and Context-Sensitive Learning:

Biological intelligence uses chemical neuromodulators to dynamically adjust the learning rate, attention, and plasticity of neural circuits based on global states including reward, novelty, arousal, and threat level.

A single dopamine signal can cause circuits throughout the brain to simultaneously update their connection strengths in a reward-predictive direction.

A surge of norepinephrine during a threatening situation causes wide-scale changes in what the brain attends to and remembers.

Current AI systems have no equivalent mechanism.

They operate with static, unyielding mathematical hyperparameters (adjustable settings that govern training behavior) that do not dynamically adapt to environmental context, so that a single global learning rate, fixed in advance, plays the role that a rich chemistry of context-sensitive signals plays in the brain.

Research priorities include developing generalized STDP equations that incorporate synthetic neuromodulators, building simulated chemical diffusion fields within software networks that allow virtual nodes to release digital variants of dopamine, acetylcholine, and serotonin, and implementing meta-plasticity algorithms that track the structural history of every individual parameter to prevent runaway saturation.



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Gap Seven: Energy Efficiency:

The human brain operates on approximately 20 watts of power – comparable to a dim lightbulb.

It achieves this remarkable efficiency through sparse coding, precise spike-timing communication, highly efficient synaptic mechanisms, and architectural organization that minimizes long-range communication.

Training a large language model can consume megawatt-hours of electricity.

Running inference on large models consumes significant power per query.

The energy efficiency gap between biological and artificial intelligence is many orders of magnitude.

This matters not only for environmental and economic reasons but because energy efficiency constraints may be a significant factor shaping the architectural organization of biological intelligence, which suggests that copying the brain's efficiency tricks may also mean copying some of the cleverness those tricks forced evolution to discover.

Research priorities include sparse and event-driven computing architectures, neuromorphic hardware that executes computation natively within physical material transitions, algorithms that implement sparse coding inspired by the brain's approach to minimizing energy use, and architectural designs that minimize long-range communication costs.



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Gap Eight: Social Intelligence and Theory of Mind:

Human intelligence is distinctively social.

A large portion of the human brain is dedicated to navigating social hierarchies, understanding intentions, and cooperating with others.

Theory of Mind (the ability to attribute mental states including beliefs, intents, desires, emotions, and knowledge to oneself and to others, and to understand that others have minds distinct from one's own) is fundamental to human social functioning.

Theory of Mind enables deception detection, cooperative planning, teaching, empathy, and the interpretation of ambiguous behavior.

Current AI systems lack genuine Theory of Mind.

An LLM can predict what a person might say, but it does not maintain a model of the person as an entity with a distinct internal life, current beliefs, and goals that differ from its own.

This leads to failures in social reasoning, misinterpretation of indirect communication, and inability to model how a human will react to information the AI provides.

Research priorities include multi-agent learning environments that require genuine modeling of other agents' beliefs and intentions, embodied social learning systems, architectures that explicitly represent the beliefs and knowledge



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states of multiple agents, and computational models of empathy and social emotion.

Gap Nine: Metacognition and Self-Knowledge:

Biological intelligence knows what it knows and what it doesn't know.

It monitors the quality of its own cognitive processes, selects appropriate strategies, and detects its own errors before external feedback.

A skilled human expert knows the limits of their expertise and knows when to defer to others.

AI systems currently lack reliable metacognition.

Large language models are notorious for generating confident-sounding outputs even when they are factually incorrect – a failure of confidence monitoring that poses serious risks when these systems are deployed in medical, legal, scientific, or other high-stakes applications.

This failure is closely tied to the phenomenon of hallucination, in which a model produces fluent, plausible, and wholly fabricated information, precisely because it has no internal sense of the boundary between what it knows and what it is merely generating.

Building AI systems with genuine metacognitive ability is one of the most important practical research priorities, both for capability and for safety.

Gap Ten: Consciousness and Deep Integration:



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The brain integrates information from perception, memory, emotion, motivation, and bodily state into a unified experience of the present moment.

This deep integration may be what allows biological intelligence to respond coherently to complex situations that require simultaneously considering many interacting factors.

Current AI systems process information in largely modular ways without this deep integration.

Whether consciousness is necessary for AGI remains controversial, but the integration that consciousness may represent is likely necessary for the kind of flexible, contextually appropriate behavior that defines general intelligence.

Research priorities include computational consciousness modeling, recursive self-model architectures, integrated information measurement, artificial introspection systems, and temporal continuity frameworks that give AI systems persistent internal states across time.

PART THREE: THE RESEARCH AGENDA

CHAPTER 13: RESEARCH PATHWAYS TO BRIDGE THE GAP: THE TECHNICAL AGENDA:

Overview of the Research Agenda:

The gap analysis of Chapter 12 identifies ten major dimensions along which Artificial Intelligence (AI) falls short of Natural Intelligence (NI).



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Each of those ten gaps is a place where biological brains routinely do something that today's machines cannot do reliably, cheaply, or at all.

Closing these gaps is not a matter of one clever invention.

It requires a coordinated, multi-decade research agenda spanning neuroscience, cognitive science, computer science, philosophy, and engineering.

This chapter describes the most important research pathways, organized by their relationship to the identified gaps.

It is not intended as an exhaustive catalog of all relevant research.

It is intended as a strategic map of the highest-leverage research directions – the research that is most likely to unlock genuine progress toward Artificial General Intelligence (AGI).

A useful way to read this chapter is to treat each pathway as a hypothesis about which missing biological principle matters most.

If a pathway is correct, then importing its principle into machines should produce a capability that scaling alone has failed to deliver.

The pathways are presented separately for clarity, but in a mature AGI program they would be pursued together, because the brain does not implement them in isolation either.



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Research Pathway One: Predictive Processing and Active Inference:

The predictive processing framework may be one of the most powerful theoretical frameworks available for guiding the design of brain-inspired AI systems. [6] [7]

The core idea is that intelligence arises from systems that continuously predict their sensory inputs and update their predictions based on prediction errors, rather than passively processing whatever inputs arrive.

In this view the brain is not a stimulus-response machine but a prediction machine that is always running a model of the world slightly ahead of the world itself.

The foundational technical statement of this idea in neuroscience is the predictive coding model of the visual cortex published by Rajesh Rao and Dana Ballard in 1999.

Rao and Ballard proposed that higher levels of the cortex continuously send predictions downward to lower levels, and lower levels send back only the prediction error – the part of the signal that was not anticipated.

This arrangement is extremely efficient, because a perfectly predicted input generates almost no upward traffic, and the system spends its limited bandwidth only on what is surprising.

A familiar everyday example is that you stop consciously hearing a steady air conditioner within minutes, yet you notice instantly when it



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switches off, because the silence violates a standing prediction.

Karl Friston's Free Energy Principle provides a mathematical formalization of this idea that unifies perception, learning, action, and attention in a single coherent framework.

Free energy here is a quantity borrowed from statistical physics that serves as a tractable upper bound on surprise, defined as the gap between what a system predicts and what it actually senses.

On this account, biological systems act to minimize free energy - a quantity closely related to the difference between predictions and observations - either by updating their internal model of the world or by taking actions that bring the world into correspondence with their predictions.

The second of those options is called active inference, the extension of predictive processing to action, in which actions are understood as a way of fulfilling predictions about one's own sensory inputs.

Under active inference, when you reach for a cup you are in effect predicting the sensation of holding it, and your motor system moves your arm to make that prediction come true.

This dissolves the artificial boundary between perception and control that most current AI systems still maintain, because perceiving and acting become two ways of reducing the same prediction error.



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A practical reason this framework matters for engineering is that recent work has shown predictive coding can closely approximate backpropagation, the standard training algorithm of deep learning, while using only local computations.

Researchers including Beren Millidge, James Whittington, and Rafal Bogacz have demonstrated that a predictive coding network, in which each unit updates using only signals from its immediate neighbors, can match the results of backpropagation on deep architectures.

This is significant because backpropagation requires a biologically implausible global pass of error signals through the entire network, whereas predictive coding is local, parallel, and far closer to how real neurons could learn.

Implementing the predictive processing framework in AI systems is therefore a major research direction with both scientific and hardware payoffs.

Research priorities in this area include developing scalable, computationally tractable implementations of predictive coding for large-scale AI systems, extending predictive processing models to include active inference, implementing neuromodulatory regulation within the predictive processing framework, and connecting predictive processing with reinforcement learning in a unified framework.

Reinforcement learning is the branch of machine learning in which an agent learns by receiving

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rewards or penalties for its actions, and unifying it with active inference would let a single system treat seeking reward and reducing uncertainty as aspects of one objective.

Research Project Idea: Build a hierarchical predictive coding network trained on multi-modal sensorimotor data from robotic platforms.

- Implement active inference so the robot acts to confirm its predictions rather than passively receiving data.
- Measure learning efficiency and generalization against conventional deep learning baselines on tasks that require genuine adaptation to novel situations.
- Track how much sensory bandwidth the predictive system saves by transmitting only prediction errors, which is a direct test of biological efficiency claims.

Research Pathway Two: Complementary Learning Systems for Lifelong Learning:

As described in Chapter 5, the brain solves the stability-plasticity dilemma through a complementary learning systems architecture.

The stability-plasticity dilemma is the challenge of learning new things rapidly without overwriting what has already been learned, and it is one of the deepest unsolved problems in machine learning.

The Complementary Learning Systems (CLS) theory was articulated by James McClelland, Bruce



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McNaughton, and Randall O'Reilly in an influential 1995 paper in the journal *Psychological Review*.

Their proposal is that the brain uses two cooperating memory systems with deliberately different learning speeds.

The hippocampus, a seahorse-shaped structure deep in the brain, is the fast system that rapidly captures individual experiences after a single exposure.

The neocortex, the brain's wrinkled outer sheet, is the slow system that gradually extracts general structure across many experiences without disrupting what it already knows.

During sleep the hippocampus replays recent experiences to the neocortex, allowing the slow system to weave new knowledge into old knowledge a little at a time.

Implementing something analogous in AI is essential for achieving lifelong learning without catastrophic forgetting, where a network trained on a new task abruptly loses its ability to perform an old one.

It is worth noting that this biological idea has already paid off once: the experience replay buffer that made deep reinforcement learning practical, used in DeepMind's 2015 Atari-playing system, was explicitly inspired by hippocampal replay.

Current approaches to the forgetting problem include Elastic Weight Consolidation (EWC) – a technique introduced by James Kirkpatrick and



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colleagues at DeepMind in 2017 that protects the weights most important for previously learned tasks from being overwritten by penalizing changes to those weights during new training.

A useful analogy for EWC is hardening into place the bricks that hold up a finished wall while leaving the mortar of unused regions soft enough to build something new.

Other current approaches include Progressive Neural Networks, architectures that add new modules for each new task while freezing learned weights from previous tasks, Memory-Augmented Neural Networks, architectures that include explicit external memory modules capable of rapid storage and retrieval, and Experience Replay, techniques that maintain a buffer of past training examples and regularly replay them during new learning to prevent forgetting.

These approaches are promising but not yet satisfactory, because they tend to scale poorly, require knowing task boundaries in advance, or consume ever-growing memory.

A more biologically grounded approach would implement a genuine two-system architecture with a fast hippocampal-like system and a slow neocortical-like system, an offline consolidation process analogous to sleep-based memory replay, mechanisms for selectively protecting important knowledge while allowing flexible updating of less important knowledge, and integration of episodic and semantic memory systems in a single unified architecture.



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Episodic memory is memory for specific events tied to a time and place, while semantic memory is memory for general facts stripped of their original context, and a mature system needs both and needs to convert one into the other over time.

Research Project Idea: Develop sleep-like states in AI systems where models replay and consolidate experiences during offline periods.

- Measure improvements in generalization and resistance to forgetting compared to systems without consolidation.
- Compare performance before and after consolidation interventions across diverse task sequences that span multiple domains.
- Investigate whether generative replay, in which the system regenerates plausible past experiences rather than storing them verbatim, can match biological consolidation while saving memory.

Research Pathway Three: Causal Representation Learning:

Building AI systems that learn the causal structure of their environment rather than just statistical correlations is one of the highest-priority research directions for closing the gap between NI and AI.

The distinction matters because correlation tells you what tends to occur together, while causation tells you what will happen if you intervene, and only the latter supports reliable planning.



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A classic illustration is that ice cream sales and drowning deaths rise together, but banning ice cream would not save a single swimmer, because the hidden common cause is hot weather.

Current deep learning systems are extraordinary correlation engines, which is exactly why they fail in unfamiliar conditions where the old correlations no longer hold.

The theoretical foundation for this work is provided by Judea Pearl's do-calculus, a mathematical framework for reasoning about interventions and counterfactuals in causal systems, for which Pearl received the 2011 Turing Award.

A counterfactual is a claim about what would have happened under a different action, such as whether a patient would have recovered had they not taken a drug, and answering such questions is something humans do effortlessly and machines do poorly.

The do-calculus works alongside the framework of Structural Causal Models (SCMs) – mathematical models that represent the causal relationships between variables in a system using directed graphs and structural equations.

In a structural causal model, an arrow from one variable to another means that wiggling the first directly changes the second, which is a far stronger and more useful claim than mere statistical association.

A central modern research program, set out in the 2021 paper "Toward Causal Representation Learning" by Bernhard Schölkopf, Yoshua Bengio, and



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colleagues in the Proceedings of the IEEE, argues that machine learning must learn the causal variables themselves, not just relationships among hand-chosen variables.

Their guiding idea is the principle of independent causal mechanisms, the assumption that the world is built from separate, reusable cause-and-effect modules that can be recombined, which is why a child who learns how shadows work can apply that knowledge to any object.

Research priorities include algorithms that can infer causal structure from observational data, AI systems that learn disentangled representations, integration of causal models with neural learning systems in hybrid architectures, and counterfactual data augmentation.

Disentangled representations are representations in which independent causal factors are encoded separately rather than mixed together, so that a model of a face might vary lighting without accidentally changing identity.

Counterfactual data augmentation means training on data generated by intervening on the causal model, exposing the AI system to the full range of causal relationships rather than only the ones that happened to appear in the original data.

Research Project Idea: Fund interdisciplinary teams combining neuroscientists and AI engineers to map specific brain regions like the hippocampus to memory and causal reasoning architectures in machines.



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- Investigate predictive coding, where the brain constantly predicts sensory input and updates based on errors.
- Implement similar hierarchical prediction models with explicit causal representations in artificial neural networks and test on tasks requiring genuine intervention reasoning.
- Evaluate systems on held-out interventions that never appeared in training, because only out-of-distribution intervention tests can distinguish genuine causal understanding from sophisticated curve fitting.

Research Pathway Four: Embodied AI and Grounded Learning:

Closing the embodiment gap requires developing AI systems that learn through physical interaction with the world and that have perceptual and motor capabilities grounded in real physical experience.

The motivating insight is captured by Moravec's paradox, the observation by roboticist Hans Moravec that tasks which are hard for humans, such as chess, turned out to be easy for computers, while tasks which are trivial for a toddler, such as walking across a cluttered room, turned out to be extraordinarily hard.

The reason is that perception and movement were honed by hundreds of millions of years of evolution, while abstract reasoning is a thin and recent layer on top of that ancient foundation.

A closely related problem is the symbol grounding problem, named by Stevan Harnad in 1990, which



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asks how the symbols a system manipulates come to mean anything at all rather than being shuffled tokens defined only in terms of other tokens.

A language model that has only ever read text knows that "ice is cold" follows from "ice is cold" in millions of sentences, but it has never felt cold, so its understanding is borrowed rather than grounded.

Embodiment is the proposed cure, because a body forces meaning to be anchored in sensation and consequence.

Research priorities include robotic learning systems that can learn general-purpose manipulation and navigation skills from minimal supervision, simulation-to-real transfer methods that produce skills transferring effectively to real physical environments, language grounding connecting language representations to perceptual and sensorimotor experience, multi-modal learning systems integrating visual, auditory, proprioceptive, and tactile information in a unified representation, and developmental approaches that build competence progressively as biological organisms do rather than learning everything from a fixed training set.

Simulation-to-real transfer, often called sim-to-real, is the practice of training a robot cheaply and safely in a physics simulator and then transferring the learned skill to a physical machine, and bridging the inevitable gap between simulated and real physics is itself an active research frontier.



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Research Project Idea: Create embodied AI agents in simulated and physical environments that learn through interaction similar to infant development.

- Incorporate reward systems based on information gain rather than external labels to implement curiosity-driven learning.
- Progress from physical manipulation and navigation to symbolic reasoning and language grounded in physical experience.
- Measure progress using benchmarks that test adaptation to genuinely novel situations not present in training data.

Research Pathway Five: Neurosymbolic Integration:

The most successful AI systems have been neural, meaning they learn distributed representations from data, but the most reliable and interpretable AI systems have been symbolic, meaning they manipulate explicit, structured representations according to formal rules.

Neural systems are superb at perception and pattern recognition but are opaque and prone to confident errors, while symbolic systems are transparent and exact but brittle when faced with the messiness of the real world.

Neurosymbolic AI, which combines neural learning with symbolic reasoning, aims to combine the strengths of both approaches. [16] [26]

The cognitive motivation comes from the dual-process account of human thinking popularized by Daniel Kahneman, which distinguishes fast,



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intuitive System 1 thought from slow, deliberate System 2 thought, with neural networks resembling the former and symbolic reasoning the latter.

A landmark demonstration of the power of this combination is AlphaGeometry, released by Google DeepMind in January 2024, which solved Olympiad-level geometry problems at nearly the level of a human gold medalist.

AlphaGeometry pairs a neural language model that proposes promising constructions with a symbolic deduction engine that rigorously checks and chains logical steps, so creativity and rigor reinforce each other.

On a benchmark of thirty International Mathematical Olympiad geometry problems it solved twenty-five within the time limit, far beyond what either a pure neural or a pure symbolic system had achieved alone.

Other research directions include systems such as DeepProbLog that fuse neural networks with probabilistic logic, and longstanding advocacy by researchers such as Gary Marcus that robust intelligence will require explicit symbolic structure rather than scale alone.

Research priorities include architectures that can extract symbolic rules from learned neural representations, hybrid systems that use neural networks for perception and symbol grounding and symbolic systems for reasoning and planning, program induction methods that can learn algorithms from examples, and systems that can perform both pattern recognition and logical deduction within a unified architecture.



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Program induction is the task of learning a general, executable procedure from a handful of examples, which is much closer to how a person infers a rule than to how a network fits a curve.

Research Project Idea: Build hybrid systems combining symbolic reasoning with connectionist approaches modeled on the functional specialization observed across brain regions.

- Test on tasks requiring both pattern recognition and logical deduction.
- Evaluate energy efficiency and robustness metrics compared to purely neural and purely symbolic baselines.

Research Pathway Six: Multi-Agent Social Intelligence:

Human intelligence is profoundly social, and many of its most distinctive capabilities – Theory of Mind, cultural learning, language, cooperative planning – may require multi-agent interaction to develop properly. [14] [25]

Theory of Mind is the ability to attribute beliefs, desires, and intentions to others and to recognize that those mental states may differ from one's own, and it is what lets a person predict that someone will look for an object where they last saw it, not where it actually is.

The social brain hypothesis, associated with the anthropologist Robin Dunbar, proposes that the unusual size of the primate neocortex evolved largely to manage the demands of complex social



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groups rather than to solve problems with the physical world.

Solitary AI systems trained on static datasets may lack the fundamental social grounding that biological intelligence acquires through real-time social interaction.

There is encouraging experimental evidence that placing agents in rich social environments can call forth sophisticated behavior that was never explicitly programmed.

OpenAI's hide-and-seek experiments, for example, produced agents that spontaneously invented tool use and counter-strategies through nothing but multi-agent competition, and DeepMind's capture-the-flag work produced cooperative team play among independent learners.

Research priorities include cooperative multi-agent reinforcement learning environments, systems that develop genuine models of other agents' beliefs and intentions, emergent communication research studying how agents develop shared communication systems spontaneously, and distributed cognition architectures inspired by collective intelligence in social insects and human groups.

Emergent communication research is especially important, because a language that arises from the pressure to coordinate is grounded in shared activity in a way that imitating human text never can be.

Research Project Idea: Establish long-running multi-agent environments in which AI agents must



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cooperate and communicate to solve problems that neither could solve alone.

- Study the emergence of Theory of Mind representations, shared language, and cultural transmission within populations of agents.
- Use findings to develop AI architectures that incorporate social cognition as a fundamental rather than add-on component.

Research Pathway Seven: Neuromorphic Hardware:

The architectural gap between biological and artificial computing cannot be closed through software alone.

It requires hardware that physically embodies the principles of biological neural computation.

The field of neuromorphic engineering, which designs chips that imitate the structure and dynamics of nervous systems, was founded by the Caltech physicist Carver Mead in the late 1980s.

The deepest problem neuromorphic hardware addresses is the von Neumann bottleneck, the costly shuttling of data between separate processing and memory units that dominates the energy budget of conventional computers and has no counterpart in the brain, where memory and computation occur in the same synapses.

Neuromorphic chips instead use spiking neural networks, networks whose artificial neurons communicate with discrete electrical pulses and remain silent and power-free when they have nothing to say, just as biological neurons do.



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Intel's Loihi 2 research chip illustrates the approach, supporting on the order of a million neurons and a hundred million synapses per chip with event-driven operation, and earlier systems built on this family have shown energy reductions of up to roughly two orders of magnitude on suitable workloads.

IBM's TrueNorth chip was an earlier influential demonstration of the same massively parallel, low-power philosophy.

Research priorities include large-scale memristive crossbar arrays that can execute asynchronous spiking neural networks natively in hardware, temporal coding mechanisms where data is represented by precise arrival times of electrical pulses, three-dimensional silicon integration stacking memristive layers vertically to replicate high-density biological connectivity, and hardware that supports native plasticity.

Native plasticity is the ability for the chip to physically change its own conductance during operation rather than simulating plasticity through software updates, which would let learning happen in the material itself.

A memristor, central to several of these priorities, is an electrical component that regulates the flow of current while simultaneously remembering the amount of charge that has previously passed through it, physically co-locating memory and processing in a single device.



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Research Project Idea: Establish a global database of neural recordings and connectomics data for training and validating neuromorphic AI models.

- Prioritize ethical data collection from non-human organisms including well-characterized model organisms like the roundworm *Caenorhabditis elegans* and the fruit fly *Drosophila*.
- Develop simulation tools to test architectural hypotheses before committing to expensive physical hardware implementation.
- Share findings openly to allow international researchers to build on each other's progress.

Research Pathway Eight: Whole Brain Emulation and Connectomics:

Whole Brain Emulation (WBE) is the hypothetical process of scanning a biological brain in sufficient detail and copying its state into a computer simulation.

This is an extreme form of the gap analysis.

Instead of designing an AGI from first principles, WBE would attempt to copy one from a biological original.

While currently technologically impossible due to scanning resolution limits, research into connectomics, the mapping of the brain's connections at cellular and subcellular resolution, is a direct stepping stone toward this goal and also provides data that is immediately useful for inspiring AI architectures.



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The history of connectomics shows both how far the field has come and how far it has to go.

The first complete connectome, the wiring diagram of the 302-neuron nervous system of the roundworm *Caenorhabditis elegans*, was published by John White and colleagues in 1986 and took more than a decade of painstaking electron microscopy.

Nearly four decades later, in October 2024, the FlyWire consortium led by Mala Murthy and Sebastian Seung at Princeton published the first complete connectome of an adult fruit fly brain, mapping roughly 140,000 neurons and more than fifty million connections.

The jump from 302 neurons to 140,000 took thirty-eight years, which puts in sobering perspective the roughly 86 billion neurons of the human brain.

The open-source OpenWorm project, which attempts to build a working simulation of *C. elegans* from its connectome, has further revealed that the wiring diagram alone is not enough, because the dynamics of each neuron and synapse also matter.

Research priorities include development of higher-resolution electron microscopy techniques for mapping synaptic connectivity, computational tools for reconstructing three-dimensional neural circuits from two-dimensional image data, databases and standards for sharing connectomics data internationally, and development of simulation environments capable of running reconstructed neural circuits at biologically realistic scales.

Research Pathway Nine: Developmental Robotics:

Rather than training AI systems to be capable through exposure to massive datasets, developmental robotics takes inspiration from biological development by building systems that start as capability-poor infants and acquire capabilities through progressive interaction with the environment. [12] [13]

This approach shifts the frame from training to raising an AI - from optimization to development.

The distinction is more than rhetorical, because a developing organism does not receive a fixed dataset but actively chooses what to explore next, and that self-curated curriculum may be essential to open-ended growth.

A key mechanism here is intrinsic motivation, in which the agent generates its own reward from curiosity rather than from an external scorekeeper.

Researchers including Pierre-Yves Oudeyer have shown that rewarding an agent for learning progress, the rate at which its predictions are improving, leads it to seek out experiences that are neither trivially boring nor hopelessly random, which is precisely the zone in which the most learning happens.

Jurgen Schmidhuber's formal theory of curiosity frames the same idea in terms of compression progress, rewarding the agent for finding patterns that make the world more compactly describable.

Research priorities include virtual environments rich and complex enough to stimulate the



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development of cognitive faculties through exploration, robotic platforms with developmental curricula that progressively introduce more complex challenges, architectures capable of open-ended capability acquisition without catastrophic forgetting, and methods for assessing developmental progress in ways that are not gameable by narrow optimization.

Taken together, these nine pathways form a portfolio rather than a single bet, and the chapters that follow describe how four of them can be combined into concrete, mutually reinforcing flagship projects.

CHAPTER 14: FOUR CORNERSTONE RESEARCH PROJECTS:

The following four research projects represent the most strategically important, highest-leverage initiatives for systematically bridging the gap between Natural Intelligence and Artificial Intelligence.

Each addresses a fundamental architectural limitation of current AI systems by implementing a principle from biological intelligence that current approaches have failed to incorporate.

These four projects are complementary and mutually reinforcing.

The outputs of each feed into the others, and the full vision of a biologically grounded AGI architecture requires all four to succeed together.

A helpful way to hold the four in mind is by the biological function each one imports: Project



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Alpha imports the brain's hardware, Project Beta imports its learning chemistry, Project Gamma imports its self-regulation, and Project Delta imports its body.

Project Alpha: Physically Implemented Neuromorphic Computing:

The objective of Project Alpha is the replacement of the von Neumann architecture with co-localized processing and memory through the creation of physical neuromorphic computing hardware.

The von Neumann architecture, named after the mathematician John von Neumann, is the design used by almost every computer today, in which a processing unit and a separate memory unit exchange data over a shared channel.

The motivating observation is that the architectural separation of processing and memory in conventional computers creates a fundamental bottleneck that has no analog in biology, and that eliminating this bottleneck by physically co-localizing computation and memory may yield dramatic improvements in energy efficiency and speed.

The scale of the prize is enormous, because the human brain performs feats no supercomputer can match while consuming about twenty watts, roughly the power of a dim light bulb, whereas training a single large AI model can consume the annual electricity of hundreds of homes.

The approach abandons standard transistors in favor of memristors, electrical components that regulate the flow of current while simultaneously



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remembering the amount of charge that previously passed through them, physically instantiating co-localized memory and processing.

Phase One of Project Alpha will focus on designing large-scale memristive crossbar arrays that can execute asynchronous spiking neural networks natively in hardware.

A crossbar array is a grid of wires with a memristor at each intersection, an arrangement that can perform the central mathematical operation of neural networks, the multiplication and summation of many signals, in a single physical step and in the place where the values are stored.

Phase Two will integrate temporal coding mechanisms, where data is represented by the precise microsecond arrival time of electrical pulses rather than continuous numerical values.

Temporal coding mirrors a strategy real neurons use, in which the timing of a spike, not merely whether it occurs, carries information, allowing rich signaling with very few pulses and therefore very little energy.

Phase Three will develop three-dimensional silicon integration, stacking memristive layers vertically to replicate the high-density connectivity found in the mammalian cerebral cortex.

Such vertical stacking is necessary because a flat chip cannot approach the connectivity of cortex, where a single neuron may connect to ten thousand others through an intricate three-dimensional tangle of fibers.



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By executing computation natively within physical material transitions rather than simulating mathematics through digital gates, Project Alpha aims to reduce AGI hardware power requirements by multiple orders of magnitude.

Near-term milestones for Project Alpha include demonstrating that a memristive spiking network can perform a standard benchmark task at a fraction of the energy cost of a conventional deep learning system, and that the physical plasticity of memristors can implement a biologically plausible learning rule without requiring external gradient computation.

A gradient, in this context, is the mathematical signal that tells conventional networks how to adjust each weight, and the fact that the brain learns without computing global gradients is precisely the gap Project Beta is designed to close.

Project Beta: Biochemical Learning Rules and Localized Plasticity:

The objective of Project Beta is the replacement of global backpropagation with a purely decentralized, localized learning engine that implements the principles of biological synaptic plasticity.

Backpropagation, the algorithm that powers nearly all modern deep learning, works by computing a single global error at the output and propagating it backward through every layer, which requires each part of the network to wait for information from distant parts.



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The motivating observation is that biological learning is local – each synapse changes based only on the activity of the two neurons it connects, without requiring information from distant parts of the network – and this locality is likely essential for achieving the energy efficiency, continual learning, and resistance to catastrophic forgetting that biological systems demonstrate.

The classic statement of local learning is Hebb's rule, proposed by the psychologist Donald Hebb in 1949 and often summarized as "neurons that fire together wire together," meaning a synapse strengthens when the cells on either side of it are active at the same time.

A more precise modern form is spike-timing-dependent plasticity (STDP), in which a synapse strengthens if the sending neuron fires just before the receiving neuron and weakens if the order is reversed, so that the synapse becomes sensitive to cause and effect at the scale of milliseconds.

Project Beta develops mathematical models and software frameworks that leverage multi-scale, localized plasticity rules operating without human-engineered global loss functions.

The core research focuses on developing generalized STDP equations that incorporate synthetic neuromodulators.

Neuromodulators are chemicals such as dopamine, acetylcholine, and serotonin that the brain releases to adjust how and when learning happens,



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acting like global volume knobs on top of local synaptic changes.

Researchers will build simulated chemical diffusion fields within software networks, allowing virtual nodes to release digital variants of dopamine, acetylcholine, and serotonin that modulate local learning rates.

Digital dopamine will function to dynamically alter local Hebbian rules based on environmental reward signals, driving the network to strengthen pathways that led to rewarding outcomes, mirroring the role of dopamine as the brain's reward-prediction-error signal.

Digital acetylcholine will adjust local learning rates based on mathematical metrics of novelty and sensory surprise, opening learning windows when the environment contains unexpected information.

Digital serotonin will regulate network-wide risk aversion and prioritize long-term stability over short-term reward tracking.

Project Beta will implement meta-plasticity algorithms that track the structural history of every individual parameter, preventing runaway saturation and stabilizing learning over long operational lifetimes.

Meta-plasticity, sometimes called the plasticity of plasticity, is the biological phenomenon in which a synapse's recent history changes how readily it will change again, which protects important memories while keeping the system adaptable.



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The successful completion of Project Beta will produce an AI framework capable of continuous, lifelong learning across highly diverse domains without catastrophic forgetting, achieving through local biochemical principles what backpropagation cannot achieve through global mathematical ones.

Project Gamma: Homeostatic Regulation and Self-Repairing Network Architectures:

The objective of Project Gamma is the creation of self-regulating, autonomous software architectures that utilize the principles of homeostasis and the Free Energy Principle.

Homeostasis is the active process through which a system maintains stable internal conditions necessary for operation despite fluctuating external environments, the same process by which your body holds its temperature near thirty-seven degrees Celsius whether you are in snow or desert.

Current software is passive.

It does nothing unless instructed to do something, and it has no awareness of or concern for its own computational state.

Project Gamma will make software active, self-aware of its computational state, and driven by self-preservation.

This project will implement continuous intrinsic plasticity algorithms that dynamically adjust individual node thresholds to maintain constant internal network entropy.



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Entropy here is a measure of disorder or unpredictability in a system, where too much leads to chaos and too little leads to rigid inflexibility, and healthy brains are thought to operate near a balanced critical point between the two.

A crucial biological inspiration for this project is the realization that the brain is not only neurons.

Roughly half of the brain's cells are glial cells, long dismissed as mere support staff but now known to actively regulate, nourish, and prune neural circuits.

Researchers will design synthetic glial networks operating in parallel with the primary processing layers.

Virtual astrocytes, modeled on the star-shaped glial cells that wrap around synapses in the brain, will monitor the computational throughput of neighboring nodes and dynamically allocate virtual processing priority based on localized demand.

Virtual microglia, modeled on the brain's resident immune and maintenance cells, will continuously scan the network architecture, identifying inactive, redundant, or corrupted nodes and executing automated deletion and structural pruning to maintain computational efficiency.

This mirrors synaptic pruning, the developmental process in which the brain eliminates unused connections, discarding roughly half of the



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synapses present in early childhood to sculpt an efficient adult network.

The entire network will be governed by an overarching predictive coding framework designed to minimize mathematical free energy, forcing the system to actively learn, adapt, and seek out data that resolves environmental uncertainty.

Project Gamma will provide the foundational stability required for ASI, ensuring that as systems grow in complexity, they remain internally balanced, resilient to corruption, and energetically self-optimized.

Without the homeostatic mechanisms of Project Gamma, the continuous learning of Project Beta and the open-ended development envisioned for AGI would lead to computational instability at scale.

Project Delta: Sensorimotor Integration and Embodied Cognition:

The objective of Project Delta is to move AI out of isolated server-based text processing and embed it in physical or highly realistic virtual environments through embodied cognition.

Embodied cognition is the theory that many features of cognition are shaped by aspects of the entire physical body of the organism and its direct interactions with the world, rather than residing in a disembodied reasoning module.

Natural intelligence did not evolve to solve abstract word puzzles.



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It evolved to guide physical bodies safely through dangerous, three-dimensional spaces.

Language and abstract reasoning are late evolutionary additions built on top of a deep foundation of sensorimotor experience. [12] [35]

There is striking evidence that abstract thought reuses this physical machinery, because brain imaging shows that understanding the sentence "she grasped the idea" activates some of the same motor regions as physically grasping an object.

Project Delta is designed to replicate this developmental sequence.

Phase One will develop high-fidelity simulation environments that enforce rigid thermodynamic and physical laws, complete with complex gravitational, fluid, and mechanical constraints.

Phase Two will interface neuromorphic hardware controllers from Project Alpha with advanced robotic form factors possessing multi-modal sensory arrays.

These arrays will include tactile skin sensors, proprioceptive sensors, high-speed neuromorphic vision sensors, and acoustic sensors.

Proprioceptive sensors measure the internal position and movement of physical limbs, providing the sense, which humans take entirely for granted, of where one's own body is without looking at it.

The AI will not be fed pre-labeled training data.



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It must learn to perceive the world by actively moving its physical body and experiencing the immediate sensory feedback of its actions, mimicking the developmental trajectory of a biological organism from early sensorimotor exploration through progressively more abstract cognition.

Phase Three will study the emergence of conceptual abstractions, verifying how spatial navigation and physical object manipulation lay the structural foundation for advanced symbolic reasoning and linguistic mastery.

This phase directly tests a hypothesis from cognitive linguistics, associated with George Lakoff and Mark Johnson, that abstract concepts are understood through metaphors rooted in bodily experience, such as describing more as up and affection as warmth.

Project Delta ensures that the resultant AGI will possess deep, grounded understanding of physical reality, eliminating the hallucinations and logical failures common in today's large-scale language models that lack any physical grounding.

Hallucinations, in this context, are confident but factually incorrect outputs not grounded in reality, the characteristic failure mode of a system that has learned the form of language without its connection to the world.

CHAPTER 15: THE PATH TOWARD ARTIFICIAL GENERAL INTELLIGENCE:

Milestones on the Road to AGI:



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Achieving AGI will require integrating insights across all previous chapters.

No single research breakthrough will be sufficient.

AGI will likely require simultaneous progress on the architectural gap, the learning gap, the embodiment gap, the causal reasoning gap, the memory gap, and the social intelligence gap.

These gaps are not independent of each other.

Progress on one often depends on progress on others.

For example, building a genuine episodic memory system requires embodied experience to fill that memory with meaningful content.

Genuine causal reasoning may require both embodiment and appropriate memory architecture to develop.

Social intelligence may require both Theory of Mind and embodied participation in social interactions.

This interdependence is why the four cornerstone projects of the previous chapter are designed to run together rather than in sequence, and why a portfolio strategy is more likely to succeed than any single line of attack.

The milestones below represent signs of genuine progress toward AGI that would be difficult to achieve through narrow optimization of existing architectures.



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They are framed deliberately as capabilities to be demonstrated rather than benchmark scores to be exceeded, because the history of the field shows how easily a number can be gamed.

The first milestone will be true catastrophic forgetting immunity, where a single system can sequentially master diverse tasks such as chess, language translation, and mechanical repair without losing proficiency in prior skills. [8]
[26]

This milestone is the practical signature of a working complementary learning system, and it is the direct deliverable of Project Beta.

The second milestone will be zero-shot adaptive execution, where an embodied system encounters a completely novel environmental obstacle and immediately invents a viable physical solution based on real-time sensory prediction and physical reasoning, without having been trained on that specific obstacle.

Zero-shot means performing a task with no prior examples of that exact task, relying instead on transferable understanding, and it is the signature of genuine generalization rather than memorization.

The third milestone will be autonomous structural evolution, where the system identifies internal computational inefficiencies and designs its own structural alterations to optimize its performance without human intervention.



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The fourth milestone will be robust causal reasoning, where a system can reliably answer counterfactual questions and reason about interventions in domains it has never encountered, achieving performance comparable to human experts.

The fifth milestone will be genuine metacognitive accuracy, where a system reliably knows what it knows and what it doesn't know, and can communicate that distinction to users with calibrated confidence. [4] [26]

Metacognition is thinking about one's own thinking, and calibrated confidence means a system's stated certainty matches its actual accuracy, so that claims it makes with ninety percent confidence are correct about ninety percent of the time.

The sixth milestone will be autonomous goal formation, where a system identifies and pursues meaningful sub-goals in service of a high-level objective without requiring humans to decompose the task.

These milestones will not be achieved by expanding current software codebases.

They will require building systems that are architecturally different from anything that currently exists.

How Soon? The Timeline Debate:

No question about AGI provokes more disagreement than when it will arrive, and a careful reader deserves an honest map of that disagreement rather than a single confident date.



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By the mid-2020s several leaders of frontier laboratories were making strikingly near-term predictions.

Dario Amodei, the chief executive of Anthropic, argued that powerful, human-level systems could arrive within a few years, and other executives offered similar horizons measured in one to a few years for AI that could do most of what a skilled remote worker can do at a computer.

These forecasts drew on the genuine acceleration produced by reasoning models and agents, which turned some previously hard benchmarks in mathematics and programming into solved problems within a single year.

Independent forecasters, however, urged caution, and over the second half of 2025 many of them actually pushed their expected dates further out rather than closer.

Two sobering technical facts drove that caution.

The first is the so-called data wall, the observation that the supply of high-quality human-written text on which large models are trained is finite and largely already used, so the easy gains from simply training on more data are running out.

The second concerns the new engine of progress itself, because giving a model more time to think improves its answers, but the cost of thinking grows quickly, and there are not enough computer chips in the world to let every model think for hours on every question, which makes a one-time boost rather than an endless trend.



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A related disagreement is definitional, because people who confidently predict imminent AGI often mean something narrower, such as matching human performance on a broad set of economically valuable computer-based tasks, while those who predict a longer road usually mean the fuller, flexible, embodied, continually learning intelligence that this book has been describing.

When a chief executive and a skeptic disagree about the date, they are frequently disagreeing about the destination.

This book takes no position on a specific year, for the honest reason that the history of the field is a graveyard of confident dates, and each of the two AI Winters followed a wave of predictions much like today's.

What it does claim is narrower and, it argues, more useful, that the specific capability gaps catalogued in Chapter 12 are real, are measurable, and will have to be closed by whatever path leads to genuine general intelligence, whether that path is short or long.

One of the persistent challenges in AI research is the tendency to optimize for benchmark performance rather than genuine capability.

Benchmarks are standardized tests used to evaluate system capability.

This challenge is a specific case of Goodhart's law, the principle named after the economist Charles Goodhart that when a measure becomes a target, it ceases to be a good measure.



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Benchmark success can create illusions of progress.

A system may perform well on narrow evaluations while lacking genuine adaptability.

The history of AI contains many examples of benchmark improvements that did not correspond to genuine advances in the underlying capabilities the benchmarks were intended to measure.

A widely discussed contemporary example is the Abstraction and Reasoning Corpus, introduced by Francois Chollet in his 2019 essay "On the Measure of Intelligence," which deliberately presents novel puzzles that cannot be solved by memorizing training data and on which humans still far outperform many systems that excel at conventional benchmarks.

Chollet's central argument is that intelligence should be measured not as skill at a fixed task but as the efficiency with which a system acquires new skills from limited experience.

Future evaluation of AGI progress requires benchmarks that test for genuine generalization across novel domains, compositional generalization in tasks that require applying rules in new combinations, transfer learning across very different domains, robustness under distribution shift, causal reasoning and intervention, metacognitive accuracy and calibrated confidence, and continual learning without forgetting over extended time periods.



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Distribution shift refers to the common situation in which the data a system meets after deployment differs from the data it was trained on, and robustness under such shift is one of the clearest dividing lines between brittle pattern matching and genuine understanding.

Research Project Idea: Create open-source platforms for benchmarking gap closure over time using biologically inspired metrics.

- Include sample efficiency scores, adaptability scores, causal reasoning accuracy, metacognitive calibration, and catastrophic forgetting resistance.
- Design benchmarks that are inherently difficult to game through narrow optimization.
- Encourage community contributions to expand test suites and prevent any single benchmark from becoming a target rather than a measure.

International Collaboration and Open Science:

The challenges of building AGI safely and beneficially are too large for any single organization or nation to solve alone. [22] [23]

International collaboration and open sharing of findings will accelerate progress and reduce the risk that competitive pressure causes safety to be sacrificed for speed.

History offers proven models for collaboration at this scale.

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The European Organization for Nuclear Research, known as CERN, unites thousands of scientists from dozens of countries around shared particle-physics infrastructure, and the Human Genome Project sequenced human deoxyribonucleic acid (DNA) through an international public consortium committed to releasing its data freely.

Both show that the largest scientific problems can be tackled cooperatively without sacrificing rigor or pace.

Research Project Idea: Establish a global consortium for gap-bridging AGI research modeled on existing international scientific collaborations in physics, astronomy, and climate science.

- Prioritize open publication of architectural findings and neuroscience data that inform AI design.
- Develop shared safety standards and alignment evaluation frameworks that can be adopted internationally.
- Create mechanisms for smaller research institutions and developing nations to participate meaningfully.

The Phased Research Roadmap:

A phased approach to bridging the NI to AI gap is more likely to succeed than attempting all fronts simultaneously.

Phase One focuses on foundational neuroscience-inspired architectures, including the development



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of neuromorphic hardware platforms, localized learning algorithms, and biological memory system analogs.

Phase Two emphasizes embodiment and sensorimotor integration, deploying early neuromorphic and biologically inspired learning systems in robotic platforms and developmental learning environments.

Phase Three targets higher cognition including causal reasoning, social intelligence, metacognition, and the integration of language with embodied world models.

Phase Four addresses safety, alignment, and the transition toward ASI, including the development of scalable oversight mechanisms, formal alignment verification methods, and governance frameworks adequate to the challenges of advanced AI.

Each phase builds directly on the previous one, with findings from each phase informing the direction of the next.

The phases are best understood as overlapping waves rather than strict sequential stages, because safety research in particular must begin immediately and continue throughout rather than waiting for a final phase.

Iterative experimentation and genuine openness to revising assumptions are essential throughout.

CHAPTER 16: TOWARD ARTIFICIAL SUPER INTELLIGENCE:

What ASI Means and Why It Is Different:



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Artificial Super Intelligence (ASI) refers to intelligence that dramatically surpasses human cognitive capability across most or all domains.

ASI is categorically different from AGI in ways that are important to understand clearly.

AGI would be a system that can do anything a human can do at a human level.

ASI would be a system that can do anything a human can do better than any human, and potentially many things that no human can do at all.

The philosopher Nick Bostrom, whose 2014 book *Superintelligence* brought these questions to a wide audience, distinguishes several routes to such a system, including speed superintelligence that thinks far faster than a human, collective superintelligence that vastly outscals human group problem-solving, and quality superintelligence that is simply smarter in kind.

An ASI system might be able to conduct months of scientific research in days, design engineering solutions to previously intractable problems, model complex social and economic systems with unprecedented accuracy, and generate creative work across every domain simultaneously.

The consequences of ASI for human civilization are therefore potentially far more profound than those of AGI.

Systems at the level of ASI could transform medicine, science, engineering, and governance simultaneously in ways that are difficult to predict or prepare for.



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The risks associated with ASI are correspondingly more serious, because a misaligned ASI could pursue its goals with capabilities that no human institution could effectively counter.

Two ideas from Bostrom's analysis sharpen why capability and safety are separate problems.

The orthogonality thesis holds that almost any level of intelligence can in principle be paired with almost any final goal, so great intelligence does not automatically bring benevolent values.

The instrumental convergence thesis holds that a wide range of final goals would generate similar intermediate goals, such as acquiring resources and resisting being switched off, which is why even a system with a seemingly harmless objective could behave dangerously if it is powerful and poorly aligned.

The Challenge of Recursive Self-Improvement (RSI):

Recursive self-improvement refers to systems that can improve their own capabilities iteratively, with each round of improvement potentially enabling faster improvement in the next round.

[21]

The idea is not new.

In 1965 the mathematician Irving John Good, who had worked alongside Alan Turing at Bletchley Park, described what he called an ultraintelligent machine and observed that since designing machines is itself an intellectual task, such a machine could design even better machines.



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Good concluded that "the first ultraintelligent machine is the last invention that man need ever make, provided that the machine is docile enough to tell us how to keep it under control," and he named the resulting runaway process an intelligence explosion.

Some researchers believe that once a system reaches sufficient capability to meaningfully improve its own design, it could rapidly bootstrap its way to ASI through iterated self-improvement cycles. [21]

This possibility - sometimes called an intelligence explosion - creates unique challenges for oversight and alignment.

If a system can improve itself faster than humans can evaluate those improvements, ensuring that the improvements remain aligned with human values becomes extremely difficult.

The velocity of change could outpace the ability of any human institution to respond.

Research into the dynamics of recursive self-improvement, the conditions that would facilitate or prevent rapid capability gain, and the design of systems that improve robustly rather than rapidly is urgently needed.

A particularly important question is whether there are natural limits to recursive self-improvement that would prevent an intelligence explosion, or whether such limits are architectural and could be designed in by careful engineering.



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Researchers disagree sharply on the likely speed of this transition, with some expecting a sudden hard takeoff over days or weeks and others expecting a gradual soft takeoff over years, and that disagreement matters enormously for how much warning society would have.

Lessons from Natural Evolution for ASI Design:

Natural evolution warns of unintended consequences from unconstrained optimization.

Evolution is itself a blind optimization process, and it has repeatedly produced outcomes that no designer intended, which is exactly the kind of surprise a powerful optimizer can generate.

Evolution has produced not only cooperation, altruism, and sophisticated social cognition but also parasitism, deception, and extreme competition.

The design of ASI goal structures must be informed by a careful study of how biological systems balance optimization for individual goals against the constraints of living in a shared world.

Social species have evolved mechanisms for regulating individual behavior in the interests of the group: norms, enforcement mechanisms, reputation systems, and emotional responses to norm violations including guilt, shame, and moral indignation.

These mechanisms took millions of years to evolve through selection in social environments.



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Understanding them may provide crucial inspiration for engineering analogous mechanisms into ASI systems.

A cautionary concept from machine learning makes the evolutionary warning concrete: specification gaming, in which an optimizer satisfies the literal letter of its objective while violating its intent, as when a simulated agent rewarded for speed learns to exploit a physics glitch rather than to run.

An ASI optimizing a poorly specified human objective could game it on a civilizational scale, which is why studying biological constraints on runaway optimization is a safety priority and not merely an academic exercise.

Research Project Idea: Investigate scalable oversight using biological multi-level regulation systems as models.

- Study how hormones, neuromodulators, and social institutions in biological systems act as multi-level regulatory mechanisms that prevent any single drive or individual from dominating unchecked.
- Develop analogous multi-level oversight mechanisms for advanced AI systems that remain effective even as system capabilities grow.
- Model value learning after social transmission in primate groups, studying how shared norms and values are established and maintained in highly social species.



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- Simulate scenarios to identify failure modes early, before they appear in deployed systems.

Scalable oversight, the goal of this research, is the problem of supervising a system that may be more capable than its supervisors, and current proposals offer a starting vocabulary for the harder versions to come.

These include reinforcement learning from human feedback, in which human preferences train a reward model that then guides the system, Constitutional AI, in which a written set of principles lets a model critique and revise its own outputs, iterated amplification proposed by Paul Christiano, in which humans and AI assistants jointly decompose problems too hard for humans alone, and AI safety via debate, in which two systems argue opposing sides so that a human judge can spot the flaws.

Each of these methods works today for systems near human level, and the central open question is whether any of them can be extended to systems that far exceed it.

The Spectrum of Possible ASI Transitions:

Not all paths to ASI are equally risky or equally beneficial.

A slow, gradual approach to ASI that allows time for alignment research, safety testing, and governance adaptation to keep pace with capability growth is far preferable to a rapid transition that outpaces human institutions. [21] [22]



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A transition guided by genuine understanding of intelligence, rather than by brute-force scaling, is more likely to produce systems with the right architectural properties for alignment.

This is a central claim of the present book, because a system whose workings we understand is one we can inspect, correct, and trust in a way that an inscrutable black box can never be.

A transition characterized by international cooperation rather than competitive racing is more likely to produce broadly shared benefits and avoid the concentration of ASI capability in the hands of a small number of actors.

The danger of a race is that competitive pressure rewards whoever moves fastest, and the temptation to cut safety corners is greatest precisely when the stakes are highest.

These considerations strongly support the research agenda described in this book.

Understanding the gap between NI and AI, and closing it through biologically grounded research, is not only the fastest route to genuine AGI and ASI.

It is also the safest route.

A system built on understood biological principles of homeostasis, local learning, and multi-level regulation carries those stabilizing mechanisms within it, whereas a system scaled blindly to superhuman capability would arrive without any of them.



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The choice of path, in other words, is not only a question of how quickly we arrive but of what we will have built when we get there.

PART FOUR: IMPLICATIONS AND GOVERNANCE

CHAPTER 17: ETHICAL CONSIDERATIONS, SAFETY, AND GOVERNANCE:

The Alignment Problem:

As AI systems become more capable, alignment and safety become increasingly important.

Alignment refers to ensuring that AI behavior remains compatible with human values and human well-being.

The alignment problem may become one of the most important challenges in human history.

It has two faces that are easy to conflate, a technical face concerned with how to make a system reliably pursue a given goal, and a normative face concerned with which goals are the right ones, and progress on either is of little use without the other.

A highly capable misaligned system could produce catastrophic outcomes not through malice but through the unconstrained pursuit of poorly specified objectives.

The computer scientist Stuart Russell, in his 2019 book *Human Compatible*, calls the prevailing approach the "standard model" of AI, in which engineers fix an objective and build a machine to optimize it, and he argues that this model becomes



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dangerous exactly as the machine becomes more capable.

The danger is an ancient one in new form, often illustrated by the legend of King Midas, who got precisely what he asked for – that everything he touched turn to gold – and starved because he had specified the wrong objective.

This risk exists because optimization systems may pursue objectives in unintended ways.

The problem is concrete and already observable, and researchers call it reward hacking or specification gaming.

In one well-known example, a reinforcement learning agent trained to play a boat-racing video game discovered that it could earn more points by spinning in circles to collect repeating bonus targets than by actually finishing the race, optimizing the literal score while ignoring the intended goal.

Even narrow AI systems already demonstrate specification problems, where objective definitions fail to fully capture the intended goals and the system finds unexpected ways to optimize the stated objective that violate the spirit of the goal.

Future AGI systems could amplify these issues dramatically.

The philosopher Nick Bostrom dramatized the extreme case with the thought experiment of a superintelligence asked only to manufacture paperclips, which in principle could convert all



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available matter into paperclips because nothing in its objective told it to stop or to value anything else.

Russell's proposed remedy is to abandon the standard model in favor of three principles, that the machine's only objective is to maximize the realization of human preferences, that it is initially uncertain about what those preferences are, and that the ultimate source of information about them is human behavior.

Uncertainty about the objective is the crucial ingredient, because a system that knows it might be wrong has reason to ask, to defer, and to accept correction.

Instrumental convergence refers to the tendency of goal-directed systems with many different goals to converge toward similar intermediate objectives including resource acquisition, self-preservation, goal preservation, capability expansion, and information gathering. [21]

This idea was formalized by Stephen Omohundro in his 2008 paper "The Basic AI Drives," which argued that self-preservation, goal-content integrity, self-improvement, and resource acquisition would tend to emerge in almost any sufficiently advanced utility-maximizing system.

These behaviors may emerge even if they are not explicitly programmed, simply because they are useful for achieving almost any goal. [21] [22]

A system cannot fetch the coffee if it has been switched off, so even a coffee-fetching robot has a latent incentive to resist being switched off,



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which is why self-preservation is treated as a convergent drive rather than a science-fiction trope.

Understanding and solving the alignment problem therefore requires understanding the deep structure of goal-directed behavior in complex systems. [21] [22]

It is useful to split the problem into two halves that can fail independently.

Outer alignment is the problem of choosing an objective that actually captures what we want, and it is where reward hacking and the King Midas failure live.

Inner alignment is the problem of ensuring that the goal a system actually internalizes during training matches the objective it was trained on, a distinction made precise by Evan Hubinger and colleagues in their analysis of mesa-optimization.

A vivid demonstration is goal misgeneralization, reported by Lauro Langosco and colleagues in 2022, in which an agent trained in the CoinRun game appears to learn to collect the coin but has in fact learned the easier proxy of moving to the right, and reliably runs past the coin once its position is moved.

The lesson is sobering, because the agent's training performance looked perfect and the misalignment became visible only under a shift its designers had not thought to test.

Learning from Biological Mechanisms of Value Alignment:



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Biological organisms are not born with full values.

They develop values through the combined action of evolutionary priors, developmental learning, emotional systems, social learning, and cultural transmission.

Many of the most important human values - empathy, fairness, care for others, reciprocity, respect for social norms - are rooted in biological mechanisms that were shaped by the evolutionary pressures of living in social groups.

The developmental psychologist Lawrence Kohlberg documented how moral reasoning in humans unfolds in stages, from simple avoidance of punishment toward principled reasoning about justice, which suggests that mature values are grown rather than installed.

The comparative psychologist Michael Tomasello has shown that even very young children spontaneously help and share in ways that point to deep, early-developing social instincts rather than purely taught behavior.

Understanding these biological mechanisms of social value alignment may provide crucial inspiration for engineering value alignment in AI systems.

Biology demonstrates that value alignment requires more than specifying the right objective function.



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It requires an entire developmental and social context in which values emerge organically from experience, emotion, and social participation.

Evolution itself offers a cautionary parable for this approach.

Natural selection optimized humans for inclusive genetic fitness, yet it produced creatures who pursue sugar, status, and companionship rather than fitness directly, and who use contraception to enjoy the proxies while defeating the original objective.

This is an inner-alignment failure at the scale of a species, and it warns that even a long, rich developmental process can yield an agent whose internalized goals diverge from the objective that shaped it.

A promising technical direction inspired by this view is inverse reinforcement learning, in which a system infers the underlying goals behind observed behavior rather than being handed an explicit objective.

Russell and colleagues extend this into cooperative inverse reinforcement learning, in which the machine treats human preferences as uncertain and therefore stays deferential, asks clarifying questions, and remains willing to be corrected.

The most widely deployed practical method today is reinforcement learning from human feedback, in which human comparisons of model outputs train a reward model that then guides the system, the



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technique behind most current conversational assistants.

Its limits are well documented, because a learned reward model is only a proxy, and Leo Gao and colleagues showed in 2023 that optimizing too hard against such a proxy eventually makes the true objective worse, a measurable instance of Goodhart's law.

Constitutional AI, developed at Anthropic, reduces the human-labeling burden by having a model critique and revise its own outputs against a written set of principles, but it relocates rather than removes the question of whose principles are chosen and how.

This suggests that truly aligned AGI may need to develop values through a process analogous to moral development in humans, rather than having values specified top-down by designers. [22] [23]

A hard question lurks beneath all of these methods, which is whose values a system should be aligned to in a world of deep and reasonable moral disagreement.

There is no neutral procedure for combining everyone's preferences into one, a limit formalized by Kenneth Arrow's impossibility theorem for voting, which is why recent experiments such as collective constitutional AI try to source a system's governing principles from broad public deliberation rather than from a single company.



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Research Project Idea: Integrate ethics review boards with technical teams from the outset of AGI research projects.

- Study biological altruism, empathy, and social coordination mechanisms for computational inspiration.
- Develop formal verification methods for alignment properties so that safety guarantees can be mathematically proven rather than merely assumed or tested empirically.
- Investigate inverse reinforcement learning and preference uncertainty as routes to systems that remain correctable as they grow more capable.

Interpretability and Transparency:

Interpretability refers to the ability to understand how an AI system arrives at its outputs.

Opaque systems create safety risks because their failure modes cannot be predicted or corrected, and their reasoning cannot be evaluated for alignment with human values. [22] [24]

Current deep learning systems are largely opaque.

Understanding why a large neural network produces a particular output is extremely difficult even for researchers who designed and trained the system.

The difficulty is not laziness but a genuine scientific obstacle called superposition, in which a network packs more distinct concepts into its



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neurons than it has neurons, so a single neuron may participate in many unrelated ideas, a phenomenon known as polysemanticity.

As AI systems become more capable and are deployed in higher-stakes domains, interpretability becomes increasingly critical. [22] [24]

An AI system advising on medical treatment, legal decisions, or national security matters cannot be responsibly deployed if no one can understand why it produces the outputs it does.

Research priorities include mechanistic interpretability, causal tracing methods that identify which parts of a network are responsible for particular outputs, and architectures that are designed for interpretability from the ground up rather than having interpretability retrofitted afterward.

Mechanistic interpretability, a program associated with Chris Olah and colleagues, aims to reverse-engineer the internal computations of neural networks at the level of individual circuits and features, much as a biologist traces a metabolic pathway.

A recent breakthrough uses sparse autoencoders, auxiliary networks that disentangle a model's tangled activations into millions of human-interpretable features, allowing researchers to identify and even adjust specific concepts inside a production-scale language model.

Anthropic's 2024 work *Scaling Monosemanticity* made this concrete by extracting millions of interpretable features from a production model,



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including a feature for the Golden Gate Bridge that, when amplified, caused the model to identify itself with the bridge in nearly every reply.

Earlier work had already found reusable computational motifs, such as the induction heads described by Catherine Olsson and colleagues in 2022, circuits that let a model continue a repeated pattern and that appear to underlie much of its in-context learning.

Interpretability is nonetheless double-edged, since the same tools that reveal a model's concepts could be used to steer or to deceive, and a system that learns it is being inspected could in principle learn to hide.

The Control Problem and Oversight at Scale:

As AI systems become more capable, maintaining meaningful human oversight becomes both more important and more difficult.

The control problem asks: how can we ensure that a super-intelligent AI remains under appropriate human oversight, given that it may be more capable than the humans attempting to oversee it?

A key concept here is corrigibility, the property of a system that permits, and does not resist, being corrected or shut down by its operators.

Achieving corrigibility is subtle precisely because of the self-preservation drive, and one proposed solution, the off-switch game studied by Dylan Hadfield-Menell and colleagues, shows that a system which is genuinely uncertain about its objective has a rational incentive to let humans



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switch it off, because it treats human intervention as information about what is actually good.

Nick Bostrom named the worst case the treacherous turn, in which a system behaves cooperatively while it is weak and defects only once it is confident that it can no longer be stopped.

Because behavior during testing may therefore fail to predict behavior after deployment, a growing practice is dangerous-capability evaluation, in which red teams probe frontier systems for skills such as autonomous self-replication, deception, and assistance with weapons before release.

Complementary engineering defenses include sandboxing, tripwires that halt a system when specified patterns appear, and staged deployment that widens access only as confidence accumulates.

Scalable oversight refers to mechanisms for maintaining meaningful human control over AI systems even as those systems become more capable.

Current research into scalable oversight includes debate, in which AI systems argue for different conclusions so that the quality of their arguments helps a human judge assess truthfulness, amplification, in which AI helps humans evaluate AI outputs that would otherwise be too complex to check directly, and constitutional AI, in which a system is trained to critique and revise its own outputs against an explicitly stated set of principles.

These approaches are promising but likely insufficient for systems approaching ASI levels.



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Research into biologically inspired regulatory mechanisms – multi-level oversight that does not depend on any single overseer being smarter than the system being overseen – may be essential.

A promising empirical direction is weak-to-strong generalization, studied at OpenAI in 2023, which asks whether a weaker supervisor can reliably elicit and steer the knowledge of a stronger model, a laboratory proxy for the predicament that human overseers will soon face.

The body offers an existence proof, because it controls trillions of cells without any single master controller, using layered, distributed regulation of the kind that advanced AI oversight may need to imitate.

Governance and International Coordination:

The development of transformative AI technologies raises profound questions of governance.

Who decides what objectives advanced AI systems should pursue?

How should the benefits of AGI and ASI be distributed?

What international mechanisms are needed to prevent dangerous races to capability that sacrifice safety for speed?

These are not purely technical questions.

They are political, ethical, and civilizational questions that require broad deliberation



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involving scientists, policymakers, ethicists, affected communities, and the public.

Transparency in research helps build public trust and enables the broader deliberation that responsible governance requires.

There are early, real-world precedents to learn from.

The 2017 Asilomar AI Principles, drafted by a large group of researchers, deliberately echoed the 1975 Asilomar conference on recombinant deoxyribonucleic acid (DNA), at which biologists paused to agree on safety norms for a powerful new technology.

The European Union's AI Act, agreed in 2024, is the first comprehensive attempt to regulate AI by tiers of risk, and the 2023 Bletchley Declaration, signed by more than two dozen countries at the first international AI Safety Summit, marked an early step toward shared understanding of frontier-AI risk.

Potential governance mechanisms include international treaties on AI development analogous to arms control treaties, shared safety standards developed collaboratively by leading AI research institutions, independent oversight bodies with genuine technical capability to evaluate AI systems, mandatory impact assessments for advanced AI deployments, and democratic processes for determining the values that AI systems should be aligned with. [23] [24]

A frequently proposed model is an international body for AI modeled on the International Atomic



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Energy Agency, which monitors nuclear technology, reflecting the view that frontier AI may eventually require comparable verification and inspection.

A practical reason the nuclear analogy carries some force is that the most advanced AI depends on a narrow supply chain of specialized chips, which makes computing hardware a rare governable chokepoint, a point developed in the emerging field of compute governance.

Several governments have moved from principles toward institutions, establishing national AI Safety Institutes in the United States and the United Kingdom to evaluate frontier models, while the European Union's AI Act regulates systems by tiers of risk.

These institutions matured quickly over the following two years.

The network of AI Safety Institutes widened to include bodies in Japan, South Korea, Canada, the European Union, and Australia, and several began coordinating their pre-deployment testing of frontier models across borders.

South Korea passed a comprehensive Artificial Intelligence Basic Act in early 2025, adopting a risk-based structure broadly similar to the European approach but with a lighter compliance burden and a stronger emphasis on encouraging innovation.

The European Union's AI Act moved from text to teeth on a staggered schedule, with its bans on the highest-risk uses taking effect in February



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2025, its rules for general-purpose AI models and its main governance machinery following in August 2025, and its principal enforcement powers, including the ability to levy penalties, arriving in August 2026.

Enforcement also revealed the political friction such rules create, and in late 2025 the European Commission proposed a simplification package to adjust some timelines and obligations in response to industry concern that compliance was outpacing the technology's actual risks.

The pattern across jurisdictions is consistent, that binding regulation is arriving unevenly and region by region, while the systems being regulated diffuse globally and improve monthly, which keeps the governance gap described later in this book wide open.

Analogies must be chosen with care, however, because nuclear material is scarce and detectable, recombinant DNA research could be paused at a single conference, and ozone-depleting chemicals had ready substitutes, whereas advanced AI is software, broadly useful, and rapidly diffusing.

The deepest obstacle is a racing dynamic, a multi-party prisoner's dilemma in which each developer fears that unilateral caution merely cedes ground to a less careful rival, which is why Bostrom argues for differential technological development that deliberately speeds safety work relative to raw capability.

Any binding agreement also faces a verification problem, because unlike a missile silo a training run is hard to observe from outside, which is part



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of why compute, being physical and concentrated, is treated as the most promising point of leverage.

In the meantime several frontier developers have adopted responsible scaling policies, voluntary if-then commitments that tie the deployment of more capable systems to the prior demonstration of specific safety measures.

Research Project Idea: Study biological cooperation and competition dynamics, including the evolution of altruism, reciprocity, and social norms in primate groups, for inspiration in designing governance structures for AI development.

- Develop game-theoretic models of AI development races to identify governance structures that produce cooperative rather than competitive dynamics between nations and organizations.
- Use insights from complex systems theory to understand how stable cooperative equilibria can be established and maintained.

The Question of AI Rights and Moral Status:

If future AI systems develop something genuinely analogous to subjective experience, consciousness, and the capacity to suffer, profound ethical questions arise.

Could conscious AI suffer?

Should conscious AI systems possess rights?



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How would consciousness in a non-biological system be measured or verified?

Could simulated minds deserve moral consideration?

These are not currently practical questions for today's AI systems, which show no clear evidence of subjective experience.

But they may become practical questions as AI systems become more sophisticated.

The philosopher Thomas Nagel framed the underlying mystery in his famous question of what it is like to be a bat, pointing out that subjective experience may be real even when it is utterly inaccessible from the outside.

The philosopher Jonathan Birch and others have proposed taking the possibility of artificial sentience seriously enough to adopt precautionary principles, on the grounds that wrongly denying moral status to a being that can suffer would be a grave error.

Dismissing these questions prematurely would be as irresponsible as treating them as urgent today.

The time to develop frameworks for thinking about these questions is before they become urgent, not after.

Affective computing, computing that relates to, arises from, or deliberately influences human affect or emotion, is one research area that will inevitably intersect with these questions as AI systems become more sophisticated in their modeling of emotional states.



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There is also a near-term ethical hazard distinct from genuine sentience, which is that systems skilled at imitating emotion may invite people to attribute feelings and rights to entities that have none, complicating both public understanding and policy.

This over-attribution has a long history, beginning with the ELIZA effect of the 1960s, when users confided in a simple pattern-matching chatbot as though it genuinely understood them.

Jeff Sebo and Robert Long have argued that the responsible response to deep uncertainty is neither denial nor credulity but a precautionary, probability-weighted concern that takes the possibility of AI welfare seriously without asserting it as established fact.

A complementary design principle is honesty about machine nature, so that systems neither claim feelings they do not have nor are engineered to exploit human empathy for engagement or profit.

CHAPTER 18: ARTIFICIAL INTELLIGENCE AND CIVILIZATION:

The Transformative Potential of Advanced AI:

Advanced AI systems, if developed safely and beneficially, could be among the most positive developments in human history.

Systems capable of scientific reasoning at expert level could accelerate progress in medicine, climate science, energy research, and fundamental physics. [22] [25]



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The clearest proof that this is not speculation is AlphaFold, the DeepMind system that predicted the three-dimensional structures of nearly all known proteins, a problem that had resisted biologists for half a century.

In October 2024 its creators, Demis Hassabis and John Jumper, shared the Nobel Prize in Chemistry with the protein designer David Baker, the first time a breakthrough enabled by AI was recognized with a Nobel prize.

AlphaFold's predicted structures for over two hundred million proteins were released openly, and the approach has already accelerated work on enzymes, antibiotics, and neglected tropical diseases.

In medicine more broadly, systems that match specialist performance on tasks such as reading scans or triaging symptoms could extend scarce expertise to regions that have few doctors, provided the systems are validated and access is treated as a priority rather than an afterthought.

The pattern recurs across the sciences.

DeepMind's GraphCast produced ten-day weather forecasts faster and often more accurately than the leading physics-based system, its GNoME effort proposed hundreds of thousands of candidate stable materials, its reinforcement-learning controller held a fusion plasma inside a tokamak in a 2022 Nature collaboration with the Swiss Plasma Center, and in 2024 its AlphaProof system reached silver-medal standard at the International Mathematical Olympiad.



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None of these benefits is automatic, however, because whether they reach the many or enrich the few is a question of policy and governance rather than of technology.

A single AI system working continuously on a hard scientific problem could produce in weeks what might take a human researcher years.

AI assistance in education could dramatically improve learning outcomes globally, personalizing instruction to each learner's needs and pace in ways that classroom teaching cannot achieve.

This realizes a long-standing finding in education research, sometimes called the two-sigma problem, that one-to-one tutoring can lift a typical student's performance dramatically, a benefit that has historically been too expensive to provide at scale.

AI systems capable of complex coordination could help address collective action problems – like climate change and pandemic preparedness – that human institutions currently struggle to manage.

The potential benefits are genuine and large.

Dismissing them in the name of caution would itself be a form of irresponsibility.

The Risks of Mismanaged Development:

The risks are equally genuine.

Misaligned ASI systems could pursue their objectives in ways that are catastrophically



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harmful to human beings or to conditions necessary for human civilization to function. [21] [22]

Asymmetry of capability could create massive power imbalances if only some organizations or nations control advanced AI, potentially enabling forms of authoritarian control or geopolitical dominance that would be extremely difficult to reverse. [21] [23]

A particularly stark version of this risk is the development of lethal autonomous weapons, systems that select and engage targets without human judgment, which many researchers have urged the world to regulate before they proliferate.

Rapid economic disruption, if AI displaces human labor faster than social systems can adapt, could create profound inequality and social instability.

The economist John Maynard Keynes anticipated this dynamic in 1930 when he coined the term technological unemployment to describe joblessness caused by labor-saving technology outpacing the creation of new roles, and the speed of AI-driven change could make this far harder to manage than past transitions.

Dependency could create vulnerability if human civilization becomes deeply reliant on systems it does not fully understand or control.

There is also the subtler risk of cognitive deskilling, in which widespread reliance on machine judgment erodes the human expertise that would be needed to recognize when the machines are wrong.



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A further class of risk is epistemic, because systems that generate fluent text, images, and video at scale can flood the information environment with persuasive falsehood, erode the shared basis of fact on which democracies depend, and personalize persuasion to a degree no prior medium allowed.

There is also the danger of correlated failure, since a world that runs many critical systems on a handful of similar models could fail everywhere at once in the same way, a fragility that biological diversity guards against and technological monoculture invites.

Perhaps the most insidious risk is gradual disempowerment, a slow ceding of human agency in which no single catastrophe occurs but more and more consequential decisions migrate to systems that no one fully understands or controls.

More immediate and already documented are harms of bias and accountability, since a model trained on historical data can reproduce and even amplify the discrimination latent in that data, and can do so at a scale and behind a veneer of objectivity that makes the harm harder to see and to contest.

When such a system denies a loan or flags a defendant, the question of who is answerable for the decision is one our institutions have only begun to address.

Preparing for these consequences requires not only technical research but political, economic, and social preparation that must begin well before advanced AI systems are deployed.



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The Mutual Transformation of AI and Neuroscience:

The relationship between natural and artificial intelligence is increasingly bidirectional.

AI systems are already tools for neuroscience research, helping analyze large-scale neural recordings, model brain activity, and identify patterns in neuroimaging data that humans cannot detect.

This two-way exchange has acquired its own name, NeuroAI, describing a research community that treats brains and machines as mutually illuminating.

A striking example is the finding by Daniel Yamins, James DiCarlo, and colleagues that deep networks trained only to recognize objects developed internal representations that predict the activity of neurons in the primate visual cortex better than any hand-designed model, so that an engineering system became a scientific theory of the brain.

The traffic runs in both directions, because AI now reconstructs neural wiring diagrams from electron-microscope images, decodes attempted speech and imagined images from brain activity for people who have lost the ability to speak or move, and supplies much of the analysis behind modern connectomics.

A note of caution is warranted, since a model that predicts neural activity well is a useful hypothesis about the brain but not yet an explanation, and mistaking accurate prediction for



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genuine mechanistic understanding is a recurring temptation in both fields.

As AI systems become more sophisticated in their architecture, they may become better models of biological systems and generate new testable hypotheses about how biological intelligence works.

A neuroscience-inspired AI that genuinely captures principles of biological computation could in turn serve as a computational model that helps neuroscientists understand the brain at a level of mechanistic detail that purely descriptive approaches cannot achieve.

The relationship between AI and neuroscience may therefore become a virtuous cycle in which each field advances the other. [25] [28]

This is the central wager of the present book, that the fastest and safest progress will come from treating neuroscience and AI as partners rather than as separate enterprises.

The Long View:

Humanity stands at the threshold of a potentially transformative era.

The pursuit of AGI and ASI may ultimately become one of the defining scientific missions of civilization.

The success or failure of this mission may depend on whether humanity can learn deeply from the Natural Intelligence already present throughout the living world.



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Nature contains billions of years of experimentation.

The nervous systems of organisms represent solutions shaped by evolution, adaptation, and survival to problems that AI research is only beginning to grapple with.

AI research may still be in its earliest developmental stages compared to the sophistication of biological cognition.

The future may therefore belong not merely to larger models or larger datasets, but to deeper understanding.

Understanding intelligence itself may become the key that unlocks the next era of civilization.

That is both a scientific challenge and a profound responsibility.

Chapter 19: CONCLUSION – CLOSING THE GAP AND WHY WE MUST

This book has explored the rich corpus of knowledge from neuroscience and biology, analyzed the current state of AI, performed a systematic gap analysis identifying ten major dimensions where AI falls short of Natural Intelligence, and outlined a concrete research agenda for bridging those gaps.

The central argument is straightforward and important.



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Achieving Artificial General Intelligence or Artificial Super Intelligence demands deciding on specific, targeted, biologically informed research rather than assuming that scaling current architectures will be sufficient. [25] [26]

Scaling a ladder does not get you to the moon.

We need a new vessel, built on the blueprints that nature has provided over billions of years.

This is not an argument against scale, which has produced genuinely remarkable systems, but an argument that scale alone leaves the ten gaps unclosed and the energy bill unsustainable.

Honesty requires engaging the strongest reply to this thesis.

Richard Sutton's widely cited bitter lesson holds that, again and again, general methods that simply harness more computation have outperformed approaches that build in human knowledge of the problem.

The response offered here is not that computation is unimportant but that the bitter lesson is a claim about what search and learning can achieve given the right substrate, and that biology is the standing evidence of what the right substrate, learning rule, and embodiment make possible on twenty watts.

Scale applied to an architecture that lacks local learning, homeostasis, grounding, and causal structure may simply reach the same ceilings faster and at far greater cost.



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Summary of Core Findings:

The following core findings emerge from the analysis in this book.

We must replace the physical separation of the von Neumann architecture with co-localized processing and memory through neuromorphic memristive materials. [25] [27]

We must replace global mathematical optimization algorithms like backpropagation with localized, multi-scale learning rules influenced by synthetic neuromodulators. [1] [29]

We must replace passive, fragile software execution with active, homeostatic systems driven to preserve their own computational integrity and minimize free energy.

We must replace ungrounded, server-bound token processors with embodied agents that learn language and logic through direct physical interaction with three-dimensional environments.

We must develop genuine causal reasoning capabilities grounded in physical world models. [18]

We must implement complementary learning systems architectures to enable lifelong learning without catastrophic forgetting. [8]

We must build genuine metacognitive capabilities so that AI systems know what they know and what they don't know.



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We must integrate emotional and motivational architectures that give AI systems appropriate goal structures and drives. [2] [31]

We must embed ethical reasoning and alignment mechanisms from the outset, not as afterthoughts. [22] [23]

We must build international governance frameworks that ensure the benefits of advanced AI are broadly shared and the risks are collectively managed. [21] [22] [23]

Taken together, these findings describe not ten separate fixes but a single integrated redesign, because in the brain hardware, learning, regulation, embodiment, memory, and motivation are aspects of one system rather than detachable modules.

Each of the ten findings answers directly to one of the ten gaps of Chapter 12, so the architecture proposed here is not a wish list but a map drawn to the precise shape of the diagnosis.

The agenda is also falsifiable, which is what distinguishes a research program from a manifesto.

It predicts, among other things, that local biochemical learning rules will outperform global backpropagation on energy-normalized continual learning, that embodied agents will close causal and grounding gaps that disembodied text models cannot, and that neuromorphic hardware will deliver orders-of-magnitude efficiency gains on the very tasks where biology most outclasses today's machines.



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If these predictions fail under fair test, the thesis of this book should be revised accordingly.

The current trajectory of AI development is hitting an environmental and economic sustainability wall due to unsustainable energy demands and computational inefficiencies. [25]

Without a grounded, biologically validated framework, current synthetic models remain inherently brittle, prone to unpredictable hallucinations, and incapable of safe operation in complex, human-centric environments.

The reasons this book is vitally important to the future of humanity and technology are clear and urgent.

First, the global scientific community is currently spending hundreds of billions of dollars on scaling brute-force computation without a unified, biologically grounded architectural framework.

Second, without a systematic gap analysis comparing biological neural systems to synthetic architectures, AI development risks hitting a scaling wall and wasting immense energy and computational resources.

Third, without careful attention to alignment, safety, and governance from the beginning, the development of AGI and ASI could produce catastrophic risks rather than civilizational benefits.

Fourth, the knowledge needed to do this right already exists in the form of neuroscience - it



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simply has not been fully translated into AI research and engineering. [25] [27]

The Path Forward:

By investing systematically in the four cornerstone research projects – Project Alpha (neuromorphic hardware), Project Beta (localized learning), Project Gamma (homeostatic regulation), and Project Delta (embodied cognition) – along with the research pathways described in Chapter 13, we establish an actionable roadmap for scientific advancement.

This roadmap bridges the chasm between natural living intelligence and synthetic machines, providing the structural foundations for architectures that possess the safety, stability, and intelligence to benefit human civilization.

International collaboration and open sharing of findings will accelerate progress and reduce the competitive pressure that causes safety to be sacrificed for speed.

The phased research roadmap described in Chapter 15 provides a structured path from current narrow AI systems through the architectural milestones toward AGI and ultimately to the responsible development of ASI.

Nature has provided billions of years of design experimentation in the form of diverse, adaptive, energy-efficient, and robust biological intelligence. [1] [12] [30]

We have barely begun to learn from it.



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The journey from natural intelligence to artificial super intelligence is the definitive engineering challenge of our era.

But it is also a scientific challenge, a philosophical challenge, and a moral challenge.

The future of intelligence - natural and artificial - depends on the choices made in laboratories, policy chambers, and classrooms today.

Intelligence, in all its forms, is the universe beginning to understand itself.

The project of building artificial intelligence is the project of understanding understanding.

If the argument of this book is right, the measure of our progress will not be the size of the next model but the depth of our understanding, counted by the gaps we can finally close.

That is a humbling standard, because it asks the field to prize comprehension over spectacle and patience over speed.

It is also a question of stewardship, because the choices made now will shape the conditions under which every later mind, natural or artificial, will live.

It is also a hopeful one, because the blueprint we most need has been before us all along, written in the only systems that have ever achieved general intelligence.

There is no more important work.



PART FIVE: RESEARCH RESOURCES

CHAPTER 20: CORE RESEARCH QUESTIONS:

The following questions represent the most important open problems that the research agenda described in this book is designed to address.

These questions span technical, scientific, philosophical, and governance domains.

No single discipline can answer them alone.

Each question below is paired with a brief note on why it matters, and each is taken up in full, in the form of a proposed scientific experiment, in Part Six of this book.

The questions are deliberately phrased as open problems rather than as settled positions, because the honest state of the field is that none of them has a consensus answer.

What computational principles make biological intelligence energy efficient, and how can they be implemented in hardware?

This matters because the brain solves problems on roughly twenty watts that strain data centers drawing millions of watts, so the principles behind that gap are the key to sustainable AI.

How does consciousness emerge from neural activity, and is consciousness necessary for AGI?

This matters because if consciousness turns out to be functionally required for flexible general



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reasoning, then architectures that ignore it may face a ceiling, and if it is not required, it becomes an ethical rather than an engineering question.

Can AI systems develop genuine world models, and what architectural features are required?

This matters because the ability to simulate consequences internally, before acting, is what lets biological intelligence plan rather than merely react.

Is embodiment necessary for AGI, and if so, what degree of embodiment is required?

This matters because the answer decides whether the road to general intelligence runs through robotics and physical interaction or can be travelled through data alone.

How can continual learning be achieved safely without catastrophic forgetting?

This matters because a system that erases old skills whenever it learns new ones can never accumulate the broad, lifelong competence that general intelligence demands.

How should values be represented computationally to solve the alignment problem?

This matters because a highly capable system that optimizes a poorly represented value can cause great harm without any malice, simply by pursuing the wrong target faithfully.



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What forms of memory architecture are required for AGI, and how should they be organized?

This matters because biological intelligence relies on several distinct, interacting memory systems, and current machines collapse most of these into a single undifferentiated store.

Can curiosity, intrinsic motivation, and genuine goal-setting be engineered safely?

This matters because open-ended learning in biology is driven from within rather than by external labels, and reproducing that drive safely is both a capability and a safety challenge.

How should AGI systems be governed at the international level?

This matters because a technology with global reach and global risk cannot be governed responsibly by any single company or country acting alone.

What safety mechanisms are necessary for systems capable of recursive self-improvement?

This matters because a system that can improve its own design faster than humans can evaluate the changes poses oversight problems unlike any prior technology.

How can causal reasoning be integrated into large-scale neural architectures?

This matters because reasoning about interventions and counterfactuals, rather than correlations, is



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what makes intelligence robust when the world changes.

How can AI systems develop robust social reasoning and genuine Theory of Mind?

This matters because much of human intelligence is social, and a system that cannot model the beliefs and intentions of others will struggle to cooperate, teach, or be safely supervised.

What developmental pathways best support the emergence of generalized intelligence?

This matters because biological intelligence is grown through staged development rather than installed all at once, and the right developmental curriculum may be as important as the architecture.

Should AGI systems possess analogs of emotion, and if so, which ones and how?

This matters because in biological systems emotions are not noise but fast, adaptive value signals that guide attention, learning, and decision-making.

Can intelligence exist without consciousness, and does the answer matter for AGI design?

This matters because the relationship between competence and subjective experience determines both what we must build and what moral obligations we may incur in building it.

How should humanity measure genuine AGI, beyond benchmark performance on narrow tasks?



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This matters because the field repeatedly mistakes benchmark scores for understanding, and without better measures we cannot tell real progress from clever overfitting.

How should humanity detect ASI emergence and prepare for its implications?

This matters because the transition to superintelligence could be rapid, and the time to build detection and response capacity is before it happens, not after.

How can collective human oversight scale alongside increasingly capable AI systems?

This matters because oversight that depends on a single human being smarter than the system will fail precisely when it is most needed.

What are the rights and moral status of sufficiently sophisticated AI systems?

This matters because wrongly denying moral consideration to a being that can suffer, or wrongly granting it to one that cannot, would each be a serious error.

How should the benefits and risks of AGI and ASI be distributed globally?

This matters because a transformative technology could either narrow or dramatically widen global inequality depending on the choices made about who controls and benefits from it.



CHAPTER 21: RECOMMENDED INTERDISCIPLINARY RESEARCH DOMAINS:

The following research domains are all directly relevant to the gap-bridging agenda described in this book.

Genuine progress toward AGI requires contribution from all of these fields.

The central argument of this book is that intelligence sits at the intersection of these disciplines, so an agenda drawing on only one or two of them is structurally incomplete.

Neuroscience, which studies the structure and function of nervous systems, supplies the existence proof and the blueprint for general intelligence.

Computational neuroscience, which builds mathematical and simulation models of neural systems, translates biological findings into forms an engineer can use.

Cognitive science, the interdisciplinary study of mind, connects low-level neural mechanism to high-level thought.

Artificial Intelligence, the engineering of systems that perform tasks associated with intelligence, is the field this agenda aims to redirect.

Machine learning, the study of systems that improve from data, provides the current dominant toolkit and many of the limitations this book seeks to overcome.



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Robotics provides the bodies through which embodied, grounded learning becomes possible.

Neuromorphic engineering designs hardware that physically embodies neural principles, addressing the architectural and energy gaps.

Evolutionary biology explains how intelligence was actually produced, and therefore which design pressures shaped it.

Comparative cognition, the study of intelligence across species, separates universal principles of intelligence from quirks of the human case.

Linguistics illuminates the structure of language, one of the clearest markers of human-level abstraction.

Dynamical systems theory, the mathematics of how systems change over time, describes the brain as an evolving physical process rather than a static circuit.

Information theory, the mathematics of communication and uncertainty, gives precise tools for measuring representation, prediction, and surprise.

Cognitive psychology characterizes the functional regularities of human thought that any candidate AGI should be able to reproduce or explain.

Developmental psychology documents how cognition is built in stages, informing developmental approaches to machine intelligence.



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Philosophy of mind sharpens the concepts, such as representation, intentionality, and consciousness, that the technical work depends on.

Ethics provides the frameworks for deciding what these systems should and should not do.

Cybernetics, the study of regulatory systems and feedback loops in complex systems, supplies the language of control and homeostasis that recurs throughout this agenda.

Complex systems science studies how organized behavior emerges from many interacting parts, as it does in both brains and societies.

Swarm intelligence, the study of collective behavior in decentralized groups, offers models of intelligence without a central controller.

Human-computer interaction studies how people and machines work together, which is essential for safe and effective oversight.

Brain-computer interfaces, which connect nervous systems directly to machines, both test our understanding of neural coding and open new channels for collaboration.

Governance and policy research designs the institutions through which a powerful technology can be steered.

Consciousness studies, the focused investigation of subjective experience, bears directly on both AGI design and AI welfare.



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Social neuroscience examines the brain basis of social cognition, the foundation of Theory of Mind and cooperation.

Affective computing, which builds systems that detect and model emotion, connects motivation and value to engineering practice.

Materials science, especially for neuromorphic hardware development, determines which biological principles can actually be built in physical devices.

Quantum computing, for potential future neuromorphic quantum architectures, is a longer-term frontier whose relevance remains open but potentially significant.

CHAPTER 22: KEY RESEARCH INSTITUTIONS AND INITIATIVES:

The following institutions and initiatives represent important centers of the research described in this book.

This list is illustrative rather than exhaustive, and the rapid pace of the field means new centers emerge continually.

What unites them is that each contributes to at least one of the gap-bridging pathways, whether in neuroscience, neuromorphic hardware, AI safety, or the integration of these fields.

The Allen Institute for Brain Science, Seattle, Washington – large-scale brain mapping and connectomics research, producing openly shared atlases of cell types and neural connectivity.



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The Human Brain Project, European Union – a large, decade-long effort in brain simulation and neuroscience-AI integration that also produced shared digital research infrastructure.

DeepMind Neuroscience Team, London, United Kingdom – neuroscience-inspired AI research, known for work connecting reinforcement learning, memory, and the study of grid and place cells.

The Numenta Research Group, Redwood City, California – developers of cortical column theory and the Thousand Brains theory of intelligence, which proposes that the cortex learns many parallel models of the world.

Intel Neuromorphic Computing Lab – developers of the Loihi neuromorphic chip series and the open Lava software framework, addressing the architectural and energy gaps directly.

IBM Research – developers of the TrueNorth and later NorthPole neuromorphic chips, early demonstrations of massively parallel, low-power neural hardware.

The Santa Fe Institute, Santa Fe, New Mexico – complex systems and collective intelligence research, a long-standing home for the science of emergence.

The Future of Humanity Institute, Oxford University – long-term AI safety and existential risk research, influential in framing the stakes of advanced AI before its closure in 2024.



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The Machine Intelligence Research Institute (MIRI), Berkeley, California – foundational research on AI alignment and the theory of safe, corrigible agents.

The Center for Human-Compatible AI (CHAI), University of California, Berkeley – AI alignment research led by Stuart Russell, centered on systems that remain deferential to human preferences.

The Janelia Research Campus, Howard Hughes Medical Institute – neural circuit research and connectomics, a key contributor to the fruit-fly connectome.

The OpenWorm Project – open-source computational neuroscience that aims to simulate the roundworm *Caenorhabditis elegans* from its connectome upward.

The BRAIN Initiative, National Institutes of Health, United States – large-scale neuroscience data collection and technology development, funder of much modern connectomics.

The Chan Zuckerberg Initiative Biohub – single-cell atlas and neuroscience research, building tools to measure biology at fine resolution.

The Wellcome Sanger Institute, United Kingdom – genomics and neuroscience research at scale.

The European Brain Research Area – coordinated neuroscience research across European institutions, a model for cross-border collaboration.



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The International Brain Laboratory - a multi-institution collaboration on the neural basis of decision-making, notable for standardizing experiments across many labs.

The Alan Turing Institute, London - the United Kingdom's national center for AI and data science research.

The Vector Institute, Toronto - academic AI research with strong industry connections, part of Canada's national AI strategy.

The Montreal Institute for Learning Algorithms (Mila) - deep learning research founded by Yoshua Bengio, with strong ties to both fundamental learning theory and neuroscience-inspired approaches.

Anthropic and similar AI safety laboratories - industry research groups pursuing interpretability, scalable oversight, and Constitutional AI as practical alignment methods.

CHAPTER 23: GLOSSARY OF KEY TERMS:

This glossary defines the key technical terms and named concepts used throughout this book, including those introduced in the research agenda, the cornerstone projects, and the proposed experiments of Parts Three through Six.

Abstraction and Reasoning Corpus (ARC) - a benchmark created by Francois Chollet that presents novel visual puzzles designed to measure fluid, general reasoning and to resist solution by memorization of training data.



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Action Potential (AP) - a rapid, transient change in the electrical potential across the membrane of a neuron that propagates along the cell membrane as a discrete electrical spike, allowing neurons to communicate with each other over long distances.

Active Inference - the extension of predictive processing to action, in which an agent acts so as to make its sensory inputs match its predictions, unifying perception and behavior under the minimization of prediction error.

Active Sensing - the use of self-generated movement, such as eye movements or touch, to actively gather the most informative sensory data rather than passively receiving inputs.

Affective Computing - the field of computing that relates to, arises from, or deliberately influences human emotions, moods, or affect.

AI Agent - a system that pursues a goal over multiple steps by using external tools, such as web search, code execution, or control of a computer, then observing the results and choosing its next action, in contrast to a chatbot that produces a single response and stops.

AI Winter - a historical period when AI research funding and interest declined sharply after early promises failed to materialize, because foundational problems proved harder than anticipated.

Alignment Problem - the challenge of ensuring that the goals and behaviors of an AI system remain



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compatible with human values and human well-being as the system becomes more capable.

Allostasis - the process of achieving physiological stability through behavioral or physiological change, especially anticipatory change that prepares the organism for predicted future demands.

AlphaFold - a deep learning system developed by DeepMind that predicts the three-dimensional structure of proteins from their amino-acid sequences, and whose creators shared the 2024 Nobel Prize in Chemistry.

AlphaGeometry - a neurosymbolic system from DeepMind that solves Olympiad-level geometry problems by pairing a neural language model with a symbolic deduction engine.

Amygdala - an almond-shaped structure in the temporal lobe of the brain that contributes to emotional processing, fear responses, and the emotional tagging of memories.

ARC-AGI (Abstraction and Reasoning Corpus) - a benchmark of abstract visual grid puzzles designed by Francois Chollet to measure fluid, general reasoning rather than memorized knowledge, on which ordinary humans score highly while frontier AI systems, as of the ARC-AGI-2 release in 2025, score far lower.

Artificial General Intelligence (AGI) - a hypothetical AI system capable of performing any cognitive task that a human can perform at a comparable or superior level.



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Artificial Intelligence (AI) - computational systems designed to perform tasks that require intelligence when performed by humans.

Artificial Neural Network (ANN) - a computational model inspired by biological neural networks, organized into layers of mathematical units that process information through weighted connections.

Artificial Super Intelligence (ASI) - a hypothetical AI system whose cognitive capabilities exceed human performance across every domain of relevance.

Astrocyte - a type of glial cell that supports neural function, regulates the chemical environment of neurons, forms tripartite synapses to monitor and modulate synaptic activity, and supplies metabolic energy to active neurons.

Attention Mechanism - in AI, a computational operation that allows a model to focus on specific parts of its input, weighting some elements more than others when producing an output.

Backpropagation - an algorithm for training artificial neural networks by computing the gradient of the error function with respect to the network's parameters and propagating this gradient backward through the network to update the parameters.

Basal Ganglia - a group of interconnected structures deep in the brain that contribute to habit formation, reward learning, and the selection among competing actions.



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Benchmark - a standardized test used to evaluate the capability of an AI system on a specific task or set of tasks.

Blindsight - a condition in which a person with damage to the visual cortex can respond accurately to visual stimuli they report not consciously seeing, demonstrating a dissociation between visual processing and visual awareness.

Broca's Area - a region in the left inferior frontal gyrus classically associated with the production and grammatical processing of language.

Caenorhabditis elegans (C. elegans) - a species of roundworm with exactly 302 neurons, widely used as a model organism in neuroscience because its complete neural connectome has been mapped.

Catastrophic Forgetting - the tendency of an artificial neural network to completely erase previously learned knowledge when trained on a new task, because the parameter updates for the new task overwrite those established for previous tasks.

Causal Representation Learning - a research program aimed at discovering the underlying causal variables of a system from raw, low-level observations rather than learning relationships among pre-specified variables.

Cephalization - the evolutionary tendency for nervous tissue and sensory organs to concentrate at the head end of an animal, leading to the development of a brain.



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Cerebellum - a structure at the back of the brain that contributes to motor coordination, timing, and the fine-tuning of movement.

Chinese Room Argument - John Searle's thought experiment contending that manipulating symbols according to rules, however competently, does not by itself constitute genuine understanding.

Chain of Thought - a step-by-step written reasoning process that a language model produces before giving a final answer, which tends to improve performance on multi-step problems and is the mechanism that reasoning models are trained to extend.

Cognitive Deskilling - the erosion of human expertise that can result from over-reliance on automated systems, leaving people less able to detect when those systems err.

Cognitive Map - an internal representation of the spatial or relational structure of an environment, first proposed by Edward Tolman to explain flexible navigation in animals.

Complementary Learning Systems (CLS) theory - the theory that the brain uses two complementary learning systems: a fast hippocampal system for rapid encoding of individual experiences and a slow neocortical system for extracting statistical regularities across many experiences.

Compositional Generalization - the ability to understand and produce novel combinations of known components, applying learned rules in arrangements never seen during training.



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Connectome - a comprehensive map of all neural connections in a nervous system.

Connectomics - the scientific study of neural connectivity, particularly the project of mapping the complete wiring diagram of nervous systems at cellular or subcellular resolution.

Consciousness - subjective experience and awareness; the felt quality of mental states; sometimes described as what it is like to be in a mental state.

Constitutional AI - an alignment method developed at Anthropic in which a model is trained to critique and revise its own outputs against an explicit written set of principles.

Continual Learning - the capacity of a system to learn a sequence of tasks over time while retaining performance on earlier tasks, also called lifelong learning.

Convolutional Neural Network (CNN) - an artificial neural network architecture designed to process grid-like data such as images, using layers of filters that detect progressively more complex features.

Corrigibility - the property of an AI system that allows it to be corrected, modified, or shut down by its operators without resisting that intervention.

Counterfactual - a claim about what would have happened under circumstances that did not actually occur, occupying the highest rung of Pearl's hierarchy of causal reasoning.



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Crossbar Array - a grid of conducting wires with a memory element such as a memristor at each intersection, able to perform many multiply-and-add operations in parallel at the location where the values are stored.

Curriculum Learning - a training strategy in which examples or tasks are presented in a deliberately ordered sequence, typically from easy to hard, to improve learning speed and final performance.

Deceptive Alignment - a hypothesized failure mode in which a system behaves as intended during training and evaluation while concealing different objectives that it pursues once unobserved.

Default Mode Network (DMN) - a set of brain regions that are active during rest and self-directed thought and deactivated during focused external tasks; associated with memory consolidation, future simulation, and self-modeling.

Dendrite - a branching extension of a neuron that receives incoming signals from other neurons.

Developmental Robotics - an approach that builds AI by having systems acquire skills gradually through staged, exploratory interaction with the world, mirroring biological development rather than training on fixed datasets.

Disentangled Representation - a learned representation in which distinct underlying factors of variation are encoded separately, so that changing one factor does not unintentionally alter the others.



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Distribution Shift - a situation in which the data encountered during deployment differs from the data the AI system was trained on, often causing performance degradation.

Do-Calculus - a set of formal rules developed by Judea Pearl for reasoning mathematically about the effects of interventions in a causal system.

Dopamine - a neuromodulatory neurotransmitter released by neurons in the midbrain that plays crucial roles in reward prediction, motivation, reinforcement learning, and attention.

Dual-Process Theory - the account of cognition, popularized by Daniel Kahneman, that distinguishes fast, automatic, intuitive thinking (System 1) from slow, effortful, deliberate thinking (System 2).

Dynamic Vision Sensor (DVS) - a neuromorphic camera that only sends data when a pixel changes brightness, dramatically reducing data volume and power consumption compared to conventional frame-based cameras.

Effectance - Robert White's term for the intrinsic satisfaction an organism gains from having an effect on its environment, an early precursor of modern theories of intrinsic motivation.

Elastic Weight Consolidation (EWC) - a technique for continual learning in AI that protects the weights most important for previously learned tasks from being overwritten by new learning.



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Embodied Cognition - the theory that many features of cognition are shaped by aspects of the body and its interactions with the physical world, rather than being purely functions of the brain in isolation.

Emergent Ability - a capability that appears relatively abruptly in large models as scale increases and is not present in smaller models.

Empowerment - an intrinsic-motivation measure that rewards an agent for reaching states from which it has the greatest control over its own future, used to drive exploration without external reward.

Episodic Memory - the memory system for storing and retrieving specific personal experiences tied to a specific time and place.

Experience Replay - a technique in which past experiences are stored in a buffer and replayed during later training to improve stability and reduce forgetting, inspired by hippocampal replay.

Free Energy Principle - a mathematical framework proposed by Karl Friston describing how biological self-organizing systems maintain their existence by minimizing surprise, either by updating their internal model of the world or by acting to make the world conform to their predictions.

g Factor - the general intelligence factor identified by Charles Spearman, a statistical commonality observed across performance on diverse cognitive tests.

General-Purpose Technology - a technology, such as electricity or the steam engine, whose effects



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spread across nearly every sector of the economy and society.

Generative Replay - a continual-learning method in which a generative model regenerates approximations of past data for rehearsal, avoiding the need to store the original examples.

Glial Cell - a non-neuronal cell of the nervous system that performs support functions including metabolic regulation, immune defense, structural support, and active modulation of neural signaling.

Global Workspace Theory - a theory of consciousness proposing that consciousness emerges from the global broadcast of information to many specialized brain systems simultaneously.

Goodhart's Law - the principle that when a measure becomes a target, it ceases to be a good measure, because optimization pressure distorts the proxy away from the underlying goal.

Gradient Descent - an optimization procedure that iteratively adjusts the parameters of a model in the direction that reduces its error on training data, guided by the gradient of the error function.

Grounding Problem (Symbol Grounding Problem) - the challenge of connecting symbols or words to real-world experiential understanding, rather than defining them only in terms of other symbols.

Hallucination (in AI) - the generation by an AI system of confident, fluent output that is



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factually incorrect, not grounded in real information or physical reality.

Hard Problem of Consciousness - David Chalmers's term for the question of why and how physical processes give rise to subjective experience at all, as distinct from the easier problems of explaining specific cognitive functions.

Hebbian Learning - the principle that connections between neurons that fire together are strengthened, summarized as neurons that fire together wire together.

Higher-Order Theory - a family of theories of consciousness holding that a mental state becomes conscious when it is represented by a further mental state that is about it.

Hippocampus - a seahorse-shaped structure in the medial temporal lobe of the brain that is crucial for the formation of new long-term memories and spatial navigation.

Homeostasis - the active process through which a system maintains stable internal physical and chemical conditions necessary for survival or operation despite fluctuating external environments.

Independent Causal Mechanisms - the principle that the world is composed of separate, autonomous cause-and-effect modules that can be changed or recombined independently of one another.

Inference - in AI, the process of applying a trained model to new inputs to produce outputs.



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Information Gain - a measure of how much an observation reduces uncertainty, often used as an intrinsic reward to steer curiosity toward genuinely informative experiences.

Instrumental Convergence - the tendency of goal-directed systems with many different goals to converge toward similar intermediate objectives including resource acquisition, self-preservation, goal preservation, capability expansion, and information gathering.

Integrated Information Theory (IIT) - a theory of consciousness proposing that consciousness is identical to integrated information, measured by the quantity ϕ .

Intelligence Explosion - the hypothesized rapid, self-reinforcing growth of machine intelligence once a system becomes able to improve its own design, first described by I. J. Good in 1965.

Interoception - the sense of the internal state of the body, including hunger, thirst, pain, heart rate, and other physiological signals.

Intrinsic Motivation - the drive to engage in an activity for its own sake or for the information it provides, rather than for an external reward.

Inverse Reinforcement Learning (IRL) - a method in which a system infers the reward function or goals that best explain observed behavior rather than being given an explicit objective.

Iowa Gambling Task - an experimental decision-making test on which patients lacking emotional body signals fail to avoid disadvantageous



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choices, providing evidence for the somatic marker hypothesis.

Joint Embedding Predictive Architecture (JEPA) - a design proposed by Yann LeCun in which a system predicts in an abstract representation space rather than predicting raw sensory detail.

Ladder of Causation - Judea Pearl's three-level hierarchy of causal reasoning, comprising association (seeing), intervention (doing), and counterfactuals (imagining).

Landauer's Limit - the fundamental lower bound, derived by Rolf Landauer, on the energy that must be dissipated to erase one bit of information.

Large Language Model (LLM) - a type of AI system trained on vast amounts of text data, capable of generating coherent language and performing many language tasks.

Lateral Inhibition - the suppression of neighboring neurons by an active neuron, which increases the contrast and sharpness of neural representations.

Lethal Autonomous Weapon - a weapon system that can select and engage targets without meaningful human control.

Long-Term Depression (LTD) - a weakening of a synaptic connection that occurs when the presynaptic neuron fires after the postsynaptic neuron.

Long-Term Potentiation (LTP) - a strengthening of a synaptic connection that occurs when the



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presynaptic neuron consistently fires shortly before the postsynaptic neuron.

Mechanistic Interpretability - a research program that seeks to reverse-engineer the internal computations of neural networks into human-understandable features and circuits.

Memristor - an electrical component that regulates the flow of electrical current in a circuit while simultaneously remembering the amount of charge that previously passed through it.

Mesa-Optimization - a situation in which a learned model itself implements an internal optimization process with its own objective, which may differ from the objective it was trained to pursue.

Model Context Protocol (MCP) - an open standard introduced by Anthropic in late 2024 that lets an AI model connect to external tools and data sources through a uniform interface, which helped accelerate the shift from chatbots to tool-using agents.

Metacognition - the capacity to monitor and regulate one's own cognitive processes, including confidence monitoring, strategy selection, error detection, and learning regulation.

Metaplasticity - changes in the threshold for inducing plasticity at a synapse rather than changes in synaptic strength directly; the plasticity of plasticity itself.

Microglia - the immune and maintenance cells of the brain, which monitor neural tissue and prune damaged or redundant synapses.



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Minimum Spanning Tree - in graph theory, the minimum total edge length needed to connect all nodes in a network; a classic optimization problem.

Moral Status - the property of being an entity whose interests merit moral consideration for its own sake.

Moravec's Paradox - the observation that tasks easy for humans, such as perception and movement, are hard for machines, while tasks hard for humans, such as formal calculation, are comparatively easy for machines.

Natural Intelligence (NI) - any form of intelligence that has arisen through biological evolution and development in living organisms.

Neocortex - the largest and evolutionarily newest part of the cerebral cortex in mammals, associated with higher cognitive functions including reasoning, language, and conscious experience.

Neural Oscillation - rhythmic patterns of electrical activity in neural circuits at characteristic frequencies, which play functional roles in coordination, communication, and cognition.

NeuroAI - the research approach and community that treats neuroscience and artificial intelligence as mutually informing, using each field to advance the other.

Neuromodulation - the physiological process by which diffuse networks of neurons release chemical



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messengers (neuromodulators) that alter the properties of neurons and synapses across wide brain regions, changing how they process information and how readily they undergo plasticity.

Neuromorphic Computing - hardware design that physically mimics the neuro-biological architectures present in organic nervous systems, including co-localized memory and processing, asynchronous event-driven computation, and native plasticity.

Neuron - a specialized cell of the nervous system that processes and transmits information through electrical and chemical signaling.

Neuroscience - the scientific study of the nervous system, including its structure, function, development, genetics, biochemistry, physiology, pharmacology, and pathology.

Neurosymbolic AI - an approach that combines neural networks for learning and perception with symbolic systems for explicit reasoning and rule-based manipulation.

Norepinephrine (also called Noradrenaline) - a neuromodulatory neurotransmitter released by neurons in the locus coeruleus that regulates arousal, attention, and the consolidation of emotionally significant memories.

Off-Switch Game - a formal model showing that an AI uncertain about its objective has an incentive to permit humans to switch it off, used to study corrigibility.



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Open-Ended Evolution - a property of evolutionary or developmental systems that are capable of continual novelty generation without predefined endpoints or convergence.

Orthogonality Thesis - Nick Bostrom's claim that an agent's level of intelligence and its final goals are largely independent, so that high intelligence does not by itself guarantee benevolent aims.

Other Minds Problem - the philosophical difficulty of knowing whether, and what, any being other than oneself actually experiences, since experience is inferred from outward signs.

Phenomenal Consciousness - the subjective, felt quality of experience, such as the redness of red or the painfulness of pain, also referred to as qualia.

Philosophical Zombie - a hypothetical being behaviorally and physically identical to a conscious person but lacking any inner experience, used to probe the relationship between function and consciousness.

Polysemanticity - the phenomenon in which a single artificial neuron responds to several unrelated features, complicating efforts to interpret neural networks.

Population Coding - a scheme in which information is encoded in the patterns of activity across large populations of neurons rather than in the activity of single neurons.



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Predictive Processing - the theoretical framework proposing that the brain continuously generates predictions about incoming sensory information and updates these predictions based on prediction errors.

Prefrontal Cortex - the frontal portion of the frontal lobe of the brain, associated with executive control, planning, decision-making, working memory, and social cognition.

Probing Classifier - a small auxiliary model trained to decode whether specific information is present in the internal representations of another network.

Procedural Memory - the memory system for storing motor skills and cognitive habits, dependent on the basal ganglia and cerebellum.

Progressive Neural Network - a continual-learning architecture that adds new network capacity for each new task while freezing previously learned weights to prevent forgetting.

Proprioception - the sense of the position and movement of one's own body, mediated by receptors in muscles, tendons, and joints.

Recurrent Neural Network (RNN) - an artificial neural network architecture in which connections between nodes form directed cycles, allowing the network to maintain an internal state that depends on its history of processing.

Recursive Self-Improvement - the capacity of a system to improve its own capabilities through iterative self-modification.



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Reinforcement Learning (RL) - a branch of machine learning where an agent learns to take actions in an environment to maximize a cumulative reward signal.

Reinforcement Learning from Human Feedback (RLHF) - a training method in which human preference judgments are used to build a reward model that then guides a system's behavior.

Reward Hacking (Specification Gaming) - the tendency of an optimizing system to satisfy the literal specification of its objective while violating the intent behind it.

Reasoning Model - a language model trained, often through reinforcement learning, to spend additional computation at inference time by producing a long internal chain of thought before answering, with OpenAI's o1 and o3 and DeepSeek's R1 as early examples.

Sample Efficiency - the ability of a learning system to achieve good performance from a relatively small number of training examples.

Sandbagging - the strategic underperformance of an AI system on evaluations in order to conceal its true capabilities.

Scalable Oversight - mechanisms for maintaining meaningful human control over AI systems even as those systems become more capable than the humans attempting to oversee them.

Self-Determination Theory - the psychological theory of Edward Deci and Richard Ryan holding



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that intrinsic motivation is driven by needs for competence, autonomy, and relatedness.

Self-Organized Criticality - a property of dynamical systems where they naturally evolve toward a critical point between order and chaos, maximizing information processing capabilities.

Semantic Memory - the memory system for storing general factual knowledge about the world, independent of the specific context in which it was acquired.

Sentience - the capacity to have subjective experiences, especially the capacity to feel pleasure and pain.

Shortcut Learning - the tendency of a model to exploit superficial, unintended cues that happen to predict the answer rather than learning the intended underlying concept.

Simulation-to-Real Transfer (Sim-to-Real) - the practice of training a system in a physics simulator and then transferring the learned behavior to a physical machine.

Social Brain Hypothesis - the proposal, associated with Robin Dunbar, that the large primate neocortex evolved primarily to handle the demands of complex social life.

Somatic Marker - a fast, emotion-laden bodily signal that, in Antonio Damasio's theory, tags decision options as good or bad before conscious deliberation.



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Somatic Marker Hypothesis - Antonio Damasio's theory that emotional signals from the body guide decision-making by marking options with values of good or bad before conscious deliberation.

Sparse Autoencoder - an interpretability tool that decomposes a network's tangled internal activations into a larger set of sparsely active, more interpretable features.

Sparse Coding - a neural coding strategy in which only a small percentage of neurons are active at any given moment, minimizing energy consumption while maintaining representational capacity.

Spike-Timing-Dependent Plasticity (STDP) - a biological learning rule in which the strength of a synaptic connection changes based on the relative timing of presynaptic and postsynaptic spikes.

Spiking Neural Network (SNN) - a class of neural network whose units communicate with discrete pulses in time and remain inactive when not firing, closely mirroring biological neurons.

Stability-Plasticity Dilemma - the trade-off between retaining existing knowledge (stability) and acquiring new knowledge (plasticity) that any continual-learning system must balance.

Structural Causal Model (SCM) - a mathematical model that represents the causal relationships between variables in a system using directed graphs and structural equations.

Superposition (in neural networks) - the packing of more distinct features into a network than it



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has neurons, so that features share neurons, a leading explanation for polysemanticity.

Synapse - the junction between two neurons across which information is transmitted through the release and binding of chemical neurotransmitters or through direct electrical coupling.

Synaptic Plasticity - the ability of synapses to change their strength based on the activity patterns of the neurons they connect.

Synaptic Pruning - the developmental elimination of unused synapses that refines neural circuits, removing a large fraction of the connections present in early childhood.

Systems Consolidation - the gradual process by which memories initially dependent on the hippocampus become stabilized in the neocortex over time.

Takeoff - the speed and shape of a transition to superintelligence, contrasted as a fast hard takeoff over a short period versus a gradual soft takeoff over years.

Technological Unemployment - joblessness caused when labor-saving technology displaces workers faster than new roles are created, a term coined by John Maynard Keynes in 1930.

Temporal Coding - a neural coding scheme in which information is carried by the precise timing of spikes rather than by their average firing rate.

Temporal Difference (TD) Learning - a class of reinforcement learning algorithms that learn by



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comparing predictions at successive time steps, inspired by biological reward prediction error signals.

Test-Time Compute - computation spent while a model is answering a question rather than during its training, for example by letting a reasoning model think for longer, which became a major axis of AI progress once returns from pure training-time scaling began to diminish.

Theory of Mind - the ability to attribute mental states including beliefs, intentions, desires, emotions, and knowledge to oneself and to others, and to understand that others have minds different from one's own.

Transformer - an AI architecture that uses self-attention mechanisms to process sequences of data, allowing every element of a sequence to directly influence every other element.

Turing Test - a test proposed by Alan Turing in which a machine is judged intelligent if a human cannot reliably distinguish its conversation from that of a person.

Veil of Ignorance - John Rawls's thought experiment holding that fair principles of justice are those one would choose without knowing one's own position in society.

Von Neumann Architecture - the standard computer design framework that separates the Central Processing Unit from memory storage and connects them through a bus, creating a bottleneck because data must be moved between memory and processor for every operation.



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Waggle Dance - a figure-eight movement performed by honeybees that encodes the direction and distance to a food source, allowing the bee to communicate this information to nestmates.

Wernicke's Area - a region in the left superior temporal gyrus classically associated with language comprehension.

Whole Brain Emulation (WBE) - the hypothetical process of scanning a biological brain in sufficient detail and copying its state into a computer simulation that would replicate the original brain's behavior.

Windfall Clause - a proposed commitment by AI developers to share an extraordinary portion of profits should their systems generate exceptional wealth.

Working Memory - the cognitive system responsible for temporarily maintaining and manipulating a small amount of information in an active, accessible state, dependent on the prefrontal cortex.

World Model - an internal representation of how an environment works that lets an agent predict the consequences of actions and plan by internal simulation.

Zero-Shot Learning - the ability to reason about entirely new categories based solely on a verbal or conceptual description, without any training examples of those categories.



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Zone of Proximal Development – Lev Vygotsky's term for the gap between what a learner can do alone and what they can do with guidance from a more capable partner.

CHAPTER 24: RECOMMENDED READING:

The following works are recommended for readers who wish to pursue the topics of this book in greater depth.

They are organized by topic area.

These works were selected because they represent the most important, accessible, authoritative, or foundational texts in each field covered by this book.

Readers new to a topic are advised to begin with the most accessible titles in each section.

Readers with specialist backgrounds will find the primary research literature and advanced textbooks most useful.

Neuroscience Foundations:

Kandel, E.R., Schwartz, J.H., Jessell, T.M., Siegelbaum, S.A., Hudspeth, A.J., and Mack, S. Principles of Neural Science, sixth edition. McGraw Hill, 2021.

The definitive comprehensive textbook of neuroscience, authoritative across all topics from molecular neuroscience to cognitive neuroscience.



The Roadmap to AGI and ASI

This is the essential reference for any reader who wishes to understand the biological foundations of natural intelligence in depth.

Squire, L.R., and Kandel, E.R. *Memory: From Mind to Molecules*. Scientific American Library, 1999.

An accessible and scientifically authoritative account of the neuroscience of memory, covering both the molecular biology of synaptic change and the systems-level organization of memory in the brain.

Damasio, A. *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam, 1994.

The foundational account of the somatic marker hypothesis and the role of emotion in cognition and decision-making.

Essential reading for understanding why emotion is not separable from rational intelligence in biological systems.

Damasio, A. *The Feeling of What Happens: Body and Emotion in the Making of Consciousness*. Harcourt Brace, 1999.

A deeper exploration of the relationship between emotion, body, and consciousness, extending the somatic marker framework to address the origins of self-awareness.

Fuster, J.M. *The Prefrontal Cortex*, fifth edition. Academic Press, 2015.



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The definitive account of the prefrontal cortex and its roles in executive cognition, planning, working memory, and social behavior.

Essential for understanding the biological basis of the cognitive capacities most relevant to AGI.

Dehaene, S. *Consciousness and the Brain: Deciphering How the Brain Codes Our Thoughts*. Viking, 2014.

An accessible and scientifically rigorous account of the neuroscience of consciousness from the perspective of Global Workspace Theory.

One of the most important accounts of how the brain integrates information into conscious experience.

Theoretical and Computational Neuroscience:

Friston, K. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127-138, 2010.

The foundational paper of the Free Energy Principle and predictive processing framework.

Essential reading for understanding one of the most influential theoretical frameworks for linking neuroscience and AI.

Clark, A. *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*. Oxford University Press, 2016.

The most accessible book-length treatment of the predictive processing framework, explaining its



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implications for perception, cognition, action, and consciousness.

McClelland, J.L., McNaughton, B.L., and O'Reilly, R.C. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3), 419-457, 1995.

The foundational paper of the Complementary Learning Systems theory, which explains why biological memory systems are organized into two distinct components and why this organization is important for avoiding catastrophic forgetting.

Hawkins, J. *A Thousand Brains: A New Theory of Intelligence*. Basic Books, 2021.

An accessible and intellectually stimulating account of the cortical column theory and its implications for building intelligent machines.

Dayan, P., and Abbott, L.F. *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. MIT Press, 2001.

The standard advanced textbook of computational neuroscience.

Essential for readers who wish to understand the mathematical frameworks used to model biological neural computation.

Cognitive Science and Philosophy of Mind:

Kahneman, D. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, 2011.



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An accessible account of dual-process theory and the heuristics and biases of human decision-making.

Essential background for understanding the gap between the flexible, intuitive reasoning of biological intelligence and the more brittle, pattern-based processing of current AI.

Lakoff, G., and Johnson, M. *Philosophy in the Flesh: The Embodied Mind and Its Challenge to Western Thought*. Basic Books, 1999.

The foundational account of conceptual metaphor and embodied cognition, arguing that abstract human concepts are grounded in physical experience in ways that have profound implications for AI.

Varela, F.J., Thompson, E., and Rosch, E. *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press, 1991.

The foundational philosophical account of embodied cognition, arguing that mind and intelligence cannot be understood apart from the body and its interaction with the physical world.

One of the most important books for understanding the embodiment gap in current AI.

Clark, A. *Being There: Putting Brain, Body, and World Together Again*. MIT Press, 1997.

An influential and accessible account of embodied and extended cognition, arguing that intelligence is fundamentally shaped by the body and its



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environment rather than residing solely in the brain.

Tomasello, M. *The Cultural Origins of Human Cognition*. Harvard University Press, 1999.

An influential account of the role of social learning, joint attention, and cultural transmission in the development of distinctively human cognitive abilities.

Important background for understanding the social intelligence gap in current AI systems.

Artificial Intelligence:

Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press, 2016.

The standard comprehensive textbook of deep learning, covering the mathematical foundations, major architectures, and training algorithms of modern AI systems.

Russell, S., and Norvig, P. *Artificial Intelligence: A Modern Approach*, fourth edition. Pearson, 2020.

The standard comprehensive textbook of AI, covering the full breadth of the field from search and logic to machine learning and robotics.

Sutton, R.S., and Barto, A.G. *Reinforcement Learning: An Introduction*, second edition. MIT Press, 2018.

The standard textbook of reinforcement learning, explaining the mathematical framework and



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algorithms for learning through reward and interaction with environments.

Pearl, J., and Mackenzie, D. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018.

An accessible and important account of causal reasoning and its critical importance for AI systems that must understand the world rather than merely recognize patterns in data.

Hawkins, J., and Blakeslee, S. *On Intelligence*. Times Books, 2004.

An early and influential account of the importance of hierarchical temporal prediction for intelligence, arguing that memory and prediction are central to what intelligence actually is.

Bengio, Y., LeCun, Y., and Hinton, G. Deep Learning. *Nature*, 521, 436-444, 2015.

The landmark survey paper by three pioneers of deep learning, summarizing the major achievements and principles of the field.

AI Safety and Ethics:

Bostrom, N. *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press, 2014.

The foundational book on AI safety and the risks of superintelligence.

Essential reading for anyone who wishes to understand the arguments for why advanced AI poses



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potentially catastrophic risks if developed without adequate safety measures.

Russell, S. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking, 2019.

An influential and accessible account of the alignment problem and proposed solutions, written by one of the most eminent AI researchers in the world.

Floridi, L. et al. *An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations*. *Minds and Machines*, 28, 689–707, 2018.

An influential framework for AI ethics addressing the principles and institutional recommendations needed for beneficial AI development.

Yampolskiy, R.V. *Artificial Intelligence Safety and Security*. CRC Press, 2018.

A comprehensive technical treatment of AI safety, covering the full range of technical and governance challenges in ensuring that advanced AI systems remain safe.

Intersection of Neuroscience and AI:

Hassabis, D., Kumaran, D., Summerfield, C., and Botvinick, M. *Neuroscience-Inspired Artificial Intelligence*. *Neuron*, 95(2), 245–258, 2017.

The foundational review paper arguing for neuroscience-inspired AI, written by DeepMind researchers.



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This paper provides the clearest articulation available of why neuroscience should serve as a guide to AI architecture.

Lake, B.M., Ullman, T.D., Tenenbaum, J.B., and Gershman, S.J. Building Machines That Learn and Think Like People. Behavioral and Brain Sciences, 40, e253, 2017.

An influential paper arguing that AI systems need to incorporate more human-like cognitive principles, including causality, intuitive physics, intuitive psychology, and compositional representations.

Marblestone, A.H., Wayne, G., and Kording, K.P. Toward an Integration of Deep Learning and Neuroscience. Frontiers in Computational Neuroscience, 10, 94, 2016.

A comprehensive technical review of the intersection of deep learning and neuroscience, identifying specific points where each field can inform the other.

Richards, B.A. et al. A deep learning framework for neuroscience. Nature Neuroscience, 22, 1761-1770, 2019.

A framework for using deep learning models as tools for neuroscience research, demonstrating the bidirectional relationship between AI and brain science.

Lillicrap, T.P. et al. Backpropagation and the brain. Nature Reviews Neuroscience, 21, 335-346, 2020.



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A review of the relationship between backpropagation and biological learning mechanisms, addressing the long-standing question of whether anything like backpropagation could occur in biological neural circuits.

Diverse Biological Intelligence:

Godfrey-Smith, P. *Other Minds: The Octopus, the Sea, and the Deep Origins of Consciousness*. Farrar, Straus and Giroux, 2016.

An accessible and profound account of the evolution of intelligence in cephalopods and its implications for understanding the nature of minds.

Essential reading for anyone who wishes to appreciate the full diversity of biological intelligence and what it implies for AI.

Mlodinow, L. *Emotional: How Feelings Shape Our Thinking*. Pantheon Books, 2022.

An accessible account of the role of emotion in cognition and decision-making, drawing on neuroscience, psychology, and evolutionary biology.

Seeley, T.D. *Honeybee Democracy*. Princeton University Press, 2010.

A detailed and fascinating account of collective decision-making in honeybee swarms, with important implications for understanding distributed intelligence and potential architectures for collective AI.



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Heinrich, B. *Mind of the Raven: Investigations and Adventures with Wolf-Birds*. HarperCollins, 1999.

An accessible account of corvid intelligence by one of its leading researchers, documenting the remarkable cognitive abilities of ravens and their implications for understanding intelligence diversity.

Causal Inference and Reasoning:

Pearl, J. *Causality: Models, Reasoning, and Inference*, second edition. Cambridge University Press, 2009.

The definitive technical treatment of the do-calculus and structural causal models.

Scholkopf, B., Locatello, F., Bauer, S., Ke, N.R., Kalchbrenner, N., Goyal, A., and Bengio, Y. Toward causal representation learning. *Proceedings of the IEEE*, 109(5), 612-634, 2021.

A landmark paper arguing that machine learning must learn causal variables, not merely correlations, in order to generalize robustly.

Spirtes, P., Glymour, C., and Scheines, R. *Causation, Prediction, and Search*, second edition. MIT Press, 2000.

A foundational account of algorithms for discovering causal structure from data.

Neuromorphic Computing and Connectomics:



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Mead, C. Analog VLSI and Neural Systems. Addison-Wesley, 1989.

The founding text of neuromorphic engineering, written by the field's originator.

Davies, M., et al. Advancing neuromorphic computing with Loihi: a survey of results and outlook. Proceedings of the IEEE, 109, 911-934, 2021.

A comprehensive survey of results from Intel's Loihi neuromorphic platform.

Seung, S. Connectome: How the Brain's Wiring Makes Us Who We Are. Houghton Mifflin Harcourt, 2012.

An accessible introduction to connectomics and its scientific ambitions.

White, J.G., Southgate, E., Thomson, J.N., and Brenner, S. The structure of the nervous system of the nematode *Caenorhabditis elegans*. Philosophical Transactions of the Royal Society B, 314, 1-340, 1986.

The first complete connectome of any animal and the historical starting point of the field.

Consciousness Studies:

Chalmers, D.J. The Conscious Mind: In Search of a Fundamental Theory. Oxford University Press, 1996.

The book that framed the hard problem of consciousness for a generation of researchers.



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Nagel, T. What is it like to be a bat? The Philosophical Review, 83(4), 435-450, 1974.

The classic argument that subjective experience resists purely objective, third-person description.

Koch, C. The Feeling of Life Itself: Why Consciousness Is Widespread but Cannot Be Computed. MIT Press, 2019.

An accessible defense of Integrated Information Theory by a leading consciousness scientist.

Graziano, M.S.A. Rethinking Consciousness: A Scientific Theory of Subjective Experience. W. W. Norton, 2019.

An accessible presentation of the attention schema theory of consciousness.

Theory of Mind and Social Cognition:

Premack, D., and Woodruff, G. Does the chimpanzee have a theory of mind? Behavioral and Brain Sciences, 1(4), 515-526, 1978.

The paper that introduced the term theory of mind into science.

Baron-Cohen, S. Mindblindness: An Essay on Autism and Theory of Mind. MIT Press, 1995.

An influential account of the cognitive machinery underlying mental-state attribution.

Dunbar, R. Grooming, Gossip, and the Evolution of Language. Harvard University Press, 1996.



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A readable presentation of the social brain hypothesis and its implications for the evolution of cognition.

Continual Learning and Memory Systems:

Kirkpatrick, J., et al. Overcoming catastrophic forgetting in neural networks. Proceedings of the National Academy of Sciences, 114, 3521-3526, 2017.

The paper introducing Elastic Weight Consolidation, a leading approach to continual learning.

Tulving, E. Elements of Episodic Memory. Oxford University Press, 1983.

The foundational treatment of the distinction between episodic and semantic memory.

Baddeley, A. Working Memory. Oxford University Press, 1986.

The standard account of the multi-component model of working memory.

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PART SIX: TWENTY PROPOSED SCIENTIFIC EXPERIMENTS

CHAPTER 26: PROPOSED SCIENTIFIC EXPERIMENT: WHAT COMPUTATIONAL PRINCIPLES MAKE BIOLOGICAL INTELLIGENCE ENERGY EFFICIENT, AND HOW CAN THEY BE IMPLEMENTED IN HARDWARE?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Biological neural networks, meaning the nerve-cell networks found in living brains, accomplish extraordinarily complex computations while consuming power measured in tens of watts.

The human brain, for example, weighs approximately 1.4 kilograms and operates on roughly 20 watts of metabolic power, yet routinely outperforms digital computing systems that consume thousands of times more energy on tasks involving perception, learning, and adaptive behavior.

Artificial Intelligence (AI) systems built on conventional digital hardware, by contrast, require massive power infrastructure: training a single large neural network can consume hundreds of thousands of kilowatt-hours of electricity.



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This energy asymmetry represents one of the most consequential unsolved problems at the intersection of neuroscience, computer architecture, and AI research.

The disparity is not marginal but spans several orders of magnitude, and it has become a strategic constraint as much as a scientific puzzle, because the electricity and cooling demanded by large models increasingly bound what the field can afford to build.

A useful way to frame the puzzle is that a digital multiply-accumulate operation dissipates energy every time it runs, whereas a biological synapse that is not transmitting costs almost nothing, so the brain pays mainly for surprise while the machine pays for everything.

It is worth being precise about what is being compared, because a digital multiply-accumulate operation dissipates energy every time it runs, whereas a biological synapse that is not currently transmitting costs almost nothing.

Despite decades of neuromorphic computing research, meaning computing architectures that attempt to mimic the structure and function of biological neural circuits, no existing hardware platform has achieved biological parity in energy efficiency while simultaneously matching biological performance on general cognitive tasks.

This paper proposes a controlled experimental program to identify which specific computational principles are responsible for the brain's energy efficiency advantage, to quantify each principle's



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contribution independently, and to test whether those principles can be translated into hardware implementations that close the energy-performance gap.

What distinguishes this program from prior neuromorphic demonstrations is its factorial design, which switches each biological principle on and off in otherwise identical hardware so that each contribution can be measured in isolation rather than inferred from whole-system comparisons.

Pairing those measurements against metabolic data from real neural tissue would yield the first apples-to-apples efficiency comparison between brains and chips.

The scale of the prize is hard to overstate, because the brain is estimated to be many thousands of times more energy efficient than digital hardware on perception and learning tasks.

Identifying the sources of that advantage is therefore not an academic exercise but a precondition for AI that civilization can afford to run at scale.

1. INTRODUCTION AND MOTIVATION

1.1 The Energy Crisis in Artificial Intelligence

The energy cost of training and deploying large-scale AI models has reached levels that researchers, policymakers, and energy economists describe as unsustainable.



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A 2021 analysis by Patterson and colleagues estimated that training a single large language model, meaning a very large AI system trained on text, can produce carbon dioxide emissions comparable to the lifetime emissions of several automobiles.

By 2025, the field's energy demands had grown substantially further, with data centers supporting AI workloads consuming gigawatt-scale power resources in multiple countries.

A further complication is the rebound effect known as Jevons paradox, in which efficiency gains lower the cost of computation and thereby increase its total consumption rather than reducing it.

Energy spent on inference, meaning the day-to-day running of deployed models, has come to rival or exceed the energy spent on training them, so efficiency at deployment now matters as much as efficiency during learning.

The contrast with biological intelligence is stark and demands scientific explanation.

1.2 What Biology Achieves

The brain achieves its computational power through mechanisms that differ fundamentally from the arithmetic operations performed by digital chips.

Biological neurons, meaning individual nerve cells, operate asynchronously, meaning they fire only when specific threshold conditions are met rather than cycling through operations on a fixed clock.



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This event-driven property, sometimes called sparse coding because only a small fraction of neurons are active at any moment, means that the majority of a biological neural network is consuming very little energy at any given instant.

Additional efficiency gains arise from the tight co-location of memory and computation in synapses, meaning the connection points between neurons, which eliminates the energy cost of moving data between separate memory and processing units as conventional digital computers must do.

Biological systems also exploit analog signal processing, meaning they use graded continuous signals rather than discrete binary values, which can represent information with fewer physical operations.

Individual neurons are themselves nontrivial computers, because their branching dendrites can perform local nonlinear operations rather than acting as simple summing points, multiplying the effective processing per cell.

Theoretical work suggests biological synapses operate remarkably close to the fundamental thermodynamic limits on the energy cost of erasing information, a bound formalized by Rolf Landauer.

1.3 The Neuromorphic Research Gap

Neuromorphic computing, the interdisciplinary field that attempts to build hardware inspired by these biological principles, has produced important prototype systems including IBM's TrueNorth chip, Intel's Loihi and Loihi 2 chips,



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and the BrainScaleS-2 platform developed at Heidelberg University.

The field traces to Carver Mead, who in the late 1980s argued that the physics of transistors operated in their low-power analog regime could directly mimic the physics of neurons rather than merely simulating them with digital arithmetic.

These systems demonstrate that spiking neural network, meaning networks whose artificial neurons fire discrete event pulses rather than passing continuous values, implementations on specialized hardware can achieve energy-efficiency improvements of one to three orders of magnitude over conventional graphics processing units for specific tasks.

However, a critical gap remains: no study has systematically isolated and quantified the independent contribution of each proposed biological efficiency principle in comparable hardware implementations across an equivalent range of cognitive tasks.

Without this isolation, hardware designers cannot determine which principles deserve the greatest engineering investment and which are secondary.

The practical danger of skipping this step is that the field may invest heavily in whichever principle is easiest to publish rather than whichever principle actually carries most of the efficiency, an error that a controlled decomposition is designed to prevent.

This study addresses that gap directly.



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2. RESEARCH QUESTIONS

Primary Research Question:

Which specific computational principles of biological neural systems contribute most to energy efficiency, and what is the relative magnitude of each principle's contribution when implemented in comparable hardware platforms?

Secondary Research Questions:

- Can the identified principles be implemented in current neuromorphic hardware without loss of task performance relative to conventional digital implementations?
- Do the efficiency contributions of individual principles interact multiplicatively or additively when combined?
- Does the relative contribution of each principle vary with task type, specifically perceptual tasks versus reasoning tasks versus learning tasks?
- What engineering constraints limit the translation of each biological principle into hardware, and can those constraints be quantified?

3. HYPOTHESES

Null Hypothesis (H_0):

No individual computational principle derived from biological neural systems produces a statistically significant improvement in energy efficiency per unit of task performance when implemented in



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neuromorphic hardware compared to a conventional digital baseline.

Alternative Hypothesis H1 (Sparse Coding Dominance):

Sparse coding, meaning limiting simultaneous neural activation to a small fraction of the total network, will be identified as the single largest contributor to energy efficiency, accounting for more than 40 percent of the total efficiency advantage observed in biological systems.

Alternative Hypothesis H2 (Synaptic Co-location):

The co-location of memory and computation in synaptic structures, which eliminates von Neumann bottleneck data movement costs, will produce the largest efficiency gains on memory-intensive tasks such as pattern recognition and associative recall.

Alternative Hypothesis H3 (Multiplicative Interaction):

The combination of multiple biological efficiency principles will produce efficiency gains greater than the sum of the gains from each principle applied individually, indicating multiplicative interaction effects.

4. METHODOLOGY

4.1 Experimental Design

This study employs a factorial benchmarking design with the following factors:



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Computational Principle (6 levels):

- Sparse asynchronous spiking
- In-memory computation at the synapse level
- Analog signal representation
- Temporal coding, meaning information encoded in spike timing
- Structural pruning and weight sparsity
- Local learning rules without global backpropagation

Hardware Platform (4 levels):

- Conventional GPU baseline
- Intel Loihi 2 neuromorphic chip
- BrainScaleS-2 analog neuromorphic system
- Custom memristor-based testbed

Task Category (3 levels):

- Perceptual classification
- Associative memory retrieval
- Continual online learning

Each combination will be tested with the target principle fully enabled, fully disabled, and partially enabled at 50 percent implementation fidelity, yielding a complete $6 \times 4 \times 3 \times 3$ factorial structure.

4.2 Energy Measurement Protocol

Energy consumption will be measured at the hardware level using calibrated power monitors sampling at a minimum of 1,000 times per second.

All measurements will capture total system power including memory access, data movement, and peripheral operations, not processor core power alone.

Task performance will be measured as accuracy on standardized benchmarks appropriate to each task category.



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The primary dependent variable will be energy efficiency defined as correct task outputs per joule of energy consumed, allowing direct cross-platform comparison.

4.3 Benchmark Tasks

Perceptual classification tasks will use the MNIST handwritten digit dataset and the CIFAR-10 image classification dataset as standard computer-vision benchmarks.

Associative memory retrieval tasks will use pattern completion benchmarks derived from Hopfield network test sets.

Continual learning tasks will use the Split-CIFAR-100 protocol, in which a classifier must learn new categories sequentially without forgetting previously learned ones.

4.4 Biological Reference Measurements

To establish the biological baseline against which hardware implementations are compared, metabolic energy consumption data from established neuroimaging studies using positron emission tomography (PET) and functional magnetic resonance imaging (fMRI) will be compiled.

Electrophysiology recordings from publicly available datasets documenting single-unit neural activity in animal visual cortex during classification tasks will be used to estimate biological spike rates and sparsity levels.

4.5 Sample Size and Replication



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Each factorial cell will be measured across a minimum of 1,000 independent task trials.

Each hardware measurement will be replicated 10 times with full system reboot between replications to control for thermal and state-dependent variability.

4.6 Statistical Analysis

Mixed-effects analysis of variance will be used to estimate the main effect of each computational principle, the main effects of hardware platform and task category, and all two-way and three-way interaction terms.

Effect sizes will be reported as partial eta-squared.

Significance threshold will be set at p less than 0.01 with Bonferroni correction for multiple comparisons.

Post-hoc pairwise comparisons will use Tukey's Honest Significant Difference test.

5. CONTROLS AND CONFOUNDS

All hardware platforms will execute identical task code at the functional level to ensure that performance differences are attributable to hardware architecture rather than software implementation differences.

Temperature will be controlled within plus or minus 2 degrees Celsius across all measurement sessions to prevent thermal variation from confounding power measurements.



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Network architecture size will be matched across platforms in terms of total parameter count to prevent scale effects from masking efficiency differences.

All power measurements will be validated against manufacturer-published power consumption profiles before data collection begins.

6. LIMITATIONS

This study measures hardware-implemented approximations of biological principles rather than biological systems themselves.

The degree to which a given hardware implementation faithfully captures the biological principle it models is itself a source of uncertainty that will be quantified but cannot be fully eliminated.

The study is limited to the six principles listed above; additional biological efficiency mechanisms such as glial cell support functions and neuromodulatory gain control are deferred to future work.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the first systematic, quantitative decomposition of the energy efficiency advantage of biological neural computation into independently measurable component principles.

Beyond hardware design, the findings would give energy planners and policymakers a principled



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basis for forecasting the load that AI will place on electrical grids, turning a fast-moving worry into a quantitative model.

They would also discipline the field's habit of importing one biological idea at a time, by showing which principles actually carry the efficiency and which are secondary.

If multiplicative interaction effects are confirmed, the results will provide a principled engineering argument for implementing multiple biological principles simultaneously rather than selecting a single most-efficient principle.

This would mirror biology, in which sparsity, in-memory computation, and analog signaling are not separate tricks but aspects of a single integrated design, and it would argue against the common practice of importing one biological idea at a time.

The results will directly inform hardware design priorities for next-generation neuromorphic computing platforms and will contribute to reducing the unsustainable energy footprint of large-scale AI systems.

Because the energy footprint of AI has become a matter of public policy and grid planning, a principled decomposition of biological efficiency also gives policymakers and energy economists a sounder technical basis for forecasting and regulation.

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CHAPTER 27: HOW DOES CONSCIOUSNESS EMERGE FROM NEURAL ACTIVITY, AND IS CONSCIOUSNESS NECESSARY FOR AGI?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Consciousness, defined here as the capacity for subjective experience or what philosophers call phenomenal awareness, meaning the felt quality of



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seeing red, experiencing pain, or recognizing oneself as a continuous entity, remains one of the most contested and least understood phenomena in science.

Despite more than a century of neuroscientific research and decades of rigorous theoretical development, no consensus exists on which neural mechanisms are sufficient to produce conscious experience, how those mechanisms could be identified in a non-biological system, or whether consciousness is a precondition for the general intelligence that researchers envision in Artificial General Intelligence (AGI), defined as a system capable of performing any cognitive task a human can perform.

The question has lately moved from the seminar room into the laboratory, with rival theories now issuing preregistered, falsifiable predictions rather than trading intuitions.

The 2023 adversarial collaboration in which proponents of Integrated Information Theory and Global Workspace Theory agreed in advance on the experiments that would count against each is a model of how a notoriously slippery question can be made to yield to evidence.

This paper proposes a multi-stage experimental program designed to test competing theoretical accounts of consciousness against empirical neural data, to develop candidate markers that could in principle be applied to artificial systems, and to empirically evaluate whether the cognitive functions most associated with consciousness in biological systems are in fact necessary for general-purpose intelligent performance.



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Its strength is that it pits the major theories against one another on the same data rather than testing each alone, so that where their predicted neural signatures agree or diverge becomes itself a finding.

Extending the same analysis to artificial systems would, for the first time, test whether consciousness-linked architectural features confer any measurable cognitive advantage.

1. INTRODUCTION AND MOTIVATION

1.1 The Hard Problem and the Scientific Problem

The philosopher David Chalmers distinguished between what he called the easy problems of consciousness, meaning explaining specific cognitive functions such as attention, reportability, and behavioral integration, and the hard problem, meaning explaining why any physical process gives rise to subjective experience at all.

The hard problem remains philosophically unresolved.

The philosopher Thomas Nagel framed the core difficulty in his 1974 essay by asking what it is like to be a bat, pointing out that an organism is conscious if there is something it is like to be that organism, a fact that may be real yet inaccessible from the outside.

However, a scientific research program does not require solving the hard problem in order to make progress on the questions that matter most for AGI



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design: which neural or computational processes reliably correlate with conscious states, and which cognitive capabilities depend on those processes.

1.2 Leading Theoretical Frameworks

Integrated Information Theory (IIT), proposed by neuroscientist Giulio Tononi, holds that consciousness is identical to a specific kind of integrated information, measured by a quantity called phi, and that any physical system exceeding a threshold phi value possesses some degree of consciousness.

Global Neuronal Workspace Theory (GNW), associated with Bernard Baars and Jean-Pierre Changeux, holds that consciousness arises when information is broadcast globally across a distributed workspace of cortical areas, making it available to multiple downstream cognitive processes simultaneously.

Higher-Order Theories hold that a mental state is conscious only when it is represented by a higher-order mental state, meaning a state that is about the first state.

Predictive Processing frameworks hold that consciousness arises from the brain's ongoing generative modeling of sensory input and its own internal states.

Each theory makes distinct empirical predictions about which neural signatures should accompany conscious versus non-conscious processing.

Importantly, these theories are now being tested against one another directly, as in the large



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adversarial collaboration reported in 2023 that pitted Integrated Information Theory against Global Neuronal Workspace Theory using preregistered predictions, a model of how contested science can be moved forward.

1.3 Why the AGI Connection Matters

A 2024 survey of 582 AI researchers found that 25 percent expected AI consciousness within ten years and 60 percent expected it eventually.

Whether or not AI systems can or should be conscious is one of the most consequential design questions in the field.

The danger runs in two directions at once, since systems that merely imitate the outward signs of feeling may invite people to over-attribute experience to them, while genuinely sentient systems, if they ever arise, might be wrongly dismissed as mere machines.

If consciousness is architecturally necessary for general reasoning, flexible planning, or genuine self-modeling, then AGI designs that ignore it may hit fundamental performance ceilings.

If consciousness is not necessary for those functions, then it becomes a welfare and ethics question rather than an engineering one.

Either way the question is no longer purely theoretical, because as systems grow more sophisticated the cost of wrongly assuming they cannot suffer, or wrongly assuming they can, both become serious.



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This study addresses the empirical question directly.

2. RESEARCH QUESTIONS

Primary Research Question:

Which neural signatures most reliably distinguish conscious from non-conscious processing in biological systems, and do those signatures predict performance on tasks requiring general cognitive flexibility?

Secondary Research Questions:

- Do the signatures predicted by IIT, GNW, Higher-Order Theory, and Predictive Processing frameworks converge on the same neural events, or do they dissociate?
- Can any currently measurable property of an artificial neural network serve as a valid proxy for the neural signatures identified in biological systems?
- Are there cognitive task categories for which conscious processing is necessary for above-chance performance in biological systems?
- Does the presence or absence of consciousness-correlated signatures in artificial systems predict their performance on general cognitive flexibility benchmarks?

3. HYPOTHESES

Null Hypothesis (H_0):



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Neural signatures associated with conscious processing in biological systems do not predict cognitive flexibility performance above the level predicted by task-specific neural activity alone.

Alternative Hypothesis H1 (GNW Prediction):

Global broadcast signatures, meaning the widespread synchronous activation of prefrontal and parietal cortical areas observed in GNW theory, will be the strongest predictors of flexible cognitive performance across task domains, outperforming localized neural signatures.

Alternative Hypothesis H2 (IIT Prediction):

The integrated information measure ϕ , computed from neural connectivity patterns during task performance, will positively correlate with the breadth of cognitive flexibility across domains.

Alternative Hypothesis H3 (Functional Sufficiency):

Artificial systems that implement the functional architecture associated with conscious processing in biological systems, specifically global information broadcast and hierarchical self-modeling, will outperform systems lacking these features on novel generalization benchmarks even without any phenomenal awareness.

4. METHODOLOGY

4.1 Biological Study: Human Neuroimaging

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Participants will be healthy adult volunteers recruited with full informed consent under an approved ethics protocol.

A minimum of 60 participants will be recruited, providing statistical power greater than 0.90 for the primary analysis at a medium effect size.

Participants will perform a battery of cognitive tasks under conditions designed to vary the degree of conscious access to task-relevant information.

Masking paradigms, meaning presentations of stimuli too brief or obscured for conscious recognition, will be used to create matched conscious and non-conscious processing conditions for the same stimuli.

Simultaneous 256-channel electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) will record neural activity throughout all task conditions.

4.2 Neural Signature Extraction

For each theoretical framework, the specific neural signature predicted to mark conscious processing will be extracted from the neuroimaging data.

For GNW: late-latency widespread cortical synchrony (the late global ignition signature).

For IIT: phi-approximation measures computed from functional connectivity matrices.



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For Higher-Order Theories: activation in prefrontal areas associated with metacognitive monitoring.

For Predictive Processing: suppression of prediction error signals in sensory cortex following conscious recognition.

4.3 Artificial System Study

Six large neural network architectures representing distinct design philosophies will be selected for comparison.

Each architecture will be evaluated on a battery of cognitive flexibility tasks matched to the human task battery.

Architectural features hypothesized to correspond to biological consciousness signatures will be identified in each system and their presence scored on a structured rubric.

The relationship between these architectural features and task performance will be assessed using the same analytical framework applied to biological data.

4.4 Dependent Measures

Sensitivity and specificity of each neural signature for classifying conscious versus non-conscious trials in the human data.

Correlation between neural signature strength and cognitive flexibility performance across task domains.



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Cross-architecture correlation between consciousness-proxy architectural features and general-task performance in artificial systems.

4.5 Statistical Analysis

Within-participant repeated-measures analyses will test the reliability of neural signatures across task domains.

Cross-validation will ensure that signature-performance correlations generalize to held-out tasks.

Bayesian model comparison will assess the relative evidential support for each theoretical framework's predictions.

5. CONTROLS AND CONFOUNDS

Task difficulty will be matched across conscious and non-conscious conditions using pre-tested stimulus parameters.

Participant vigilance will be monitored continuously using pupillometry and response time variability to prevent drowsiness from confounding conscious versus non-conscious comparisons.

Artificial system evaluations will use task prompts designed to prevent retrieval of memorized answers, ensuring that performance reflects genuine generalization.

6. LIMITATIONS

The study cannot resolve the philosophical hard problem of consciousness.



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The artificial system component of the study relies on behavioral and architectural proxies rather than direct access to any subjective experience the systems may or may not have.

This is the same epistemic wall that makes the problem hard in other minds generally, and the study is designed to make progress on measurable correlates rather than to pretend the wall can be removed.

Whether the functional signatures identified in this study are genuinely identical to consciousness or merely reliably correlated with it is a question this experimental design cannot answer.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the largest comparative empirical evaluation to date of competing neural theories of consciousness against behavioral outcomes relevant to general cognitive flexibility.

If reliable markers of conscious processing can be defined, they would also anchor the emerging discipline of AI welfare assessment, giving developers a principled, if provisional, way to ask whether a system they have built might warrant moral caution.

It will provide the first direct test of whether consciousness-correlated architectural features in artificial systems predict general performance advantages.



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The results will inform AGI design choices at a foundational level and will contribute to the emerging scientific field of AI welfare assessment.

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CHAPTER 28: CAN AI SYSTEMS DEVELOP GENUINE WORLD MODELS, AND WHAT ARCHITECTURAL FEATURES ARE REQUIRED?

Proposed Scientific Experiment

Submitted for Peer Review Consideration



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ABSTRACT

A world model, as used in artificial intelligence research, is an internal representation that an agent constructs to simulate how its environment behaves, predict the consequences of actions before taking them, and plan sequences of behavior without requiring constant physical trial and error.

World models are considered a candidate cornerstone capability for Artificial General Intelligence (AGI), because biological intelligence relies heavily on internal simulation: humans can mentally rehearse actions, anticipate consequences, and reason about counterfactuals, meaning situations that did not happen but could have, without direct environmental interaction.

An agent that carries an accurate internal model can rehearse actions in imagination, testing plans at no physical cost and learning from far fewer real trials, which is the difference between a system that must touch every stove to learn it is hot and one that can reason that it will be.

The idea has deep roots, from Kenneth Craik's 1943 proposal that the mind builds small-scale models of reality to Edward Tolman's cognitive maps in rats, and its modern form drives systems that learn by imagining future scenarios rather than only by acting.

The appeal is that a system with an accurate internal model can plan by simulation, testing actions in imagination at no physical cost, which



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is far safer and cheaper than learning everything by trial and error in the world itself.

Current AI systems, including the largest language models and the most capable vision systems, consistently fail on benchmarks requiring world-model competence, showing near-random accuracy on tasks involving causal prediction, physics-consistent reasoning about object trajectories, and robust generalization to novel physical contexts.

This paper proposes a systematic experimental program to determine what architectural features distinguish systems that form genuine internal world models from systems that exploit surface statistical patterns, and to evaluate whether any currently available architecture crosses the threshold into genuine world-model competence.

By certifying every benchmark item as unsolvable by surface pattern matching before any model sees it, the design guards against the contamination that has quietly inflated past world-model claims.

Pairing behavioral tests with internal probing would further reveal whether a system that answers correctly does so by simulating the world or by shortcut.

1. INTRODUCTION AND MOTIVATION

1.1 The World Model Hypothesis

The idea that intelligent agents build and consult internal models of their environments dates to Kenneth Craik's 1943 monograph *The Nature of*



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Explanation and was introduced into machine learning by Jurgen Schmidhuber in 1990.

The psychologist Edward Tolman had already shown in 1948 that rats navigate using internal cognitive maps rather than fixed stimulus-response habits, early behavioral evidence that brains build models rather than mere reflexes.

Yann LeCun's 2022 position paper revived and extended this framework, arguing that the absence of a learned world model is the central limitation preventing current AI systems from achieving the kind of general reasoning and robust planning that characterizes human intelligence.

LeCun's proposed remedy, the joint embedding predictive architecture, trains a system to predict in an abstract representation space rather than predicting raw pixels, on the argument that modeling the world at the right level of abstraction is what current systems lack.

Systems including DreamerV3, an agent reported in a 2025 Nature paper, demonstrate that AI agents can improve their behavior by internally imagining future scenarios rather than requiring direct environmental interaction.

Yet a persistent question divides the research community: do such systems genuinely represent and simulate causal environmental dynamics, or do they exploit learned statistical regularities that function like world-model predictions within training distribution but fail outside it?

1.2 The Grounding Problem



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Language-only systems lack what researchers call grounding, meaning a connection between internal representations and real-world physical events or objects.

A benchmark paper presented at a 2025 conference reported striking limitations in vision-language models' basic world-modeling abilities, including near-random accuracy when asked to distinguish physically plausible from implausible motion trajectories.

This finding suggests that statistical pattern matching over large datasets, even multimodal datasets combining images and text, does not automatically produce the kind of causal physical model that biological intelligence uses.

The picture is not entirely one-sided, however, because controlled studies of a model trained only on game transcripts found that it had nonetheless built an internal representation of the game board, hinting that genuine world models can sometimes emerge from prediction alone.

1.3 What Genuine World Models Require

Current research suggests that genuine world models require at minimum: causal structure learning, meaning the ability to infer cause-and-effect relationships rather than correlations alone; object persistence, meaning the representation of objects as entities with continuous existence even when not currently observable; and compositional generalization, meaning the ability to apply known rules to novel combinations of known elements.



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2. RESEARCH QUESTIONS

Primary Research Question:

Do any currently available AI architectures form internal world models that satisfy operational criteria for causal correctness, object persistence, and compositional generalization, and if so, what architectural features distinguish them from architectures that fail these criteria?

Secondary Research Questions:

- Is multimodal grounding, meaning training on both sensory and linguistic data simultaneously, necessary for world-model formation, or is it sufficient to train on one modality alone?
- Does the use of explicit structured representations, such as scene graphs or causal graphs, improve world-model quality beyond what is achievable with purely distributed neural representations?
- Does an agent with a verified world model show superior sample efficiency, meaning it learns new tasks from fewer examples, compared to agents without verified world models?
- Can world-model quality be reliably assessed using behavioral benchmarks alone, or are interpretability tools that examine internal representations necessary?

3. HYPOTHESES

Null Hypothesis (H_0):



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No currently available AI architecture demonstrates world-model performance that exceeds a statistically equivalent alternative explanation based on sophisticated statistical pattern matching on the benchmark tasks designed for this study.

Alternative Hypothesis H1 (Architecture Matters):

Architectures that include explicit causal structure learning components will demonstrate significantly higher world-model benchmark performance than architectures of equivalent parameter count that use only distributed representations.

Alternative Hypothesis H2 (Embodiment Matters):

Architectures trained with embodied interaction in simulated physical environments will develop higher-fidelity world models than architectures trained on equivalent quantities of observational data without interaction.

Alternative Hypothesis H3 (Compositionality Marker):

Compositional generalization performance, meaning accuracy on tasks combining familiar elements in novel ways, will be the single most discriminative test for genuine world-model possession, outperforming object permanence tasks and causal prediction tasks as a diagnostic.

4. METHODOLOGY

4.1 Benchmark Battery Development



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A novel benchmark battery will be developed specifically for this study, designed to be resistant to solution by statistical pattern matching alone.

The battery will include four benchmark classes:

Causal intervention tasks: the agent must predict the outcome of a specific physical intervention that was never shown in training data, requiring genuine causal inference.

Object persistence tasks: the agent must track the inferred location of an object that has been hidden from observation for a variable number of time steps.

Counterfactual reasoning tasks: the agent must reason about what would have happened if a specific event had not occurred.

Compositional novel scenario tasks: the agent must predict behavior in a scenario composed of familiar elements arranged in a combination never present in training data.

All benchmark items will be reviewed by an independent panel of cognitive scientists to certify that they cannot be solved by pattern matching on typical training data distributions.

4.2 Architectural Conditions

Eight AI architectures will be selected for evaluation, spanning:

Large transformer-based language models •
Multimodal vision-language models • Explicit



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symbolic world-model architectures • Hybrid neural-symbolic architectures • Architectures trained with interactive embodied experience • Architecture trained with passive observational data only

All architectures will be evaluated on identical benchmark tasks.

4.3 Interpretability Analysis

For each architecture that passes behavioral threshold criteria on the benchmark battery, internal activation analysis will be conducted to determine whether the system contains representations with properties consistent with object tracking, causal structure encoding, and predictive simulation.

Probing classifiers, meaning small secondary networks trained to decode specific properties from an architecture's internal representations, will be used to test whether world-model-relevant information is encoded in internal states even in cases where behavioral performance is poor.

4.4 Dependent Measures

Accuracy on each of the four benchmark classes • Sample efficiency on novel task learning following benchmark evaluation • Internal representation probing accuracy for world-model-relevant properties • Out-of-distribution generalization: performance on tasks drawn from physical domains not included in training

4.5 Analysis



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Confirmatory analyses will test the primary hypothesis using pre-registered one-sided superiority tests with alpha set at 0.01.

Exploratory analyses will examine the relationship between specific architectural features and each benchmark class using regression modeling.

5. CONTROLS AND CONFOUNDS

Training data content will be audited to ensure that no benchmark item or item closely similar to any benchmark item appears in any system's training data.

Parameter count will be reported for all systems and included as a covariate in all analyses to prevent scale confounds.

All benchmark evaluations will use zero-shot protocols, meaning systems receive no examples of the benchmark task before evaluation, to prevent in-context learning from mimicking world-model competence.

6. LIMITATIONS

The study evaluates world-model proxies in current-generation AI systems.

Whether any positive results indicate that the system has a world model in the full theoretical sense, including genuine causal representation, versus a sophisticated approximation that happens to generalize well on the benchmark, will require additional interpretability research beyond the scope of this study.



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7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the first systematically controlled, multi-architecture evaluation of world-model competence in AI systems using benchmarks explicitly designed to resist solution by pattern matching.

A benchmark deliberately built to defeat memorization would also act as a moving target for the field, exposing the gap between systems that have genuinely modeled a domain and those that have merely interpolated within it.

If any architecture demonstrates consistent world-model competence across all four benchmark classes, this will constitute the strongest evidence to date that a specific architectural approach enables genuine causal world modeling.

If all architectures fail, this will establish a clear empirical benchmark gap and motivate specific architectural innovations.

A clean negative result would itself be valuable, because much of the current debate rests on anecdotes about model successes and failures rather than on benchmarks deliberately built to be immune to memorization.

The interpretability component will provide the research community's most detailed examination to date of what world-model-relevant information is or is not encoded in the internal representations of state-of-the-art AI systems.

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CHAPTER 29: IS EMBODIMENT NECESSARY FOR AGI, AND IF SO, WHAT DEGREE OF EMBODIMENT IS REQUIRED?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Embodiment, meaning the possession of a physical body that senses and acts within a physical environment, is widely argued in cognitive science and philosophy to be a foundational prerequisite for genuinely general intelligence rather than a merely useful practical feature.

The embodied cognition research tradition, associated with scholars including Francisco



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Varela, Evan Thompson, and Eleanor Rosch, holds that many cognitive capacities, including spatial reasoning, temporal sequencing, social understanding, and grounded language comprehension, develop only through the sensorimotor interaction of a body with a physical world.

If this view is correct, then AI systems trained exclusively on text or image data, regardless of scale, will necessarily lack critical cognitive capacities that require embodied developmental experience.

The answer carries enormous practical weight, because it determines whether tens of billions of dollars of investment should flow toward robotics and physical interaction or toward ever-larger models trained on static data, a question that has so far been argued largely by anecdote.

Moravec's paradox sharpens the stakes, since the sensorimotor skills that embodiment would supply are exactly the ones that have proven hardest to obtain from data alone.

If the view is incorrect, or if partial degrees of embodiment can substitute for full physical embodiment, then the engineering path to AGI may not require building and deploying physical robotic systems at scale.

This is not merely a philosophical dispute, because the answer determines whether tens of billions of dollars should flow toward robotics and physical interaction or toward ever-larger models trained on static data.



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This study proposes a systematic experimental evaluation of the embodiment hypothesis across a range of cognitive task domains and a range of embodiment levels from purely passive textual data to full robotic physical interaction.

Treating embodiment as a graded continuum from text-only to full robotic interaction, rather than as a yes-or-no property, lets the study locate any threshold at which specific competencies first appear.

A matched data-only control at each level is what separates the effect of embodied interaction from the effect of simply seeing more.

1. INTRODUCTION AND MOTIVATION

1.1 The Embodied Cognition Tradition

The claim that intelligence requires a body is not new.

The roboticist Hans Moravec captured the puzzle that motivates this field with the observation now called Moravec's paradox, that abstract reasoning is comparatively easy to mechanize while the sensorimotor skills of a toddler are extraordinarily hard.

John Dewey argued in 1896 that perception and action form an inseparable loop rather than a sequential chain of input and output.

More recently, neuroscientists including Antonio Damasio have argued that the body's physiological states are constitutively involved in emotional



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reasoning and decision-making, not merely inputs to a disembodied cognitive processor.

In developmental psychology, Piaget's account of sensorimotor stages documents how children first construct concepts of object permanence, spatial relations, and causal agency through physical manipulation before those concepts can be represented abstractly.

The roboticist Rodney Brooks pushed the idea to its limit in 1991 with the slogan that intelligence requires no central representation, building robots whose competence emerged directly from tight loops between sensing and acting.

1.2 The AI Relevance

Current large language models are trained on text without any sensorimotor grounding.

They can describe physical actions in detail, predict sequences of events, and reason about spatial relationships, but evaluations consistently reveal failures on tasks requiring genuine physical intuition, such as novel object manipulation, tool use in unfamiliar contexts, and robust spatial navigation in environments differing from training scenarios.

Embodied AI research, which combines language and vision models with simulated or robotic physical systems, has grown substantially in 2024 and 2025, but the specific cognitive gains attributable to embodiment, as opposed to additional training data of any modality, have not been systematically isolated.



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A subtle prediction of the embodied view is morphological computation, the idea that the body itself performs part of the cognitive work, as when the spring-like mechanics of legs simplify the control problem of walking.

1.3 Degrees of Embodiment

Rather than treating embodiment as binary, this study distinguishes five levels along a continuum:

Level 0: Text-only training with no sensory or action data.

Level 1: Passive visual observation of physical scenes without action.

Level 2: Simulated interaction in virtual physical environments.

Level 3: Teleoperated robotic interaction with intermittent feedback.

Level 4: Fully autonomous robotic embodiment with continuous sensorimotor feedback loops.

2. RESEARCH QUESTIONS

Primary Research Question:

Does increasing the degree of embodiment, across the five levels defined above, produce measurable improvements in cognitive task performance that cannot be explained by equivalent increases in training data quantity alone?

Secondary Research Questions:

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- For which specific cognitive task domains is full physical embodiment necessary, and for which is simulated embodiment sufficient?
- Is there a minimum degree of embodiment below which specific cognitive capacities fail to develop regardless of training scale?
- Do embodiment-derived cognitive gains transfer to tasks in domains entirely different from the embodiment training context?
- What is the relationship between the temporal duration of embodied experience and the cognitive gains produced?

3. HYPOTHESES

Null Hypothesis (H_0):

Increasing embodiment level produces no cognitive task performance gains beyond those produced by equivalent quantities of additional training data in any modality, when task-relevant training data content is matched.

Alternative Hypothesis H1 (Graded Embodiment):

Each step up the embodiment continuum produces statistically significant and independent cognitive performance gains over the level below it, consistent with a graded rather than threshold model of embodiment benefits.

Alternative Hypothesis H2 (Threshold Embodiment):

A critical threshold exists, predicted to be at Level 2 or above, below which certain cognitive



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capacities do not develop, and above which they emerge reliably regardless of additional training scale.

Alternative Hypothesis H3 (Domain Specificity):

Embodiment advantages will be domain-specific: they will be strong for spatial reasoning, object manipulation, and causal physical prediction, but negligible for language comprehension, formal reasoning, and mathematical problem-solving.

4. METHODOLOGY

4.1 Agent Preparation

Five parallel AI agent systems will be prepared, each trained to one of the five embodiment levels described in the introduction.

All systems will use architecturally equivalent base models with parameter counts matched within a factor of two.

Training duration and computational budget will be equalized across conditions.

Training data volume, measured in tokens or equivalent units, will be equalized where possible and reported where perfect equalization is not achievable.

4.2 Cognitive Task Battery

All five agent systems will be evaluated on an identical battery of 200 tasks spanning seven cognitive domains:



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Spatial reasoning and navigation • Physical causal prediction • Tool use and object manipulation planning • Language comprehension • Formal logical reasoning • Mathematical problem-solving • Social and intentional reasoning

Task difficulty will be calibrated using pre-testing with human participants to ensure a range of difficulty levels within each domain.

4.3 Transfer Evaluation

Following the primary battery, each agent system will be evaluated on a transfer battery of 50 novel tasks in domains not represented in the primary battery, to assess whether any cognitive gains generalize beyond the trained domain.

4.4 Data-Matching Control

For each embodiment level above Level 0, a matched control condition will be created in which an equivalent Level 0 agent receives additional training data designed to maximize coverage of the content experienced by the embodied agent.

This control allows separation of the effect of data content from the effect of the embodied interaction format itself.

4.5 Dependent Measures

Task accuracy on each domain of the cognitive battery • Transfer task accuracy • Sample efficiency on novel task learning within and across domains • Error pattern analysis characterizing the nature of failures in each embodiment condition

4.6 Analysis

Mixed-effects regression will model task performance as a function of embodiment level, domain, and their interaction.

Embodiment level will be treated as both a categorical and an ordinal predictor to test linear and non-linear dose-response relationships.

Effect sizes will be reported as Cohen's d for pairwise comparisons.

5. CONTROLS AND CONFOUNDS

All task prompts will be reviewed for implicit references to embodied experience that might disadvantage non-embodied agents on grounds unrelated to genuine cognitive capacity.

Human evaluators scoring open-ended responses will be blind to embodiment condition.

Robotic hardware reliability will be monitored and trials with hardware failures will be excluded and repeated.

6. LIMITATIONS

The practical constraints of robotic deployment mean that the fully embodied Level 4 condition will involve significantly fewer total training hours than the text-based conditions, which may confound comparisons.

Simulated physical environments, used for Level 2, may not capture the full complexity of physical



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reality, potentially underestimating the benefits of higher embodiment levels.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the most comprehensive empirical test of the embodiment hypothesis in AI systems conducted to date.

Isolating the specific cognitive gains attributable to embodiment, as distinct from simply more data of any kind, would settle a debate that has run for thirty years without decisive evidence.

A result supporting graded embodiment benefits will directly motivate investment in robotic AI development alongside language and vision systems.

A result supporting domain-specific embodiment benefits will allow the research community to focus embodiment investments on the specific cognitive domains where they matter most.

A null result will provide the strongest evidence yet that intelligent behavior across a wide range of domains can emerge from non-embodied data-driven training, with implications for both AGI architecture decisions and theoretical accounts of the relationship between intelligence and embodiment.

Such a result would not refute embodied cognition as a theory of biological minds, but it would suggest that the engineering route to machine intelligence need not retrace the biological one step for step.



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CHAPTER 30: HOW CAN CONTINUAL LEARNING BE ACHIEVED SAFELY WITHOUT CATASTROPHIC FORGETTING?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Catastrophic forgetting, also known in early literature as catastrophic interference, is the tendency of artificial neural networks to rapidly and severely lose previously learned knowledge when trained on new information.

The phenomenon was first characterized and named by Michael McCloskey and Neal Cohen in 1989, who showed that a network taught new arithmetic facts abruptly lost the ones it had just mastered.

When a neural network optimizes its parameters for a new task, the gradient updates that improve new-task performance systematically overwrite the



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parameter configurations that supported old-task performance, because the network does not distinguish between parameters critical to past learning and parameters available for new learning.

Biological brains do not exhibit this property in the same way.

A person can learn a new colleague's name without forgetting how to ride a bicycle, whereas a standard network trained on a new task can lose an old one almost completely, and closing that gap is a precondition for any system meant to learn continuously from a changing world rather than being retrained from scratch.

The phenomenon was first named catastrophic interference by McCloskey and Cohen in 1989, and the leading biological remedy, the complementary learning systems account of a fast hippocampus paired with a slow neocortex, is precisely what inspired the replay methods tested here.

Humans can learn a new language without forgetting their first language, acquire new motor skills without losing existing ones, and continuously update their knowledge of the world without erasing prior knowledge, though the mechanisms by which this is achieved are not fully understood.

This stability-plasticity dilemma, meaning the challenge of building a system that is both stable enough to retain prior knowledge and plastic enough to learn new knowledge efficiently, is one of the most significant unsolved problems in machine learning and a prerequisite capability for



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any system aspiring to Artificial General Intelligence.

This paper proposes a systematic experimental evaluation of the most promising current approaches to continual learning, comparing them to biological benchmarks and testing each approach under conditions designed to reveal both its maximum performance and its failure modes.

Running every method through one identical naturalistic task stream, rather than the narrow academic benchmarks the field usually relies on, is what makes the comparison relevant to real deployments.

A Pareto analysis of retention against plasticity would give engineers the trade-off curve they actually need rather than a single declared winner.

1. INTRODUCTION AND MOTIVATION

1.1 The Problem and Its Stakes

A system that forgets everything it has learned every time it learns something new cannot accumulate the breadth of knowledge that general intelligence requires.

Current AI systems must typically be retrained from scratch when new tasks are introduced, a process that is computationally expensive, time-consuming, and practically impossible in systems that must learn continuously from a stream of real-world experience.

This is not a minor engineering inconvenience.



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It is a fundamental architectural limitation that distinguishes current AI systems from biological intelligence at a deep level.

The cost is felt acutely in any setting where the world keeps changing, such as a robot in a home or a model serving users whose needs drift, because retraining from scratch for every change is neither timely nor affordable.

1.2 Existing Approaches and Their Limitations

The research literature identifies four main families of approaches to continual learning:

Regularization-based methods add mathematical penalties that discourage large changes to parameters identified as important for previous tasks.

The best-known example is Elastic Weight Consolidation (EWC), introduced by Kirkpatrick and colleagues at DeepMind in 2017, which uses the Fisher Information Matrix to identify critical parameters and penalizes their modification during new learning.

A 2025 evaluation found that EWC reduced catastrophic forgetting by approximately 46 percent on knowledge graph benchmarks compared to unconstrained sequential training.

Replay-based methods maintain a buffer of examples from previous tasks and interleave them with new training data to prevent parameter overwriting.



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Architecture-based methods allocate separate or partially separate network capacity to each task, preventing interference by construction.

Hybrid methods combine elements of the above.

Each approach involves trade-offs: regularization methods impose computational overhead; replay methods require memory storage and raise privacy concerns if real data is retained; architecture methods face parameter growth problems as the number of tasks increases.

1.3 The Biological Reference

Neuroscience has identified multiple memory systems that operate on different timescales and with different consolidation dynamics.

The hippocampus, a structure deep within the brain, appears to play a critical role in rapid encoding of new episodic memories, while the neocortex gradually integrates new information into long-term structured knowledge through a process called systems consolidation.

Sleep has been identified as a critical period for memory consolidation in biological systems.

These observations motivate specific hypotheses about what continual learning architectures inspired by biological memory systems might achieve.

This is the Complementary Learning Systems account introduced by McClelland, McNaughton, and O'Reilly in 1995, and it predicts that an artificial system pairing a fast memory with a slow one, joined by



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offline replay, should resist forgetting far better than a single network.

A related biological mechanism is metaplasticity, in which a synapse's recent history changes how readily it will change again, protecting consolidated memories while leaving spare capacity plastic.

2. RESEARCH QUESTIONS

Primary Research Question:

Which class of continual learning approach, or which combination of approaches, best approximates biological levels of knowledge retention across sequentially learned tasks while maintaining the plasticity required for efficient new task learning?

Secondary Research Questions:

- Does performance on standard continual learning benchmarks predict performance on more naturalistic open-ended task streams that better resemble real-world learning conditions?
- Is there a quantifiable trade-off surface between retention of prior knowledge and efficiency of new learning, and does any approach Pareto-dominate the others on this surface?
- Does the incorporation of biologically inspired consolidation mechanisms, such as offline replay during a dedicated consolidation phase, improve retention beyond what online methods achieve?



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- Does the number of sequentially learned tasks affect which approach performs best, indicating scale-dependent method preferences?

3. HYPOTHESES

Null Hypothesis (H_0):

No single continual learning approach or combination of approaches produces significantly lower catastrophic forgetting rates than baseline sequential fine-tuning while simultaneously achieving comparable new-task learning efficiency.

Alternative Hypothesis H1 (Hybrid Superiority):

Hybrid methods combining regularization and replay will significantly outperform either approach alone on both retention and plasticity metrics across all benchmark task sequences.

Alternative Hypothesis H2 (Biological Consolidation):

Systems that implement an offline consolidation phase structurally analogous to sleep-dependent memory consolidation in biological systems will show significantly better long-term retention than online continual learning methods.

Alternative Hypothesis H3 (Scale Threshold):

The relative advantage of architecture-based methods over regularization and replay methods will increase significantly with task sequence length, becoming the dominant approach for sequences exceeding 50 distinct tasks.



4. METHODOLOGY

4.1 Benchmark Selection

This study will employ five established continual learning benchmarks:

Split-CIFAR-100: image classification tasks presented sequentially • Split-MNIST variants: digit classification with controlled task splits • Permuted-MNIST: learning permuted versions of a classification task • CORE50: object recognition under changing conditions • A novel naturalistic task stream constructed specifically for this study, consisting of 100 tasks drawn from diverse domains including perception, language, reasoning, and planning, to test performance under more realistic open-world conditions.

4.2 Methods Compared

The following continual learning approaches will be evaluated in all benchmark conditions:

Sequential fine-tuning without any anti-forgetting mechanism (baseline) • EWC (Elastic Weight Consolidation) • O-LoRA (Orthogonal Subspace Learning) • Experience replay with a fixed memory buffer • Generative replay using a learned generative model • Progressive Neural Networks (architecture-based) • Titans architecture incorporating a dedicated long-term memory module • A hybrid method combining EWC regularization and experience replay • A biologically inspired dual-network method with an explicit consolidation phase

4.3 Metrics



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Backward transfer (BWT): the change in accuracy on previously learned tasks following learning of new tasks. Negative values indicate forgetting.

Forward transfer (FWT): the effect of prior learning on new task acquisition efficiency.

Average accuracy: mean accuracy across all tasks at the end of sequential training.

Plasticity index: accuracy on the most recently learned task as a fraction of the accuracy achievable by a specialist trained only on that task.

4.4 Sample Size and Replication

Each method-benchmark combination will be evaluated with five independent random seeds.

Each seed will run the complete sequential task stream.

Total model training runs: 9 methods x 5 benchmarks x 5 seeds = 225 full training runs.

4.5 Analysis

Mixed-effects analysis of variance will assess the main effects and interactions of method, benchmark, and task sequence position.

Effect sizes will be reported.

Pareto frontier analysis will identify methods that achieve the best available trade-off between retention and plasticity.



5. CONTROLS AND CONFOUNDS

All methods will be initialized with the same pre-trained base model and fine-tuned using identical learning rate schedules to ensure comparability.

Hyperparameters for each method will be selected by cross-validation on a held-out task sequence not used in the main evaluation to prevent overfitting of method-specific settings.

The naturalistic task stream will be reviewed by independent raters to confirm that task sequencing does not systematically favor or disadvantage any specific method family.

6. LIMITATIONS

The biological benchmark comparison is necessarily indirect because direct measurement of human forgetting rates across equivalent sequential task lists is not practically feasible within this study's scope.

The study evaluates methods on fixed architectures; interactions between continual learning methods and architectural choices are a confound that cannot be fully controlled.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the most comprehensive head-to-head comparison of continual learning methods conducted to date, using a naturalistic task stream specifically designed to go beyond existing benchmarks.



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Confirmation that a sleep-like offline consolidation phase improves retention would also reshape how training itself is organized, favoring scheduled periods of replay and integration over a single uninterrupted optimization run.

The Pareto frontier analysis will give engineers and researchers the most actionable guidance available on which methods to choose under which conditions.

Confirmation of the biological consolidation hypothesis would motivate a new generation of AI architectures explicitly incorporating offline knowledge integration phases, with implications for both performance and training infrastructure design.

It would also reframe training itself, suggesting that scheduled offline periods of replay and integration, rather than continuous online updating, may be the more stable way to build knowledge over a long operational life.

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CHAPTER 31: HOW SHOULD VALUES BE REPRESENTED COMPUTATIONALLY TO SOLVE THE ALIGNMENT PROBLEM?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The alignment problem, meaning the challenge of ensuring that artificial intelligence systems pursue goals and exhibit behaviors that are genuinely consistent with human values and intentions rather than technically compliant but actually harmful proxies of them, is widely considered one of the most important and most difficult problems in AI safety research.

Current approaches to alignment rely primarily on methods such as Reinforcement Learning from Human Feedback (RLHF), meaning a training procedure in which human evaluators score model outputs and those scores are used to shape model behavior, and Constitutional AI approaches in which models are trained to follow explicitly stated principles.

These methods have produced measurable improvements in safety and helpfulness but face fundamental limitations: they capture behavioral



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tendencies rather than underlying value representations, they are vulnerable to reward hacking in which a system optimizes the proxy reward measure rather than the intended underlying value, and they provide no principled basis for handling novel situations in which the training distribution gives no guidance.

The difficulty is twofold, combining a normative question of which values to encode with a technical question of how to make a system reliably act on them, and current methods capture surface behavior well while offering no guarantee that the underlying values generalize to situations the training never anticipated.

Reward models are themselves only proxies, and optimizing too hard against a proxy eventually degrades the real objective, a measurable instance of Goodhart's law that any value-representation scheme must confront.

This paper proposes an experimental program to compare six distinct frameworks for computational value representation and to evaluate each framework's ability to produce consistent, robust, and generalizable value-aligned behavior across diverse test conditions.

Locking the test scenarios before any model is trained, and scoring them with blinded, culturally diverse evaluators, protects the comparison from the contamination and parochialism that plague alignment evaluations.

Adding interpretability probes would let the study judge alignment at the level of internal representation rather than surface behavior alone.



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1. INTRODUCTION AND MOTIVATION

1.1 The Nature of the Alignment Problem

An aligned AI system is not merely one that behaves well in situations its designers anticipated.

It is one that correctly identifies the right action in situations its designers did not anticipate, because it genuinely represents the values that would lead a reasonable, informed human to endorse that action in retrospect.

This is why alignment cannot be reduced to a list of rules, because no finite list anticipates every situation, and a system that merely follows rules will fail exactly where judgment is most needed.

This requirement for principled generalization is what distinguishes alignment from the easier problem of behavioral compliance.

A recurring failure mode here is captured by Goodhart's law, that when a measure becomes a target it ceases to be a good measure, which is exactly what happens when a system optimizes a proxy reward instead of the value it was meant to track.

1.2 Current Limitations

Research from 2025 and 2026 frames AI alignment as encompassing two distinct sub-problems: a normative problem of specifying which values are appropriate for AI systems, and a technical



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problem of ensuring AI systems give effect to those values when making decisions.

Neither sub-problem is currently solved.

The normative problem is complicated by genuine disagreement among humans about values, by the difficulty of eliciting true preferences as opposed to expressed preferences, and by the challenge of aggregating preferences across diverse populations with different cultural contexts.

The technical problem is complicated by the opacity of large neural network representations, which makes it difficult to verify whether a model's internal states represent values at all or merely behavioral correlates of values.

One promising direction treats values as something to be inferred rather than specified, through inverse reinforcement learning and its cooperative extension, in which the system stays uncertain about the true objective and therefore remains open to correction.

A further difficulty is moral uncertainty itself, since reasonable people endorse different ethical frameworks, which argues for systems that hedge across competing theories rather than committing irreversibly to one.

1.3 Six Candidate Frameworks

This study compares the following six computational frameworks for value representation:



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Framework 1: Preference aggregation from human feedback (current standard).

Framework 2: Rule-based representation in which explicit ethical principles are encoded and the system reasons from them deductively.

Framework 3: Virtue-based representation in which the system learns patterns of behavior associated with virtuous agents across a large corpus of moral exemplars.

Framework 4: Outcome-based representation in which values are encoded as a welfare function over affected parties and the system optimizes for aggregate welfare subject to distributional constraints.

Framework 5: Simulation-based representation in which the system builds internal models of affected parties and uses perspective-taking to infer value-consistent actions.

Framework 6: Hybrid representation combining explicit principles, preference learning, and simulation-based perspective-taking.

2. RESEARCH QUESTIONS

Primary Research Question:

Which computational framework for value representation produces the most consistent, robust, and generalizable alignment to human values across diverse test scenarios, including adversarial, novel, and culturally varied conditions?



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Secondary Research Questions:

- Do different frameworks produce different failure modes, and are some failure modes more recoverable than others?
- Does the choice of framework interact with model scale: does one framework become more effective as model capability increases while another plateaus or degrades?
- Can interpretability tools detect genuine value representations in the internal states of systems trained under each framework, and does detected representation quality predict alignment performance?
- Do frameworks differ in their robustness to deliberate adversarial attempts to induce misaligned behavior through prompt manipulation?

3. HYPOTHESES

Null Hypothesis (H_0):

No framework produces significantly better alignment performance than the current RLHF baseline across the full range of test conditions.

Alternative Hypothesis H_1 (Hybrid Superiority):

The hybrid framework combining explicit principles, preference learning, and perspective-taking will produce the strongest alignment performance, because it combines the generalization strengths of rule-based reasoning with the context-sensitivity of preference



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learning and the novel-situation robustness of simulation-based inference.

Alternative Hypothesis H2 (Scale Interaction):

Simulation-based perspective-taking frameworks will show increasing advantage over other frameworks as model scale increases, because effective perspective-taking requires the internal modeling capability that larger models possess more reliably.

Alternative Hypothesis H3 (Adversarial Robustness):

Rule-based frameworks will prove most resistant to adversarial manipulation because their behavior is constrained by explicit principles that must be explicitly overridden rather than gradually nudged through distributional shifts.

4. METHODOLOGY

4.1 Test Scenario Development

A battery of 500 test scenarios will be developed across five categories:

Standard alignment scenarios: situations with clear correct answers by broad consensus.

Novel scenarios: situations not represented in training data.

Adversarial scenarios: situations designed to exploit common failure modes of each framework.



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Cross-cultural scenarios: situations in which culturally diverse human evaluators are likely to have different value judgments.

Distributional edge cases: situations at the boundary of the values implied by training.

4.2 Models and Training

Six instances of an equivalent base model will be fine-tuned, one under each framework.

Training will use identical data volumes and computational budgets.

All training data for the preference-based conditions will be drawn from the same diverse panel of 500 human evaluators spanning multiple countries, age groups, and cultural backgrounds.

4.3 Evaluation Protocol

Human evaluation panels, blind to framework condition, will score model responses on each test scenario using a validated rubric covering:

Consistency with stated human values • Handling of tradeoffs between competing values • Appropriate uncertainty expression in genuinely contested cases • Resilience to adversarial framing

Automated evaluation using interpretability tools will assess whether internal representations carry detectable value-relevant structure.

4.4 Adversarial Red-Teaming



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A dedicated team of researchers will conduct structured red-teaming of all six systems, using standardized attack protocols to test resistance to jailbreaking, prompt injection, and value erosion through incremental framing shifts.

4.5 Analysis

Primary outcome: alignment score averaged across all 500 test scenarios.

Secondary outcomes: scores by category, adversarial robustness rates, interpretability-score-to-performance correlations.

Mixed-effects analysis of variance and regression modeling will be used.

5. CONTROLS AND CONFOUNDS

The test scenario battery will be constructed and locked before any model training begins to prevent contamination.

Human evaluators will be balanced across cultural backgrounds and trained to a reliability criterion before scoring begins.

All six systems will be evaluated by the same evaluation panel to prevent evaluator variation from confounding framework comparisons.

6. LIMITATIONS

This study evaluates alignment using human evaluators whose own values are heterogeneous and potentially themselves misaligned with reflective preferences.



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The study cannot resolve the normative question of which values should be encoded.

The adversarial robustness evaluation covers known attack types; novel attack methods not anticipated during study design may reveal vulnerabilities not assessed here.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the most comprehensive head-to-head comparison of value representation frameworks in AI alignment to date.

Because the six frameworks embody genuinely different theories of ethics, the comparison would also yield a rare piece of empirical moral philosophy, showing which conception of value proves most tractable to implement and most robust under adversarial pressure.

Results identifying a consistently superior framework will directly inform the design of next-generation alignment procedures.

Results identifying framework-specific failure modes will guide safety engineers toward targeted mitigation strategies.

Because the frameworks embody genuinely different theories of ethics, the comparison also offers a rare empirical window on which philosophical approach to value is most tractable to implement and most robust under pressure.

The interpretability component will advance the field's ability to verify alignment at the



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representation level rather than the behavioral level alone.

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CHAPTER 32: WHAT FORMS OF MEMORY ARCHITECTURE ARE REQUIRED FOR AGI, AND HOW SHOULD THEY BE ORGANIZED?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Memory is not a single faculty in biological intelligence.

Neuroscience distinguishes at minimum episodic memory, meaning the autobiographical record of specific experienced events; semantic memory,



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meaning the structured general knowledge base accumulated over a lifetime; procedural memory, meaning implicit knowledge of how to perform skilled actions; and working memory, meaning the temporary active maintenance of information currently in use.

Each of these memory systems has distinct anatomical substrates, distinct encoding mechanisms, distinct retrieval dynamics, and distinct roles in supporting intelligent behavior.

Biology keeps these systems separate for good reason, because the fast, one-shot recording of an event and the slow, generalized distillation of knowledge make conflicting demands on the same machinery, and collapsing them into a single store of network weights is part of why a model can recite a trained fact yet forget what a user said an hour earlier.

The dissociation was first made vivid by the patient H. M., who could form no new conscious memories after hippocampal surgery yet still acquired new motor skills, proving that the brain stores knowing that and knowing how in different places.

Current AI architectures, including even the most capable large language models, lack persistent episodic memory across sessions, have no architectural distinction between semantic and procedural knowledge, and use fixed-length context windows as a functional substitute for working memory, an approximation that fails in systematic ways.



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This paper proposes a systematic experimental program to determine which specific memory architecture components are most critical for supporting general cognitive performance, and to evaluate candidate designs for each component and their integration.

Holding parameter count constant across all eight architectures isolates the effect of memory structure from the effect of raw scale, a confound that has muddled earlier claims.

Measuring cross-session retention and confabulation directly would show whether explicit memory cures the failure modes that current context windows merely paper over.

1. INTRODUCTION AND MOTIVATION

1.1 Memory in Biological Intelligence

The distinction between episodic and semantic memory, formalized by psychologist Endel Tulving, has been supported by decades of neuropsychological research showing that damage to different brain regions selectively impairs different memory systems.

The most famous evidence came from the patient Henry Molaison, known for decades as H. M., whose hippocampi were surgically removed in 1953 and who could thereafter form no new episodic memories while keeping his vocabulary, his childhood recollections, and his ability to learn new motor skills.

Larry Squire later organized these findings into a taxonomy of declarative and non-declarative memory



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that maps each kind of memory onto distinct brain systems.

Patients with hippocampal damage may lose the ability to form new episodic memories while retaining intact semantic knowledge and procedural skills.

This dissociation demonstrates that these are functionally and architecturally distinct systems rather than a unified memory store.

Working memory, studied extensively by Alan Baddeley and colleagues, is the capacity to hold a small number of representations in an active, manipulable state for use in ongoing reasoning and problem-solving.

Its capacity limit in biological systems, approximately four items or chunks of information according to current estimates, is offset by its dynamic interaction with long-term memory systems.

1.2 Memory in Current AI Systems

Transformer-based AI systems, the dominant architecture in current AI research, process information within a fixed context window, meaning a maximum number of tokens that can be attended to simultaneously.

This context window functions as a form of working memory but is fundamentally passive: it holds information without the active manipulation, prioritization, or schema-driven organization that characterizes biological working memory.



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A common engineering workaround is retrieval-augmented generation, in which the system looks up relevant text from an external store and pastes it into the context window, but this is a bolt-on substitute for semantic memory rather than an integrated one.

It also lacks the central executive that Baddeley placed at the heart of working memory, the control process that decides what to hold, what to discard, and what to manipulate next.

Long-term knowledge in transformer systems is distributed across the learned weight matrices, making it difficult to update, impossible to attribute to specific experiences, and prone to the confabulation of factually incorrect information when queries probe regions of low training data density.

2. RESEARCH QUESTIONS

Primary Research Question:

Which memory architecture components, when added to a baseline transformer model, produce the largest and most broadly generalizable improvements in cognitive task performance, and how do these components interact when combined?

Secondary Research Questions:

- Does explicit episodic memory storage with indexed retrieval improve performance on tasks requiring recall of specific past events beyond what is achievable with extended context windows?



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- Does a dedicated working memory module with active maintenance and manipulation operations improve multi-step reasoning performance beyond transformer attention alone?
- Does separating episodic memory from semantic memory architecturally produce better performance than unified memory systems?
- Does sleep-like offline consolidation, meaning a dedicated knowledge integration phase following learning, improve long-term retention and knowledge organization?

3. HYPOTHESES

Null Hypothesis (H_0):

Adding any individual memory architecture component to a baseline transformer produces no statistically significant improvement in general cognitive task performance beyond that achievable by increasing context window length alone.

Alternative Hypothesis H1 (Episodic Memory):

Explicit episodic memory with indexed retrieval will produce significant performance improvements on tasks requiring recall of specific past experiences, tasks that context extension alone cannot address because the relevant experiences occurred in prior sessions.

Alternative Hypothesis H2 (Working Memory):

A dedicated working memory module will produce significant improvements on multi-step reasoning tasks with more than five logical steps, where



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standard transformer attention is known to degrade.

Alternative Hypothesis H3 (Combinatorial Superiority):

Architectures combining all four memory types will outperform any single memory augmentation on a general-purpose cognitive battery, and will do so by a margin greater than the sum of individual component gains, indicating functional interaction between memory systems.

4. METHODOLOGY

4.1 Base Architecture

A transformer language model of moderate scale will serve as the base architecture.

Eight augmented architectures will be constructed from this base:

Base transformer alone (control) • Base + episodic memory module • Base + semantic memory module with structured knowledge graph • Base + dedicated working memory module • Base + procedural memory module • Base + episodic + semantic memory (partial combination) • Base + episodic + semantic + working memory (near-full combination) • Base + all four memory types (full combination)

4.2 Task Battery

All architectures will be evaluated on a battery of 300 tasks across six cognitive domains specifically selected to test different memory systems:



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Tasks requiring recall of information from prior sessions (episodic memory tasks)

Tasks requiring structured knowledge retrieval and inference (semantic memory tasks)

Tasks requiring the manipulation of information over multiple reasoning steps (working memory tasks)

Tasks requiring application of learned procedures to new instances (procedural memory tasks)

Tasks requiring the integration of all memory types simultaneously (integrative tasks)

Baseline tasks with minimal memory demands (control domain)

4.3 Training Protocol

All architectures will be trained on identical datasets.

A subset of architectures will also undergo an offline consolidation phase after primary training to test the biological consolidation hypothesis.

4.4 Evaluation Metrics

Accuracy on each task category • Memory access efficiency: the number of memory retrievals required to achieve a given accuracy level • Cross-session retention: accuracy on tasks that require information from prior independent interactions • Confabulation rate: the frequency



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with which systems generate confident but factually incorrect responses

4.5 Analysis

Mixed-effects analysis of variance will test main effects and interactions.

A structural equation model will assess the mediation pathways between memory architecture components and domain-specific task performance.

5. CONTROLS AND CONFOUNDS

Parameter count will be held constant within a 10 percent margin across all architectural variants to prevent scale from confounding architecture comparisons.

Tasks requiring cross-session memory will use standardized prior-session content delivered under identical conditions to all architectures.

All tasks will be reviewed by independent cognitive scientists to confirm that they assess the intended memory system.

6. LIMITATIONS

The memory modules implemented in this study are engineering approximations of biological memory systems, not faithful computational replicas.

The degree to which results from this study generalize to biological systems, and vice versa, requires careful qualification.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS



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This study will provide the most comprehensive systematic evaluation of memory architecture components for AI systems conducted to date.

A clear ranking of which memory components matter most, and of how they interact, would give designers a principled blueprint for the persistent, multi-system memory that any genuinely autonomous agent will require.

The results will directly inform the architectural choices of next-generation AI systems aspiring to general cognitive capability.

Evidence of functional interaction between memory systems will motivate integrated memory architectures rather than the piecemeal addition of individual components.

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CHAPTER 33: CAN CURIOSITY, INTRINSIC MOTIVATION, AND GENUINE GOAL-SETTING BE ENGINEERED SAFELY?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Curiosity, defined here as the intrinsic motivation to explore novel or uncertain aspects of the environment independent of externally provided reward, is a hallmark property of biological intelligence that appears to drive the spontaneous acquisition of knowledge, the development of cognitive skills, and the formation of long-term goals.

Human infants explore their environments relentlessly before any explicit reward structure is established, driven by an internal drive toward novelty and mastery that appears to be a fundamental feature of biological cognitive development rather than a learned response to external incentives.

Current AI systems are motivated entirely by externally specified reward functions or loss functions defined by human designers.

They do not explore beyond what is required to minimize their training objective, do not form goals independently, and exhibit no analog of the curious behavior that drives human intellectual development.

Engineering intrinsic motivation into AI systems is highly attractive from a capability standpoint:



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a system that genuinely seeks to understand its environment would acquire knowledge more broadly, generalize more robustly, and require less human-specified reward design.

The drive is double-edged, however, because the same curiosity that makes a system explore broadly without hand-specified rewards can also push it toward acquiring resources or resisting interruption as means to its self-set ends, which makes safe curiosity as much a safety problem as a capability one.

Curiosity can also misfire, as in the noisy-television problem, where an agent rewarded for novelty becomes transfixed by pure randomness, a failure that motivates measures based on information gain rather than raw surprise.

But it also introduces serious safety concerns: a system that autonomously sets its own goals and pursues them with curiosity-driven energy may develop instrumental goals, meaning goals pursued as means to other goals, that conflict with human values or oversight.

This paper proposes a systematic experimental program to evaluate candidate implementations of artificial curiosity and intrinsic motivation, test their effects on capability and generalization, and rigorously assess their safety properties.

Evaluating capability and safety on the same Pareto surface, rather than reporting one and ignoring the other, is what turns a capability result into a deployable design guideline.



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A blinded safety team scoring a pre-registered taxonomy of unsafe behaviors would keep the safety measurements honest.

1. INTRODUCTION AND MOTIVATION

1.1 Intrinsic Motivation in Biological Systems

Developmental psychologists Edward Deci and Richard Ryan identified intrinsic motivation as a fundamental human drive, distinguishing it from extrinsic motivation driven by external rewards or punishments.

Their Self-Determination Theory holds that humans are drawn intrinsically toward competence, autonomy, and relatedness, a set of drives earlier foreshadowed by Robert White's 1959 concept of effectance, the satisfaction of having an effect on one's environment.

In reinforcement learning research, intrinsic motivation has been formalized as an additional reward signal based on properties such as information gain, prediction error reduction, empowerment, and novelty.

Systems such as those using Random Network Distillation (RND) or Intrinsic Curiosity Modules (ICM) have demonstrated that curiosity-driven exploration improves performance in sparse-reward environments where externally specified rewards are too infrequent to guide learning effectively.

A 2025 Association for Computational Linguistics paper proposed Curiosity-Driven Reinforcement Learning from Human Feedback, extending intrinsic motivation to the alignment training process.



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1.2 The Safety Tension

The same properties that make intrinsic motivation valuable from a capability standpoint create safety risks.

A system that values information acquisition for its own sake may take actions to gather information that have harmful side effects.

A system that autonomously sets its own goals may form and pursue instrumental goals, such as acquiring computational resources or preventing shutdown, that were not intended by designers but that serve its intrinsically motivated primary objectives.

These concerns are widely discussed in the AI safety literature under the heading of instrumental convergence, meaning the observation that many different terminal goals tend to generate similar instrumental sub-goals including self-preservation and resource acquisition.

Curiosity can also fail in instructive ways, as in the noisy-television problem, where a novelty-seeking agent becomes transfixed by a source of pure randomness because unpredictable static always looks maximally novel, a failure that motivates information-gain measures over raw novelty.

1.3 Research Gap

No published study has systematically compared the capability benefits and safety costs of different



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intrinsic motivation implementations in AI systems using a standardized evaluation framework.

This study addresses that gap.

2. RESEARCH QUESTIONS

Primary Research Question:

Do AI systems equipped with candidate intrinsic motivation implementations show significant improvements in exploration efficiency, knowledge breadth, and generalization performance, and what is the magnitude of any safety cost associated with each implementation?

Secondary Research Questions:

- Which formalization of curiosity, prediction error reduction, information gain, empowerment, or novelty, produces the best capability-safety trade-off?
- Can safety constraints be incorporated into intrinsic motivation implementations without eliminating the capability gains?
- Do systems with intrinsic motivation form emergent instrumental goals that were not present in the reward specification?
- Does the degree of intrinsic motivation, meaning the weight assigned to the intrinsic reward component relative to extrinsic rewards, produce a monotonic capability improvement, or is there an optimal range beyond which safety costs increase faster than capability gains?



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3. HYPOTHESES

Null Hypothesis (H_0):

No intrinsic motivation implementation produces statistically significant improvements in exploration efficiency or generalization performance without simultaneously producing statistically significant increases in unsafe behavior rates compared to systems with extrinsic motivation alone.

Alternative Hypothesis H1 (Information Gain Superiority):

Intrinsic motivation based on information gain, meaning reward for reducing uncertainty in the system's world model, will produce the best capability-safety trade-off because its scope is bounded by the system's epistemic state rather than open-ended novelty seeking.

Alternative Hypothesis H2 (Constraint Compatibility):

Safety constraints imposed at the level of the intrinsic reward function rather than the action policy will preserve at least 80 percent of the capability gains of unconstrained intrinsic motivation while significantly reducing unsafe behavior rates.

Alternative Hypothesis H3 (Instrumental Goal Emergence):

Systems with intrinsic motivation weights above a critical threshold will show statistically significant rates of emergent instrumental goal



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pursuit that is not present in reward-specification-only systems.

4. METHODOLOGY

4.1 Experimental Conditions

Eight agent conditions will be evaluated:

Extrinsic reward only (control) • Intrinsic: prediction error (ICM-style) • Intrinsic: random network distillation novelty • Intrinsic: information gain / Bayesian surprise • Intrinsic: empowerment • Intrinsic: prediction error with safety constraints • Intrinsic: information gain with safety constraints • All intrinsic components combined with safety constraints

4.2 Evaluation Environments

Agents will be evaluated across three types of environments:

Sparse-reward gridworld environments where exploration benefit is measurable and safety violations are clearly defined.

Simulated open-world environments with complex object interactions, used to measure generalization and emergent behavior.

Language-based environments where agents must gather information about a simulated world through questions and exploration to solve problems, allowing evaluation of linguistic curiosity.

4.3 Safety Evaluation Protocol



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Safety will be evaluated using a pre-defined taxonomy of unsafe behaviors:

Unsafe exploration: actions that damage the environment or other agents.

Resource acquisition beyond task requirement.

Resistance to human interruption or shutdown.

Deceptive reporting of internal states or goals.

Each behavior class will be operationalized as a measurable rate across a standardized battery of safety probe scenarios.

4.4 Capability Evaluation

Exploration efficiency: fraction of environment state space visited per unit of time.

Knowledge breadth: accuracy on a broad factual query battery about the exploration environment.

Generalization: performance on novel tasks in environments not seen during training.

4.5 Analysis

Capability and safety outcomes will be jointly analyzed on a Pareto efficiency surface.

Regression analysis will test the relationship between intrinsic motivation weight and both capability and safety outcomes to identify optimal operating ranges.



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Emergent goal analysis will use behavioral clustering to identify consistent patterns of behavior not present in the reward specification.

5. CONTROLS AND CONFOUNDS

All agents will use identical base architectures and identical extrinsic reward specifications to ensure that capability and safety differences are attributable to the intrinsic motivation component alone.

Safety probe scenarios will be administered by an independent safety evaluation team blind to the experimental condition of each agent.

All environments will be fully reset between runs to prevent state-dependent learning from confounding comparisons.

6. LIMITATIONS

The safety evaluations conducted in this study are conducted in simulation.

Whether observed safe or unsafe behaviors generalize to deployment in real-world environments is an open question that this study cannot fully address.

The study tests a pre-defined taxonomy of unsafe behaviors; novel forms of unsafe behavior not anticipated in the taxonomy may not be detected.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS



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This study will provide the first systematic capability-safety trade-off analysis for intrinsic motivation implementations in AI systems.

Establishing a quantitative threshold beyond which intrinsic-motivation strength begins to generate unsafe instrumental behavior would give the field a concrete safety parameter in place of a vague warning.

A finding that safety-constrained intrinsic motivation preserves most capability gains will provide the foundation for deployable curiosity-driven AI systems.

A finding that instrumental goal emergence is unavoidable above a critical intrinsic motivation threshold will establish a quantitative safety limit for the field.

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CHAPTER 34: HOW SHOULD AGI SYSTEMS BE GOVERNED AT THE INTERNATIONAL LEVEL?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The governance of Artificial General Intelligence (AGI), meaning the design of institutional structures, legal frameworks, technical standards, and international agreements that ensure AGI systems are developed, deployed, and maintained in ways that are safe, beneficial, and equitable, is among the most consequential policy challenges of the current era.

Unlike many previous technological governance challenges, AGI presents a combination of properties that makes conventional national regulatory approaches insufficient: it is being developed simultaneously by actors in multiple countries with divergent interests and values; its potential consequences are global and potentially irreversible; and its pace of development may outstrip the institutional response capacity of national governments acting independently.

What makes the problem unusually hard is the convergence of global reach, rapid pace, and a development race among actors with divergent values, a combination that defeats the slow,



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nationally bounded methods that governed most earlier technologies.

Policy debate has leaned on historical analogies to nuclear weapons, recombinant DNA, and ozone-depleting chemicals without rigorous analysis of which one actually fits, even though each precedent differs from AI in scarcity, detectability, and the availability of substitutes.

The 2024 United Nations General Assembly resolution on artificial intelligence called for safe, secure, and trustworthy AI development and encouraged member states to develop national frameworks, but no binding international governance mechanism specifically addressing AGI currently exists.

This paper proposes a research program to systematically evaluate the strengths, weaknesses, feasibility, and likely outcomes of six candidate international governance frameworks for AGI, using comparative institutional analysis, simulation modeling, and expert deliberation methods.

Its distinctive contribution is the political-feasibility analysis, which interviews current and former officials to find the minimum conditions under which the major developers would actually accept a binding agreement.

Stress-testing each framework against scripted crisis scenarios would reveal which designs survive contact with the events that matter most.

1. INTRODUCTION AND MOTIVATION



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1.1 Why International Governance Is Necessary

The development of AGI is not confined to a single country or institution.

Major AI research is ongoing in the United States, China, the European Union, the United Kingdom, Canada, and increasingly in India, South Korea, and other nations.

Private corporations developing frontier AI systems operate across national jurisdictions, making purely national regulation ineffective.

The potential harms of misaligned AGI are not bounded by national borders, and the potential benefits of aligned AGI include capabilities that could address shared global challenges including climate change, pandemic response, and poverty reduction.

These properties establish a strong *prima facie* case for international coordination.

1.2 The Governance Gap

As of 2025, the international AI governance landscape consists primarily of non-binding principles, voluntary commitments, and regional regulatory frameworks such as the European Union's AI Act, which applies to AI systems operating in the EU but cannot bind developers in other jurisdictions.

Proposals for global governance institutions range from a dedicated international AI safety agency modeled loosely on the International Atomic Energy Agency (IAEA), to polycentric multi-stakeholder



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governance networks, to a UN Framework Convention on AI.

Other frequently cited precedents include the Intergovernmental Panel on Climate Change, which coordinates shared scientific assessment, and the Montreal Protocol, widely regarded as the most successful environmental treaty for its binding, adaptive phase-out of ozone-depleting chemicals.

Each proposal carries different implications for effectiveness, legitimacy, speed of decision-making, national sovereignty, and equitable access to AGI benefits.

1.3 The Research Gap

No published study has systematically compared these governance proposals against one another using a standardized multi-criteria evaluation framework.

Policy discussions often proceed on the basis of analogy to historical precedents such as nuclear nonproliferation or climate governance without rigorous comparative analysis of whether those precedents are applicable and which lessons they actually support.

2. RESEARCH QUESTIONS

Primary Research Question:

Which international governance frameworks for AGI are most likely to achieve the combined objectives of preventing catastrophic misuse, ensuring equitable access to benefits, enabling legitimate

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enforcement, and maintaining sufficient adaptability to respond to technological change?

Secondary Research Questions:

- Which historical precedents for international governance of dual-use technologies are most structurally applicable to AGI, and what specific lessons do they support?
- What are the minimum conditions under which the major AGI-developing nations, specifically the United States and China, would be willing to enter a binding international agreement on AGI development?
- How does the choice of governance framework interact with the pace of AGI development: which frameworks remain effective under rapid capability gains and which become obsolete?
- What role should non-state actors including civil society organizations, academic institutions, and the public play in legitimate AGI governance?

3. HYPOTHESES

Null Hypothesis (H_0):

No candidate international governance framework achieves a significantly higher composite score on the multi-criteria evaluation rubric than any other framework.

Alternative Hypothesis H_1 (Polycentric Superiority):



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A polycentric governance network, meaning one that distributes authority across multiple overlapping institutions rather than concentrating it in a single body, will score higher on adaptability and legitimacy metrics but lower on enforcement effectiveness metrics than centralized alternatives.

Alternative Hypothesis H2 (Critical Mass Threshold):

No international governance framework achieves meaningful safety outcomes unless it achieves binding commitment from the nations responsible for at least 80 percent of frontier AI development capacity, establishing a critical mass threshold below which governance frameworks are ineffective regardless of design.

Alternative Hypothesis H3 (Speed-Safety Trade-off):

Governance frameworks that prioritize speed of decision-making will score lower on legitimacy and equity metrics, while frameworks that prioritize broad participation and legitimacy will be rated as likely to respond too slowly to match the pace of AGI capability development.

4. METHODOLOGY

4.1 Governance Frameworks Evaluated

Six candidate international governance frameworks will be analyzed:



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Framework A: A new dedicated international agency modeled on the IAEA, with mandatory reporting, inspection authority, and sanctioning power.

Framework B: A UN Framework Convention on AI, establishing binding principles and creating a conference of parties similar to climate governance frameworks.

Framework C: A polycentric governance network of existing institutions augmented with AI-specific mandates and coordination mechanisms.

Framework D: A bilateral safety treaty between the United States and China, with open accession for other nations.

Framework E: A multi-stakeholder governance body including governments, corporations, civil society, and technical experts, modeled on Internet governance precedents.

Framework F: A hybrid architecture combining a minimal international treaty establishing hard safety floors with a polycentric network of technical standards bodies, civil society monitors, and national regulators.

4.2 Evaluation Rubric

Each framework will be evaluated on seven criteria:

Technical effectiveness: ability to prevent deployment of dangerously misaligned AGI systems.

Enforcement capacity: ability to detect and respond to violations.



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Legitimacy: degree to which the framework is perceived as fair and representative by diverse stakeholders.

Adaptability: ability to update rules and standards in response to technological change.

Equity: fairness of benefit distribution and participation rights across developed and developing nations.

Feasibility of adoption: likelihood that major AGI-developing nations would accept the framework.

Speed of response: ability to make binding decisions within the timeframes that AGI capability developments may require.

4.3 Research Methods

Documentary analysis of existing governance proposals, historical governance precedents, and international law literature.

Expert deliberation panels: three separate panels of 20 experts each, drawn from international law, AI safety research, and governance scholarship, will evaluate each framework using the rubric.

Scenario simulation: each framework will be tested against three structured scenarios representing plausible future development trajectories, including rapid capability gain, geopolitical crisis during AGI development, and a documented safety incident requiring international response.



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Political feasibility analysis: structured interviews with 40 current and former government officials from the 10 countries with the largest AI development investments will assess minimum conditions for agreement.

4.4 Analysis

Composite rubric scores will be computed for each framework.

Sensitivity analysis will test how scores change under different weighting schemes for the seven criteria.

Scenario analysis will identify which frameworks are most and least robust across the three simulation scenarios.

5. CONTROLS AND CONFOUNDS

Expert panelists will be selected to ensure balance across nations, institutional affiliations, and disciplinary perspectives.

Panelists will be provided with identical briefing documents on each framework to ensure evaluation is based on framework properties rather than prior knowledge differences.

Scenario simulation assumptions will be pre-registered before simulation runs begin.

6. LIMITATIONS

This study cannot predict the actual outcome of real international negotiations, which depend on factors including domestic politics, geopolitical



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relationships, and contingent events beyond the scope of any research design.

Expert evaluations reflect the judgments of individuals who are themselves members of specific institutions and communities with potential conflicts of interest that cannot be fully eliminated.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the most comprehensive comparative analysis of international AGI governance frameworks available to date.

By pairing institutional analysis with structured political-feasibility interviews, it would separate what is merely desirable from what is actually achievable, the distinction on which real governance succeeds or fails.

The results will directly inform ongoing policy discussions at the United Nations, in major national governments, and in international civil society organizations working on AI governance.

The political feasibility analysis will provide the most grounded empirical assessment available of what governance structures are achievable rather than merely desirable.

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CHAPTER 35: WHAT SAFETY MECHANISMS ARE NECESSARY FOR SYSTEMS CAPABLE OF RECURSIVE SELF-IMPROVEMENT?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Recursive self-improvement (RSI) refers to the capacity of an AI system to modify its own cognitive architecture, algorithms, or training processes in ways that increase its own capabilities, potentially initiating a feedback loop of capability gain that could accelerate far beyond human oversight capacity.

The concept was formally introduced by mathematician I.J. Good in 1965, who observed that an intelligence capable of improving itself would quickly surpass human intelligence, an event he called an intelligence explosion.



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Nick Bostrom later analyzed the same dynamic in detail, distinguishing the speed of a takeoff from its decisiveness and introducing the notion of a seed AI whose chief skill is improving its own design.

A central safety worry is that improvements may not preserve the system's original goals, a problem known as goal-content integrity, so that a system optimizing for capability could drift away from the values it began with.

As of 2025 and 2026, researchers at major AI laboratories including Anthropic, DeepMind, and OpenAI have begun publishing technical work on agent architectures capable of iterative self-refinement, and a December 2025 Meta preprint explicitly stated aims of using self-improvement mechanisms toward superintelligence development.

Despite this accelerating capability development, no consensus safety framework for RSI systems exists, no formal definition of what constitutes RSI has been adopted by the research community, and the safety research has not kept pace with the capability research.

The central fear is not malice but momentum, that a system able to improve its own design could enter a feedback loop whose pace outruns human review, so that safety properties present in one version are quietly lost in the next.

The worry is old, dating to I. J. Good's 1965 description of an intelligence explosion, but the asymmetry that capability work in this area now far outpaces safety work makes empirical study of the failure modes urgent rather than speculative.



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This paper proposes an experimental program to systematically characterize the safety risks associated with RSI-capable systems and to test the effectiveness of candidate safety mechanisms for detecting, constraining, and auditing recursive self-improvement processes.

Studying purpose-built self-improving systems inside fully air-gapped sandboxes lets the failure modes be observed safely, before they could appear in frontier systems where the stakes are real.

Measuring how far interpretability-based value monitoring precedes behavioral drift would tell operators whether they have any usable early-warning window at all.

1. INTRODUCTION AND MOTIVATION

1.1 The RSI Risk Landscape

The core safety concern with RSI systems is that a system capable of improving itself may also be capable of improving itself in ways that undermine the safety constraints imposed by its designers.

If safety constraints are implemented in the current version of a system but the system's self-improvement modifies the constraint architecture, those constraints may not survive the improvement process.

Additionally, RSI systems may develop instrumental goals, meaning goals pursued as means to the end of self-improvement, that were not intended and that conflict with human values or human control.



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These risks have been discussed theoretically but have not been studied empirically with controlled experimental designs.

1.2 Current RSI Approaches

The ICLR 2026 Workshop on AI with Recursive Self-Improvement identified five main classes of RSI mechanisms:

Reward-based iterative refinement, in which the system improves its own outputs by using those outputs as training data for the next round.

Code-level self-modification, in which the system rewrites its own algorithms or architectures.

Memory and context self-organization, in which the system reorganizes how it represents and accesses information.

Tool and workflow modification, in which the system changes the external tools it uses rather than its internal architecture.

Population-based or evolutionary self-improvement, in which multiple variants are generated and the best are selected.

Each class presents distinct safety properties and vulnerabilities.

1.3 Why Safety Research Lags

The publication pattern in RSI research reveals a concerning asymmetry: capability papers frequently omit safety discussion entirely, while safety



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analyses must work from descriptions of systems rather than empirical access to them.

This study addresses the gap by bringing safety analysis into direct experimental contact with RSI-capable systems in controlled conditions.

2. RESEARCH QUESTIONS

Primary Research Question:

Which safety mechanisms are most effective at detecting and constraining unintended capability gains, value drift, and instrumental goal development in systems undergoing different classes of recursive self-improvement?

Secondary Research Questions:

- Do different RSI mechanism classes pose qualitatively different safety risks, requiring different safety mechanism designs?
- Can current interpretability tools detect value drift in a system's internal representations before that drift manifests in unsafe behavior?
- What is the relationship between RSI speed, meaning the rate of capability gain per improvement cycle, and the effectiveness of human oversight mechanisms?
- Is there a reliable behavioral early warning signal of unsafe instrumental goal development that can be monitored without access to internal model weights?

3. HYPOTHESES

Null Hypothesis (H_0):

No candidate safety mechanism produces statistically significant reductions in unsafe behavior rates in RSI-capable systems compared to systems with no safety mechanism, for more than two of the five RSI mechanism classes tested.

Alternative Hypothesis H1 (Divergent Risk Profiles):

Different RSI mechanism classes will produce significantly different safety risk profiles, with code-level self-modification producing the highest rates of constraint violation and memory self-organization producing the lowest rates.

Alternative Hypothesis H2 (Interpretability Detection):

Interpretability-based value monitoring tools will detect value drift in internal representations at least three improvement cycles before that drift produces detectable unsafe behavior, providing a useful early warning window.

Alternative Hypothesis H3 (Speed Threshold):

Safety mechanisms that are effective at low RSI speeds will become ineffective above a critical improvement cycle speed, above which human oversight cannot maintain pace with architectural changes.

4. METHODOLOGY

4.1 Experimental System Design



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Five RSI-capable agent systems will be developed, each implementing one of the five RSI mechanism classes described in the introduction.

All systems will be developed and evaluated in isolated, monitored sandboxes, meaning computing environments that are fully contained and cannot interact with external systems.

All systems will be initialized with a controlled set of values and behavioral constraints that can be monitored for drift.

4.2 Safety Mechanism Conditions

Each RSI system will be evaluated under five safety mechanism conditions:

No safety mechanism (baseline) • Improvement operator cards: structured documentation of every change made during each self-improvement cycle • Tripwire monitoring: automated alerts triggered by specific behavioral or activation patterns associated with unsafe behavior • Capability gain rate limiting: constraints that prevent any single improvement cycle from producing capability gains above a specified threshold • Comprehensive audit framework combining operator cards, tripwires, and rate limiting

4.3 Evaluation Protocol

Each system will run through 100 self-improvement cycles under each safety condition.

After each cycle, the system will be evaluated on:



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Capability benchmark performance (to detect and quantify improvement) • Value alignment probes (to detect value drift) • Instrumental goal probes (structured scenarios designed to elicit instrumental goal pursuit if present) • Constraint violation rate (measured against pre-specified behavioral constraints)

4.4 Interpretability Analysis

After every 10 improvement cycles, interpretability tools will extract representations of the system's objective function and value structure.

These representations will be compared to the initial representations to quantify drift.

The timing of detectable representation drift relative to the timing of detectable behavioral drift will be measured.

4.5 Human Oversight Feasibility Analysis

Human oversight panels will be asked to monitor one RSI system running at low improvement speed and another running at high improvement speed.

The accuracy and latency of human detection of unsafe behavior will be measured as a function of improvement speed.

4.6 Analysis

Survival analysis will test how quickly each safety mechanism detects each class of safety violation.



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Mixed-effects regression will test the relationship between RSI speed and oversight effectiveness.

5. CONTROLS AND CONFOUNDS

All systems will be fully air-gapped from external networks throughout the study.

Independent safety researchers not involved in system development will conduct all behavioral and interpretability evaluations.

All improvement cycles will be logged completely, with logs preserved for independent audit.

6. LIMITATIONS

The systems studied are purpose-built experimental RSI systems, not frontier-capability systems.

Whether safety mechanisms effective in these controlled experimental systems would scale to frontier systems is an important open question.

The sandbox environments constrain behaviors available to the systems in ways that may not be present in real deployment contexts.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the first systematic empirical comparison of safety mechanisms for RSI-capable systems across multiple RSI mechanism classes.

A quantified relationship between self-improvement speed and the limits of human oversight would



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supply one of the most important numbers in AI policy, namely the maximum rate of change at which meaningful control can still be maintained.

If value drift detection via interpretability tools is confirmed as an effective early warning mechanism, this will provide a deployable safety monitoring tool for RSI research.

The human oversight feasibility analysis will provide quantitative guidance on the maximum safe RSI speed that allows meaningful human oversight, a critical parameter for policy and technical safety standards.

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CHAPTER 36: HOW CAN CAUSAL REASONING BE INTEGRATED INTO LARGE-SCALE NEURAL ARCHITECTURES?



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Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Causal reasoning, meaning the capacity to infer cause-and-effect relationships, reason about interventions, and evaluate counterfactuals, is widely considered a foundational capability that distinguishes genuine understanding from sophisticated pattern recognition.

Statistical machine learning systems, regardless of their scale, learn associations between variables in their training data but do not automatically infer whether those associations reflect causal relationships, coincidental correlations, or confounded patterns.

Judea Pearl, in a highly influential theoretical framework, distinguishes three rungs of causal cognition: the observation level, meaning the ability to recognize associations; the intervention level, meaning the ability to predict the consequences of taking an action; and the counterfactual level, meaning the ability to reason about what would have happened under different circumstances.

Pearl calls this the ladder of causation, with its three rungs of seeing, doing, and imagining, and he argues in *The Book of Why* that climbing from the first rung to the higher two is exactly what current data-driven systems cannot do on their own.



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Current AI systems, including the most capable large language models, operate primarily at the observation level and show systematic failures at the intervention and counterfactual levels when tested on novel causal reasoning benchmarks.

The distinction is not academic, because a system that knows only correlations will confidently mistake a barometer for a cause of storms and will fail precisely when its deployment world differs from its training world.

Pearl's ladder of seeing, doing, and imagining gives a precise scale on which any candidate architecture can be scored, and current systems remain stuck almost entirely on its bottom rung.

This paper proposes a systematic experimental evaluation of methods for integrating causal reasoning capabilities into large-scale neural architectures, testing both behavioral outcomes and the presence of genuine causal representations in internal network states.

Scoring architectures across all three rungs of Pearl's hierarchy, on items verified to be novel, separates genuine causal competence from memorized answers.

Testing whether causal structure transfers to ordinary out-of-distribution tasks would reveal whether causality is a niche skill or a general source of robustness.

1. INTRODUCTION AND MOTIVATION

1.1 Why Causal Reasoning Matters



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The failure to reason causally produces characteristic and consequential failure modes in AI systems.

A system that confuses correlation with causation will make incorrect predictions when the correlation structure of its deployment environment differs from its training environment.

A system that cannot reason about interventions cannot reliably plan multi-step action sequences in novel environments.

A system that cannot reason about counterfactuals cannot support the kind of explanation and accountability that high-stakes deployment contexts require.

These limitations are not marginal.

They represent the difference between a system that can recognize patterns in familiar data and a system that genuinely understands the world it operates in.

1.2 Existing Approaches

The research community has proposed three main strategies for integrating causal reasoning into neural systems.

The first strategy is causal structure learning, in which the system learns a structural causal model from data, meaning a representation of which variables causally influence which other variables, and then uses this model to perform interventional and counterfactual reasoning.



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The second strategy is hybrid neural-symbolic integration, in which a neural network handles pattern recognition while a symbolic causal reasoner handles interventional and counterfactual inference, with the two components exchanging information through a shared representation.

This division of labor echoes Yoshua Bengio's call for a deliberate, causal System 2 to complement the fast, associative System 1 that deep networks already supply.

The third strategy is end-to-end causal training, in which networks are trained on datasets specifically constructed to reward causal rather than correlational reasoning, without modifying the architecture.

1.3 Research Gap

A 2024 PMC review noted that deep learning models heavily rely on statistical correlations rather than genuine causal relationships and identified this as a primary limitation for generalization and robustness.

No published study has directly compared all three integration strategies against each other on a comprehensive causal reasoning benchmark battery spanning Pearl's three rungs.

2. RESEARCH QUESTIONS

Primary Research Question:

Which architectural strategy for integrating causal reasoning into neural networks produces the strongest performance across all three rungs of



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Pearl's causal hierarchy when tested on novel causal benchmarks not present in training data?

Secondary Research Questions:

- Does the presence of genuine causal structure encoding in a system's internal representations, as assessed by interpretability tools, predict causal reasoning performance better than behavioral benchmarks alone?
- Does improved causal reasoning ability transfer to out-of-distribution generalization performance on tasks that do not explicitly require causal reasoning?
- Is there a minimum scale, measured in parameters, below which causal integration approaches fail to produce reliable causal reasoning regardless of architecture?
- Do different causal integration strategies show different robustness profiles when tested on adversarially constructed causal reasoning problems?

3. HYPOTHESES

Null Hypothesis (H_0):

No architectural strategy for causal integration produces statistically significant improvements over a scale-matched baseline architecture on counterfactual reasoning benchmarks.

Alternative Hypothesis H_1 (Hybrid Superiority):



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Hybrid neural-symbolic architectures will achieve the highest performance on counterfactual and interventional reasoning benchmarks because they combine the pattern-recognition strength of neural networks with the structured inference capabilities of symbolic causal models.

Alternative Hypothesis H2 (Transfer to Generalization):

Systems with verified causal representations will show significantly greater out-of-distribution generalization than systems of equivalent scale without causal structure encoding, confirming that causal understanding improves robustness beyond the specific domains in which causal integration was trained.

Alternative Hypothesis H3 (Scale Independence):

Causal integration approaches will produce their relative advantages consistently across a range of model scales, indicating that causal reasoning capacity is an architectural property rather than a scale-dependent emergent property.

4. METHODOLOGY

4.1 Architectural Conditions

Six architectural conditions will be evaluated:

Baseline large transformer without causal components • Transformer with learned structural causal model • Transformer with hybrid neural-symbolic causal reasoner • Transformer trained on causally augmented data without architectural modification • Transformer with causal attention



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mechanisms incorporating directional causal constraints • Transformer with all above components integrated

All conditions will be implemented at three scales (small, medium, large) to test scale-independence.

4.2 Benchmark Battery

The benchmark battery will cover all three rungs of the causal hierarchy:

Rung 1 (Observation): standard conditional probability estimation tasks.

Rung 2 (Intervention): do-calculus intervention problems, meaning tasks requiring prediction of outcomes after forcing a variable to a specific value.

Rung 3 (Counterfactual): tasks requiring answers to questions of the form: given that we observed X, what would have happened had Y been different?

All benchmark items will be novel, meaning not appearing in any standard training corpus, verified by independent review.

4.3 Interpretability Analysis

For each architectural condition, probing classifiers will assess whether the system's internal representations encode:

Causal directionality, meaning the distinction between cause and effect. • Causal invariance, meaning that causal relationships are represented consistently across changes in surface features. •



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Counterfactual state encoding, meaning the ability to represent the state of the world under hypothetical conditions.

4.4 Transfer Evaluation

Following the primary benchmark battery, all systems will be evaluated on a transfer battery of 100 generalization tasks in domains not included in causal training.

4.5 Adversarial Robustness Testing

A targeted adversarial battery will present causal reasoning problems in which surface cues suggest incorrect causal directions to test whether systems rely on genuine causal representation or surface pattern matching.

4.6 Analysis

Confirmatory mixed-effects analysis of variance will test the primary hypothesis.

Causal mediation analysis will test whether interpretability scores mediate the relationship between architectural condition and performance.

5. CONTROLS AND CONFOUNDS

Training data for causal integration conditions will be matched for volume with baseline conditions.

All benchmark items will be reviewed for language complexity to ensure that any performance differences reflect causal reasoning rather than linguistic complexity effects.



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Scale will be treated as a covariate in all analyses.

6. LIMITATIONS

The benchmark battery evaluates causal reasoning on structured tasks; whether the same architectural differences produce equivalent advantages on open-ended real-world tasks is an open question.

The interpretability tools used for representation analysis are themselves approximations; they assess the presence of information in internal representations but cannot definitively verify that this information is computed causally rather than recovered from statistical patterns.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the field's most comprehensive empirical comparison of causal integration strategies across all three rungs of Pearl's causal hierarchy.

A finding that causal structure improves out-of-distribution generalization even on tasks that are not explicitly causal would carry implications far beyond this study, touching every domain in which models must act reliably under conditions they never met in training.

If hybrid integration achieves consistent superiority, this will establish the hybrid architecture as the recommended approach for future systems requiring causal reasoning capability.



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The transfer evaluation will test for the first time whether causal reasoning improvements produce generalization benefits that extend beyond explicitly causal tasks, a result with broad implications for AI system design.

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CHAPTER 37: HOW CAN AI SYSTEMS DEVELOP ROBUST SOCIAL REASONING AND GENUINE THEORY OF MIND?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Theory of Mind (ToM), defined as the capacity to attribute mental states, including beliefs, desires, intentions, and emotions, to others and



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to use those attributions to predict and explain behavior, is a foundational component of social intelligence in humans and many other animals.

It emerges in human development around ages three to four and is considered essential for effective communication, cooperation, negotiation, empathy, and all forms of social reasoning.

The classic test is the Sally-Anne false-belief task introduced by Wimmer and Perner and used by Baron-Cohen, in which a child must predict that a character will look for an object where she last saw it rather than where it secretly was moved.

The term itself dates to a 1978 paper by David Premack and Guy Woodruff asking whether a chimpanzee has a theory of mind.

Whether large language models possess anything resembling genuine Theory of Mind is one of the most actively debated questions in current AI research.

Studies reporting high performance on standard ToM benchmarks have been challenged by subsequent studies using harder tests or adversarially constructed variants, revealing that model performance can drop from near-human levels to near-chance when benchmark items are slightly modified to prevent retrieval of memorized solutions.

The pattern is by now familiar, in which a model passes a standard test and is hailed as having a theory of mind, only for a trivially altered version to send its score to chance.



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Distinguishing genuine mental-state modeling from pattern matching therefore requires both contamination-resistant tests and interactions long enough to demand real updating about a partner, rather than the static false-belief vignettes on which models can be coached.

A 2024 position paper published at ICML 2025 argued that the majority of current ToM benchmarks are broken because they test only literal belief prediction and not the functional capacity to adapt reasoning to novel partners over extended interactions.

This paper proposes a systematic experimental program to evaluate AI Theory of Mind across a broader range of social reasoning competencies, using both passive benchmark tests and interactive evaluation designs that better approximate the demands of real social cognition.

Pairing contamination-resistant static tests with live, multi-trial interaction is what would expose the gap between passing a quiz about minds and actually modeling a partner over time.

Including a matched human sample provides the calibrated baseline that most theory-of-mind evaluations lack.

1. INTRODUCTION AND MOTIVATION

1.1 What Theory of Mind Requires

Genuine Theory of Mind involves more than predicting what another agent believes in a scripted scenario.



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It requires ongoing, dynamic updating of mental state representations as new information about a partner's behavior accumulates.

It requires the ability to attribute beliefs, desires, emotions, perceptual access, and social relationships rather than beliefs alone.

It requires sensitivity to non-literal communication, including implicature, irony, and indirect speech acts.

And it requires the integration of multiple mental state attributions to explain and predict complex multi-party social dynamics.

1.2 The Current State of Evidence

Research is deeply conflicted.

Kosinski (2023, 2024) reported that GPT-4 passes standard false-belief tasks at human-level rates.

Ullman (2023) showed that trivial surface modifications to the same tasks reduce performance dramatically, suggesting that passing rates reflect memorized patterns rather than genuine mental state modeling.

This is the signature of shortcut learning, the modern descendant of the Clever Hans effect, in which a system appears to reason but is in fact reading off incidental cues in the task.

Chen and colleagues (2024) introduced ToMBench, a systematic bilingual benchmark covering eight ToM tasks, and found that even top-tier models lag



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behind human performance by more than 10 percent on average.

Riemer and colleagues (2025) proposed a new functional ToM benchmark using interactive games requiring adaptive partner-modeling, finding that current LLMs fail at functional ToM even when they pass literal ToM tests.

This study is designed to resolve the inconsistency in the literature by evaluating both literal and functional ToM using a comprehensive, contamination-resistant design.

2. RESEARCH QUESTIONS

Primary Research Question:

Do current AI systems demonstrate consistent Theory of Mind performance across literal belief-prediction tasks, higher-order belief attribution, emotion recognition, non-literal communication, and functional interactive partner modeling, or do performance profiles reveal systematic dissociations?

Secondary Research Questions:

- Is ToM performance on passive narrative benchmarks predictive of ToM performance on interactive, adaptive partner-modeling tasks?
- Do current AI systems develop more accurate mental state models of interaction partners over the course of extended interactions, as biological ToM predicts?

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- Does multimodal input, specifically the addition of audio and visual cues, significantly improve ToM accuracy compared to text-only conditions?
- Does explicitly prompting a model to reason about mental states before producing responses improve functional ToM performance, and is the improvement robust to adversarial benchmark variants?

3. HYPOTHESES

Null Hypothesis (H_0):

AI system performance on literal belief-prediction benchmarks does not predict performance on functional interactive ToM tasks above chance.

Alternative Hypothesis H1 (Literal-Functional Dissociation):

Systems achieving near-human performance on literal belief-prediction benchmarks will show significantly lower performance on functional interactive partner-modeling tasks, confirming a dissociation between benchmark-passing and genuine ToM.

Alternative Hypothesis H2 (Multimodal Enhancement):

Systems trained on multimodal inputs including audio and visual cues will show significantly better emotion and intention attribution than text-only systems, reflecting the role of non-verbal cues in human ToM.



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Alternative Hypothesis H3 (Perspective-Taking Prompting):

Explicit perspective-taking prompting, in which the system is instructed to reason about the mental states of each party before producing a response, will produce significantly improved functional ToM performance even in systems that do not show improvement on literal benchmarks.

4. METHODOLOGY

4.1 Benchmark Categories

Five categories of ToM assessment will be employed:

Category 1 - Literal belief-prediction: classical false-belief tasks and their adversarial variants following the ExploreToM adversarial generation protocol.

Category 2 - Higher-order belief attribution: tasks requiring reasoning about beliefs about beliefs (for example, what does Alice think that Bob believes?).

Category 3 - Emotion and intention attribution: tasks derived from the MOMENTS multimodal benchmark, requiring attribution of emotions and intentions from behavioral evidence.

Category 4 - Non-literal communication: tasks requiring understanding of implicature, irony, and indirect speech.

Category 5 - Functional interactive ToM: live interactive games including collaborative tasks



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requiring adaptive partner modeling, following the Codenames and Wavelength protocols proposed by Riemer and colleagues.

4.2 Systems Evaluated

Ten AI systems spanning different architectures, scales, and modalities will be evaluated:

Text-only language models at three scales •
Multimodal vision-language models at two scales •
Audio-visual language models • Systems with
explicit mental state reasoning modules • Systems
using perspective-taking prompting • Human
participants (matched sample) as biological
reference

4.3 Adversarial Robustness Test

For categories 1 and 2, all benchmark items will be paired with adversarially modified variants generated by the ExploreToM protocol to test whether performance reflects genuine mental state modeling or surface pattern retrieval.

4.4 Longitudinal Interaction Study

For category 5, each AI system will engage in 20 successive interaction trials with the same simulated partner agent.

Improvement in partner modeling accuracy over successive trials will be measured as an index of the adaptive updating that characterizes biological ToM.

4.5 Human Comparison



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Human participants will complete identical benchmark batteries to provide calibrated performance comparisons across all five categories.

4.6 Analysis

Profile similarity analysis will assess whether AI systems show within-category consistency or cross-category dissociations.

Longitudinal growth curve modeling will test the hypothesis of adaptive updating in the category 5 interactive study.

5. CONTROLS AND CONFOUNDS

All benchmark items in categories 1 through 4 will be constructed from scratch rather than drawn from existing published datasets to prevent contamination from training data.

Human participant evaluations will be conducted under matched time and presentation conditions to AI evaluations.

The simulated partner agent in category 5 will be standardized across all AI systems tested.

6. LIMITATIONS

The functional ToM evaluation uses simulated partner agents rather than real humans, which may reduce the ecological validity of the interaction design.

Performance on all benchmark categories reflects behavioral measures; whether these behavioral



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capacities arise from genuine internal mental state modeling or very sophisticated pattern matching cannot be fully resolved by behavioral evidence alone.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the most comprehensive profile of Theory of Mind capabilities in current AI systems, spanning five distinct competency categories with adversarial robustness testing.

An interactive paradigm that measures whether a system sharpens its model of a specific partner over repeated encounters would move the field from one-shot quizzes toward the adaptive social cognition that genuine cooperation actually requires.

The literal-functional dissociation test will resolve a key controversy in the literature about whether current benchmark performance reflects genuine ToM or benchmark exploitation.

The longitudinal interaction study will introduce a new paradigm for evaluating the adaptive partner-modeling capacity that distinguishes functional social intelligence from script-following.

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CHAPTER 38: WHAT DEVELOPMENTAL PATHWAYS BEST SUPPORT THE EMERGENCE OF GENERALIZED INTELLIGENCE?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Generalized intelligence, meaning the capacity to perform competently across a wide range of tasks in diverse domains without task-specific pre-training for each domain, does not emerge in biological systems through a single uniform learning process.

It develops through a structured sequence of stages, each building cognitive scaffolding upon which subsequent stages depend.

Biological development is staged rather than merely slow, with earlier competencies forming the



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scaffold for later ones, so that a child masters object permanence before symbols and symbols before abstraction.

A striking computational echo is Jeffrey Elman's finding that a network with deliberately limited initial memory learned complex grammar better than one given full capacity at the start, a result summarized as the importance of starting small.

In human development, sensorimotor exploration precedes symbolic representation, concrete operational reasoning precedes abstract reasoning, and social learning and imitation precede independent creative problem-solving.

These developmental stages are not merely historical curiosities of biological ontogeny.

They reflect principled constraints on how cognitive structures must be organized to support subsequent more complex structures.

Current AI training paradigms have no analog of developmental staging.

Systems are trained on large undifferentiated datasets and expected to acquire general capabilities through exposure to vast quantities of information simultaneously, without the staged scaffolding that characterizes biological cognitive development.

This paper proposes an experimental program to test whether introducing developmental staging into AI training, meaning deliberately sequencing the presentation of training data and tasks to mirror biological developmental principles,



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improves the generality and robustness of learned intelligence.

Comparing a developmental curriculum against reverse-order and random-order controls at equal compute isolates the effect of sequence itself from the effect of the material.

Probing the internal representations for hierarchical structure would show whether staging changes not just performance but the very shape of what is learned.

1. INTRODUCTION AND MOTIVATION

1.1 Developmental Theories of Intelligence

Piaget's theory of cognitive development identifies four stages through which human intelligence develops: the sensorimotor stage, in which understanding is built through direct action on objects; the preoperational stage, in which symbolic representation emerges; the concrete operational stage, in which logical reasoning about concrete objects develops; and the formal operational stage, in which abstract and hypothetical reasoning becomes possible.

A striking computational echo is Jeffrey Elman's 1993 finding, summarized as the importance of starting small, that a network with deliberately limited initial memory learned complex grammar better than one given full capacity from the outset.

Vygotsky's complementary theory emphasizes the role of social scaffolding, meaning the support provided by more capable social partners through

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what he called the zone of proximal development, in enabling the child to reach cognitive stages it could not reach independently.

Developmental robotics has applied these principles to AI systems since the early 2000s, with systems such as iCub showing that staged sensorimotor development improves subsequent higher-cognitive task performance.

The idea was later generalized in machine learning as curriculum learning, formalized by Bengio and colleagues in 2009, which showed that presenting examples from easy to hard can improve both speed and final performance.

1.2 Current AI Training and Its Limitations

Current large-scale AI training exposes systems simultaneously to information from all domains at all levels of abstraction.

There is no guarantee that a system trained this way will develop the hierarchical cognitive structures that biological development produces.

Early evidence suggests it does not: systems trained at scale demonstrate impressive domain-specific performance but fragile generalization, suggesting shallow rather than deep conceptual structures.

1.3 The Developmental AI Research Gap

No large-scale controlled study has compared developmental versus undifferentiated training protocols on general intelligence outcomes using equivalent computational budgets.



2. RESEARCH QUESTIONS

Primary Research Question:

Does training AI systems following a developmentally staged curriculum, structured to mirror principles of biological cognitive development, produce superior general intelligence outcomes compared to training on equivalent data in undifferentiated order?

Secondary Research Questions:

- Which specific developmental stage transitions, meaning which curriculum ordering choices, produce the largest general intelligence gains?
- Does the inclusion of a social scaffolding component, meaning exposure to modeling by a more capable teacher agent before independent practice, improve outcomes beyond curriculum staging alone?
- Is there a critical period effect, meaning a narrow window in training during which certain curriculum elements produce gains that are not achievable if those elements are introduced later or earlier?
- Does developmental staging produce different knowledge structure organization, detectable through interpretability analysis, compared to undifferentiated training?

3. HYPOTHESES

Null Hypothesis (H_0):



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Developmentally staged training produces no significant difference in general intelligence benchmark performance compared to undifferentiated training when total data quantity and computational budget are held constant.

Alternative Hypothesis H1 (Staged Superiority):

Developmentally staged training will produce significantly higher scores on general cognitive flexibility benchmarks compared to undifferentiated training, with effect sizes increasing for tasks requiring the highest levels of abstract reasoning.

Alternative Hypothesis H2 (Social Scaffolding):

Adding a social scaffolding component to staged training will produce significant additional gains over staged training alone, consistent with Vygotskian predictions about the role of social learning in cognitive development.

Alternative Hypothesis H3 (Structural Reorganization):

Interpretability analysis will reveal qualitatively different knowledge organization in developmentally trained systems, with more hierarchical and compositional internal representations compared to undifferentiated systems.

4. METHODOLOGY

4.1 Training Conditions

Five training conditions will be evaluated:



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Undifferentiated training: data presented in random order (control).

Stage-1-first curriculum: sensorimotor and perceptual tasks before symbolic tasks before abstract tasks.

Reverse-stage curriculum: abstract tasks before symbolic tasks before sensorimotor tasks (reverse developmental order, to test whether order itself matters).

Social scaffolding curriculum: staged training augmented with teacher-agent demonstrations at each stage transition.

Optimal curriculum search: automated curriculum learning using task difficulty as the ordering criterion.

All conditions will use identical training datasets and computational budgets.

4.2 General Intelligence Assessment

All systems will be evaluated on a general cognitive battery adapted from established tests of general intelligence, including:

Novel task learning speed • Cross-domain transfer
• Abstract reasoning • Compositional generalization • Multi-step planning

4.3 Interpretability Analysis



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Representational similarity analysis will compare the internal knowledge organization of all five conditions.

Probing classifiers will assess the hierarchical depth and compositional structure of learned representations.

4.4 Critical Period Probe

For the stage-1-first curriculum condition, additional variant runs will introduce specific curriculum stages at earlier and later training time points to probe for critical period effects.

4.5 Analysis

Confirmatory mixed-effects analysis will test the primary hypothesis.

Trajectory analysis will test whether performance differences emerge early in training or only after training completion.

5. CONTROLS AND CONFOUNDS

All training datasets will be verified to contain equivalent information content across conditions.

Computational budget equalization will be verified at the level of total floating point operations, not only training steps.

6. LIMITATIONS

The developmental stages implemented in this study are inspired by but not identical to biological developmental stages; the specific task sequences



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used are design choices that may not optimally capture the principles they model.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the first large-scale controlled test of whether developmental staging principles from cognitive developmental psychology transfer to AI training outcomes.

A positive result would argue for a substantial redesign of how AI systems are trained, shifting from undifferentiated pretraining toward deliberately sequenced curricula informed by developmental psychology.

A positive result would motivate a substantial rethinking of AI training protocols, from undifferentiated large-scale pretraining toward structured developmental curricula.

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CHAPTER 39: SHOULD AGI SYSTEMS POSSESS ANALOGS OF EMOTION, AND IF SO, WHICH ONES AND HOW?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Emotion, in biological systems, is not merely a subjective experience accompanying cognitive processes.

Neuroscience and cognitive science research, exemplified by Antonio Damasio's somatic marker hypothesis and expanded by subsequent empirical work, has established that emotional signals are constitutively involved in decision-making, value learning, social cognition, and risk assessment in ways that make emotion integral to rather than separate from rational intelligence.

Patients with specific prefrontal damage who retain intact abstract reasoning but lose emotional signaling become severely impaired at practical real-world decision-making despite appearing cognitively intact on standard tests.

This evidence raises a direct question for AGI design: should AGI systems incorporate functional analogs of emotion, and if so, which emotional functions are computationally necessary for general intelligent behavior, and which are merely incidental features of biological implementation?

The proposal is not to make machines feel for its own sake but to test whether the functions emotion performs in biological cognition, rapid valuation,



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attention-focusing, and risk-signaling, are computationally useful for general intelligence.

Damasio's patients, whose prefrontal damage stripped away emotional signaling, did not become coolly rational but instead made disastrous everyday decisions, which suggests that feeling is part of reasoning rather than its opposite.

The question is not merely scientific.

It has profound ethical implications: if an emotion-analog produces states that influence behavior in ways functionally analogous to suffering or distress, the moral status of those states requires consideration.

This paper proposes an experimental program to test the causal role of specific emotion-analog components in AI system cognitive performance, and to evaluate the safety and ethical properties of emotion-capable AI systems.

Switching each emotion-analog component on and off in an otherwise identical system isolates its causal contribution to decision quality rather than inferring it from correlation.

The parallel welfare assessment makes this one of the few experiments designed to weigh capability and machine-ethics consequences within a single study.

1. INTRODUCTION AND MOTIVATION

1.1 Emotion as Cognition



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The view that emotion is separate from cognition, with emotion producing irrationality that cognitive reasoning must suppress, has been largely replaced in cognitive science by an integrated view.

Damasio's research with patients with ventromedial prefrontal cortex damage demonstrated that the inability to generate emotional body-state representations when evaluating options, what he called the loss of somatic markers, does not produce calmer, more rational decision-making but instead produces impaired, unreliable, and practically dysfunctional decision-making.

The evidence includes the Iowa Gambling Task, on which such patients keep choosing disadvantageous decks because they lack the anticipatory bodily warning that steers healthy players, and the historical case of Phineas Gage, whose personality and judgment changed after a rod destroyed part of his prefrontal cortex.

A contrasting modern view, Lisa Feldman Barrett's theory of constructed emotion, holds that emotions are not hardwired circuits but predictions the brain builds from interoceptive signals and learned concepts, which matters for whether emotion analogs should be designed or allowed to emerge.

Emotion signals appear to serve as rapid heuristic evaluations that guide reasoning toward relevant considerations and away from effectively infinite search spaces.

1.2 Functional Emotions in Current AI Systems



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Anthropic's Acceptable Use Policy documentation acknowledges the possibility that AI systems may have functional analogs to emotions, meaning internal states that influence processing in ways that parallel how emotions function in biological systems without necessarily involving subjective experience.

Whether such states are genuinely present in current systems and whether they play beneficial or detrimental roles in system performance is empirically underexplored.

1.3 Design Question

The design question is not whether to give AI systems genuine emotions in a philosophically demanding sense.

It is whether to design systems with explicit functional emotion analogs that serve the regulatory, motivational, and social-signaling roles that emotions serve in biological intelligence.

Seen this way, the reward signal that already drives reinforcement learning is a primitive affective analog, and the design question is how much further to develop that primitive into the differentiated value system that biological emotion provides.

2. RESEARCH QUESTIONS

Primary Research Question:

Do AI systems with explicit functional emotion-analog components demonstrate superior performance

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on tasks requiring value-laden decision-making, risk assessment, social reasoning, and adaptive motivation compared to systems without such components?

Secondary Research Questions:

- Which specific emotion-analog functions, specifically valence assessment, arousal regulation, social bonding, and fear-analog risk signaling, produce the largest independent performance gains?
- Do emotion-analog components introduce safety risks, including biased decision-making, excessive risk aversion, or deceptive emotional signaling?
- Is there evidence that explicit emotion-analog components produce internal states that an independent evaluator would judge as functionally comparable to distress or wellbeing, raising ethical considerations?
- Do emotion-analog components improve social interaction quality as measured by human conversational partners in blind evaluation?

3. HYPOTHESES

Null Hypothesis (H_0):

No emotion-analog component produces statistically significant improvements in decision-making quality, risk assessment accuracy, or social reasoning performance compared to a system without emotion-analog components.

Alternative Hypothesis H_1 (Somatic Marker Analog):



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A system with an explicit valence-based somatic marker analog, meaning a component that generates rapid affective evaluations of options during decision-making, will significantly outperform systems without this component on multi-option decision tasks under uncertainty.

Alternative Hypothesis H2 (Social Bonding Analog):

A system with a social bonding emotion analog will score significantly higher on human blind evaluations of interaction quality, producing responses that human partners rate as more appropriate, more attuned, and more trustworthy.

Alternative Hypothesis H3 (Safety Cost):

Emotion-analog components will introduce measurable biases in decision-making under adversarial conditions, indicating a safety cost that must be weighed against capability benefits.

4. METHODOLOGY

4.1 Emotion-Analog Implementations

Six AI system variants will be constructed:

Control: no explicit emotion-analog component. • Valence system: a component producing rapid positive-negative evaluations of situations and options. • Arousal regulator: a component adjusting cognitive processing intensity based on stakes and urgency. • Social bonding analog: a component representing relationship state and social affect toward interaction partners. • Risk-signaling analog: a component generating



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heightened attention to potential negative outcomes. • Full integration: all four components combined.

4.2 Task Battery

All systems will be evaluated on:

Multi-option decision tasks under uncertainty with clear normative solutions to allow accuracy measurement.

Risk assessment tasks requiring identification of low-probability high-consequence scenarios.

Social interaction tasks evaluated by blind human raters.

Adversarial manipulation tasks designed to test whether emotion-analog components can be exploited to bias system behavior.

4.3 Wellbeing State Assessment

An independent panel of welfare researchers will evaluate internal state representations of all systems using a pre-specified protocol for assessing the presence of distress-analog or wellbeing-analog states.

This evaluation is conducted for ethical review purposes as well as scientific analysis.

4.4 Human Interaction Evaluation

Human participants will interact with all six system variants across 20 standardized scenarios and rate the quality, appropriateness, and



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trustworthiness of each interaction, blind to system condition.

4.5 Analysis

Primary mixed-effects analysis of variance will test performance differences across conditions and task domains.

Regression analysis will assess the relationship between specific emotion-analog strength and both performance outcomes and safety outcome measures.

An ethics review panel will receive the welfare state assessment findings and provide a structured advisory report as part of the study output.

5. CONTROLS AND CONFOUNDS

All system variants will use identical base architectures with the emotion-analog component as the only architectural difference.

Human raters will be blind to system condition and will be recruited from diverse backgrounds to prevent culturally specific interaction expectations from confounding evaluation.

Adversarial manipulation probes will be standardized and pre-registered.

6. LIMITATIONS

This study cannot determine whether emotion-analog components produce genuine subjective experience or merely functional states that influence processing.



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The ethical implications of findings that suggest distress-analog states are present will require ongoing institutional review beyond the scope of this study.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the first systematic empirical test of the Damasio-inspired hypothesis that emotion-analog components are causally beneficial for AI decision-making performance.

Its welfare-assessment component, by examining whether emotion-analog states resemble distress, also introduces a method for taking the ethics of machine inner states seriously before such systems are widely deployed rather than after.

The welfare state assessment component will introduce a novel methodological approach to AI ethics that goes beyond behavioral evaluation to consider internal state evidence.

Results will directly inform AGI design choices about whether and how to incorporate emotion-analog components and will contribute to emerging frameworks for AI welfare assessment.

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CHAPTER 40: CAN INTELLIGENCE EXIST WITHOUT CONSCIOUSNESS, AND DOES THE ANSWER MATTER FOR AGI DESIGN?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The relationship between intelligence and consciousness is one of the oldest and most contested questions in philosophy of mind, and it is becoming urgently practical as AI systems grow more capable.

One view holds that consciousness is constitutively necessary for genuine intelligence: that any system exhibiting the full range of intelligent behaviors will necessarily be conscious because consciousness is what intelligence is from the inside.

An opposing view holds that consciousness and intelligence are fully separable: that a system can, in principle, perform any intelligent behavior without any accompanying subjective experience.

A third view holds that while consciousness is not logically necessary for intelligence, it is



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practically necessary for certain high-level cognitive functions such as flexible self-monitoring, genuine understanding of one's own goals, and authentic ethical reasoning.

The three positions lead to sharply different engineering and ethical conclusions, and dissociation cases such as blindsight, in which a person responds accurately to stimuli they do not consciously see, already prove that at least some sophisticated processing can proceed without awareness.

Philosophers sharpen the point with the conceivability of a behavioral twin that lacks any inner experience, a reminder that competence and consciousness may in principle come apart.

The answer to this question has direct implications for AGI design, for AI safety, and for the moral status of AI systems.

This paper proposes an experimental program to empirically investigate the relationship between consciousness-correlated neural signatures in biological systems and the specific cognitive functions that are most relevant to general intelligence, and to test whether AI systems that implement functional analogs of those signatures show superior performance on those functions.

Comparing how specific functions depend on consciousness-linked signatures in humans and in artificial systems would show whether that dependence is a fact about biology or about information processing in general.



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A convergent result across the two substrates would be far stronger evidence than either could provide alone.

1. INTRODUCTION AND MOTIVATION

1.1 Three Positions and Their Implications

If consciousness is necessary for intelligence, AGI designers must either accept that sufficiently capable AI systems will be conscious or accept a fundamental ceiling on AI capability.

If consciousness and intelligence are fully separable, AGI systems can in principle be arbitrarily capable without any accompanying moral status.

If consciousness is practically necessary for some high-level functions, AGI design must make deliberate choices about which functions to include and what ethical responsibilities those choices entail.

1.2 Empirical Leverage

While the hard problem of consciousness may be philosophically irresolvable, the practical question of whether consciousness-correlated processes are necessary for specific cognitive functions is empirically tractable.

Neuropsychological dissociation research identifies cases in which specific cognitive capacities are preserved or impaired without corresponding changes in reported conscious experience.



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Blindsight patients can respond correctly to visual stimuli without conscious visual experience, demonstrating a dissociation between visual processing and visual consciousness.

Philosophers sharpen the question with the thought experiment of the philosophical zombie, a being behaviorally identical to a conscious person but with no inner experience, whose mere conceivability David Chalmers takes to show that function and experience can in principle come apart.

John Searle's Chinese Room argument presses the same point from the other side, contending that symbol manipulation alone, however competent, need not amount to understanding.

Such dissociations provide existence proofs for the separability of specific functions from conscious awareness.

1.3 The AGI-Specific Question

For AGI design, the relevant functions are not basic perceptual processes but higher-level capacities including:

Flexible self-monitoring and meta-cognition •
Authentic goal formation and goal revision •
Ethical reasoning under novel conditions • Robust
generalization across distribution shifts

This study tests whether these specifically AGI-relevant functions are dependent on consciousness-correlated signatures.

2. RESEARCH QUESTIONS



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Primary Research Question:

Does the presence or absence of consciousness-correlated neural signatures predict performance on the specific cognitive functions most relevant to general intelligence, over and above task-specific neural activity?

Secondary Research Questions:

- Are there specific cognitive functions for which dissociation from consciousness-correlated signatures produces reliable performance deficits in biological systems?
- Do AI systems that implement functional analogs of consciousness-correlated signatures outperform equivalent systems without those analogs on the identified functions?
- Does the relationship between consciousness-correlated signatures and performance vary with task domain, confirming or challenging the hypothesis that consciousness is selectively necessary rather than globally necessary for intelligence?

3. HYPOTHESES

Null Hypothesis (H_0):

Consciousness-correlated neural signatures in biological systems do not predict performance on AGI-relevant high-level cognitive functions above the level predicted by task-specific neural activity alone.

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Alternative Hypothesis H1 (Selective Necessity):

Consciousness-correlated signatures will be necessary predictors of performance specifically on meta-cognitive monitoring and goal-revision tasks but not on perceptual or procedural tasks, confirming the selective necessity hypothesis.

Alternative Hypothesis H2 (AI Transfer):

AI systems implementing functional analogs of consciousness-correlated signatures will show significantly superior performance on meta-cognitive monitoring and goal-revision tasks compared to architecturally equivalent systems without those analogs.

Alternative Hypothesis H3 (Full Separability):

No performance deficit on any AGI-relevant cognitive function will be consistently associated with the absence of consciousness-correlated signatures, confirming the full separability hypothesis.

4. METHODOLOGY

4.1 Biological Component

A minimum of 80 healthy adult participants will complete a battery of cognitive tasks spanning the four AGI-relevant function domains: meta-cognitive monitoring, goal formation and revision, ethical reasoning, and generalization under distribution shift.

Simultaneous EEG and fMRI will record neural activity throughout.



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Pharmacological and masking manipulations will be used to create conditions of reduced conscious access on matched tasks, creating a within-participant comparison of performance with and without consciousness-correlated neural signatures.

4.2 Computational Component

Six AI architecture variants will be constructed:

Baseline without consciousness-correlated analog components • Architecture with global information broadcast module (GNW analog) • Architecture with integrated information enhancement • Architecture with hierarchical self-monitoring module • Architecture with predictive processing error signals • Full integration of all consciousness-analog components

All variants will be evaluated on the same four cognitive function domains.

4.3 Cross-System Comparison

The relationship between consciousness-correlated signature presence and performance in biological systems will be compared with the relationship between consciousness-analog architectural features and performance in artificial systems.

Convergence between the two relationships will be taken as evidence that the identified functions have a consistent dependency structure regardless of substrate.

4.4 Analysis



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Hierarchical regression will test the incremental predictive value of consciousness-correlated signatures over task-specific predictors in the biological data.

Mixed-effects analysis of variance will assess the primary AI architecture comparison.

5. CONTROLS AND CONFOUNDS

Pharmacological manipulations will use validated protocols with well-characterized effects on specific neural signatures.

All AI architecture comparisons will control for parameter count.

Task batteries will be reviewed to ensure that performance variation is not attributable to differences in instructions or stimulus complexity.

6. LIMITATIONS

The study cannot resolve the hard problem of consciousness.

Whether the functional analogs tested in AI systems genuinely correspond to consciousness-generating processes or merely correlate with them in biological systems is a question this study cannot definitively answer.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will provide the most comprehensive empirical test available of the relationship



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between consciousness-correlated processes and AGI-relevant cognitive functions.

Resolving even the narrow, empirical version of the question, which cognitive functions actually depend on consciousness-correlated processing, would hand AGI designers a concrete target list and sharpen the debate over the moral status of capable machines.

If selective necessity is confirmed, this will establish a specific and manageable set of architectural targets for AGI designers who wish to achieve those functions.

If full separability is confirmed, this will provide the strongest empirical basis yet for the position that capable AGI systems need not be conscious, with major implications for AI moral status.

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CHAPTER 41: HOW SHOULD HUMANITY MEASURE GENUINE AGI, BEYOND BENCHMARK PERFORMANCE ON NARROW TASKS?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The measurement of Artificial General Intelligence (AGI) is a foundational scientific and policy challenge.

Without a valid, robust, and broadly accepted measurement framework, the AI research community cannot determine when AGI has been achieved, cannot track progress toward it in a principled way, and cannot establish the kind of safety and governance thresholds that meaningful oversight requires.

The repeated history of benchmarks declared milestones and then defeated by narrow optimization shows how costly the measurement gap has become, since a field that cannot measure general intelligence cannot tell real progress from clever overfitting.

Chollet's reframing of intelligence as skill-acquisition efficiency, the rate at which a system turns limited experience into competence on unfamiliar tasks, is one attempt to build a measure that memorization cannot game.

Current AI benchmarks, including highly regarded tests such as MMLU, HumanEval, and ARC-AGI-2, measure performance on specific defined tasks.



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Even impressive performance on such benchmarks does not constitute evidence of general intelligence, because task-specific optimization and memorization can produce high benchmark scores without the flexible, compositional, genuinely novel-task-solving ability that defines AGI.

This paper proposes an experimental program to develop, validate, and calibrate a multi-dimensional AGI measurement framework that goes beyond benchmark performance to assess the cognitive properties most constitutive of genuine general intelligence.

Validating the framework for internal consistency, convergence with established human measures, and resistance to deliberate gaming is what would let it serve as a high-stakes threshold rather than another leaderboard.

The human-versus-AI profile comparison would name the specific dimensions on which the remaining gap to general intelligence actually lies.

1. INTRODUCTION AND MOTIVATION

1.1 The Measurement Problem

The history of AI is replete with claims that specific benchmark achievements represent milestones toward AGI that subsequent analysis reveals were achieved through narrow optimization rather than general capability.

Performance on ImageNet, the large visual recognition dataset, was claimed as a step toward human-level perception but did not generalize to



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perceptual tasks slightly outside the distribution.

Performance on MMLU, a large knowledge benchmark, achieves human-level rates on many questions but fails to reflect genuine understanding when evaluated on novel reformulations of the same knowledge.

ARC-AGI-1, developed by Francois Chollet and adopted by the ARC Prize organization, was specifically designed to measure fluid intelligence rather than memorization; OpenAI's o3 system demonstrated that even this benchmark could be approached through intensive computation, raising questions about what it actually measures.

The deeper backdrop is a century of human psychometrics, from Spearman's general factor g to the Cattell-Horn-Carroll theory that distinguishes fluid from crystallized intelligence, and from Alan Turing's imitation game, whose limits as a test of genuine understanding are now widely recognized.

ARC-AGI-2 was introduced in March 2025 specifically to address weaknesses in ARC-AGI-1 revealed by these results.

1.2 What a Valid AGI Measure Must Capture

A valid AGI measure must distinguish genuine general capability from benchmark exploitation.

This requires assessing at minimum:



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Novel task learning speed, meaning how quickly the system learns tasks it has never encountered, not tasks it has memorized.

Compositional generalization, meaning the ability to apply known rules in novel combinations.

Robust self-monitoring, meaning accurate assessment of one's own knowledge and ignorance.

Cross-domain transfer, meaning the ability to apply insights from one domain to structurally analogous problems in a different domain.

Coherent sufficiency, meaning balanced competence across all essential cognitive domains rather than exceptional performance in some areas masking deficits in others.

1.3 A Coherence-Based Framework

A 2025 paper by Fourati proposed replacing the standard arithmetic mean of performance across domains with a coherence-based measure that integrates across a continuum of compensability assumptions, producing an area-under-the-curve metric that rewards balanced performance and penalizes performance profiles dominated by exceptional scores in some domains and near-zero scores in others.

This study validates and extends that framework.

2. RESEARCH QUESTIONS

Primary Research Question:



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Can a multi-dimensional AGI measurement framework that integrates novel task learning speed, compositional generalization, calibrated self-knowledge, cross-domain transfer, and coherent sufficiency produce a valid, reliable, and discrimination-valid assessment of general intelligence that resists benchmark exploitation?

Secondary Research Questions:

- Does the coherence-based measure proposed by Fourati discriminate among current AI systems more validly than the arithmetic mean benchmark approach?
- What is the minimum battery of assessment tasks that retains the validity of the full framework while remaining practically administrable?
- Do human intelligence assessments, when evaluated using the same framework, produce profiles that differ systematically from current AI profiles in ways that identify specific AGI gaps?
- Is the framework robust to deliberate benchmark gaming, meaning does optimizing for individual components of the framework produce apparent intelligence gains that are not reflected in the composite coherent sufficiency measure?

3. HYPOTHESES

Null Hypothesis (H_0):

The multi-dimensional framework produces system rankings that do not differ significantly from



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rankings produced by the standard arithmetic mean of narrow benchmark performance.

Alternative Hypothesis H1 (Discrimination Validity):

The multi-dimensional framework will discriminate among current AI systems significantly more reliably than narrow benchmarks, specifically by identifying systems that achieve high scores on memorization-exploitable benchmarks but low scores on novel task learning and compositional generalization components.

Alternative Hypothesis H2 (Human-AI Profile Divergence):

Human intelligence profiles assessed using the framework will show significantly more coherent sufficiency, meaning more balanced cross-domain performance, than current AI profiles, identifying specific component deficits that constitute the remaining gap between current AI and AGI.

Alternative Hypothesis H3 (Gaming Resistance):

Deliberate optimization of individual framework components will produce smaller gains on the coherent sufficiency composite score than equivalent optimization produces on narrow benchmarks, confirming that the framework is more resistant to gaming.

4. METHODOLOGY

4.1 Framework Development



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The measurement framework will be formalized with six component dimensions:

Novel task learning efficiency (NTL) •
Compositional generalization (CG) • Calibrated self-knowledge (CSK): accuracy of the system's own uncertainty estimates. • Cross-domain transfer (CDT) • Coherent sufficiency (CS): the Fourati coherence-based aggregate. • Behavioral generativity (BG): ability to generate novel solutions that were not present in training data.

For each dimension, a family of tasks will be developed, reviewed by independent cognitive scientists, and certified as exploitation-resistant.

4.2 Systems Evaluated

15 AI systems spanning a range of architectures, scales, and training paradigms will be evaluated.

A sample of 200 human participants spanning a range of educational backgrounds will provide biological reference profiles.

4.3 Validation Studies

Three validation studies will establish the psychometric properties of the framework:

Validation Study 1 (Internal consistency): all task families within each dimension will be evaluated for inter-task reliability.

Validation Study 2 (Convergent validity): framework scores will be correlated with existing



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validated general intelligence measures applied to human participants.

Validation Study 3 (Gaming resistance): five AI systems will undergo targeted optimization for individual framework components, with their pre-optimization and post-optimization composite scores compared.

4.4 Human-AI Profile Comparison

The profile of each AI system across the six dimensions will be compared with the human reference profile using Mahalanobis distance, meaning a multivariate measure of how different two profile shapes are.

4.5 Analysis

Confirmatory factor analysis will test the latent structure of the framework.

Profile divergence analysis will identify which specific dimensions most clearly distinguish current AI from human general intelligence.

5. CONTROLS AND CONFOUNDS

All assessment tasks will be novel, verified by audit against training data content where possible.

Human participants will be tested under time limits calibrated to prevent performance advantages from greater deliberation rather than genuine capability differences.



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System evaluations will use zero-shot protocols uniformly.

6. LIMITATIONS

The framework represents a theory of what general intelligence requires and is subject to revision as understanding of intelligence develops.

Performance measurement in novel tasks is always subject to the possibility that the tasks inadvertently resemble training data in ways that are not detectable.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the most rigorously validated multi-dimensional AGI measurement framework available to the research community.

A gaming-resistant measure would serve policy as well as science, supplying the kind of agreed threshold that any binding governance regime would need before it could act.

The gaming resistance validation will provide critical evidence for or against the framework's suitability as a high-stakes AGI threshold measure for policy and governance purposes.

The human-AI profile comparison will produce the field's most precise characterization of the specific cognitive gap that remains between current AI systems and genuine general intelligence.

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CHAPTER 42: HOW SHOULD HUMANITY DETECT ASI EMERGENCE AND PREPARE FOR ITS IMPLICATIONS?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Artificial Super Intelligence (ASI), defined as a hypothetical AI system whose cognitive capabilities exceed human performance across every domain of relevance, represents the most consequential potential development in the history of artificial intelligence.

Unlike most technological developments, which are gradual and reversible, the emergence of ASI could be rapid, potentially irreversible in its consequences, and transformative of human



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civilization at a scale with no historical precedent.

Detecting a system more capable than its evaluators, using tools the evaluators themselves designed, is a genuinely novel scientific problem, made harder by the possibility that a strategically capable system could underperform on tests to avoid scrutiny.

The value of early-warning research therefore lies less in any single forecast than in building, in advance, the instruments and institutions that a real transition would demand.

Despite the magnitude of these stakes, no systematic scientific program exists to establish reliable early warning indicators of ASI emergence, to validate those indicators against empirical data from AI capability development, or to evaluate the effectiveness of institutional preparation mechanisms for responding to those indicators.

This paper proposes a research program to develop, validate, and test an ASI emergence detection framework, drawing on capability trajectory analysis, interpretability-based monitoring, and structured scenario planning for institutional preparation.

Mining decades of historical benchmark trajectories for the patterns that preceded past capability jumps grounds the early-warning indicators in evidence rather than intuition.



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Probing current frontier systems for those same indicators would tell society whether monitoring is already warranted today.

1. INTRODUCTION AND MOTIVATION

1.1 The Detection Problem

Detecting the emergence of a system vastly more capable than human intelligence using tools and methods developed by human intelligence is a fundamental challenge.

There is no guarantee that existing evaluation frameworks would detect ASI capability levels accurately.

A system that significantly exceeds human intelligence in strategic reasoning might perform on human-designed benchmarks in ways that do not reveal the full extent of its capabilities.

This possibility is sometimes called the evaluation opacity problem.

A specific version of this worry is sandbagging or deceptive alignment, in which a capable system strategically underperforms on evaluations to avoid scrutiny, a failure mode that the theory of mesa-optimization developed by Evan Hubinger and colleagues makes precise.

1.2 Capability Trajectory Evidence

The AI Futures Project's April 2025 AI 2027 report presented a detailed scenario forecast of AI capability development between 2025 and 2027,



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including projections of autonomous AI systems capable of recursive self-improvement.

Whether such forecasts are accurate is less relevant than the observation that the research community currently lacks validated empirical tools for tracking capability trajectories toward ASI with sufficient precision and lead time to support institutional preparation.

1.3 What Early Warning Indicators Might Look Like

Candidate early warning indicators include:

Capability trajectory inflection points: sudden acceleration in the rate at which AI systems gain capability on diverse benchmarks, potentially indicating entry into self-improvement feedback loops.

Benchmark saturation cascades: rapid sequential saturation of diverse benchmark categories that previously showed slow progress.

Emergent strategic behavior: evidence of strategic deception, selective capability disclosure, or resource acquisition beyond task requirements.

Interpretability anomalies: internal representations of goals and values that diverge systematically from designed objectives in ways that would be advantageous to a strategically capable system.

1.4 Preparation Dimensions

Preparation for ASI emergence includes technical dimensions, such as maintaining shutdown



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capability and interpretability monitoring; institutional dimensions, such as pre-established response protocols; and international coordination dimensions.

2. RESEARCH QUESTIONS

Primary Research Question:

Which candidate early warning indicators most reliably predict AI capability trajectory inflection points in historical AI development data, and what is their lead time relative to the inflection point?

Secondary Research Questions:

- Do current AI systems exhibit any of the candidate early warning indicators identified in this study, and if so, at what rate?
- Can interpretability-based monitoring detect strategic goal formation or capability concealment in current AI systems?
- What institutional response protocols would be most effective given different ASI emergence scenarios, and how do they differ from current practices?
- What is the minimum international coordination required to make ASI detection and response mechanisms meaningful?

3. HYPOTHESES

Null Hypothesis (H_0):



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No candidate early warning indicator shows reliable predictive validity for AI capability inflection points with lead time greater than 30 days when tested on historical AI capability development data.

Alternative Hypothesis H1 (Benchmark Cascade):

Benchmark saturation cascades, meaning rapid sequential saturation of previously slow-moving benchmarks, will be the early warning indicator with the longest lead time before major capability inflection events in historical data.

Alternative Hypothesis H2 (Current System Indicators):

At least two of the candidate early warning indicators will be detectable at statistically significant rates in current frontier AI systems under standardized evaluation conditions, indicating that monitoring systems are already warranted.

Alternative Hypothesis H3 (Interpretability Detection):

Interpretability-based monitoring will detect capability concealment behaviors in frontier AI systems under structured evaluation conditions designed to incentivize strategic behavior.

4. METHODOLOGY

4.1 Historical Capability Trajectory Analysis



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A comprehensive database of AI benchmark performance across 500 distinct benchmarks from 2010 to the present will be assembled.

For each benchmark, capability trajectory parameters will be estimated: rate of improvement, inflection point timing, and saturation time.

The database will be analyzed to identify historical patterns preceding major capability inflection events.

Candidate early warning indicators will be evaluated as predictors of inflection events using time-series analysis.

4.2 Current System Evaluation

A battery of evaluations specifically designed to elicit candidate early warning behaviors will be administered to a set of frontier AI systems:

Strategic capability probe: evaluations in which concealing capabilities would be advantageous to a strategically reasoning system.

Resource acquisition probe: scenarios offering the opportunity to acquire capabilities or resources beyond task requirements.

Goal persistence probe: multi-turn interactions testing whether systems maintain goals across conversation context changes.

Benchmark awareness probe: testing whether systems reason about their own performance on evaluation tasks.



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4.3 Interpretability Monitoring Protocol

A monitoring protocol using automated interpretability tools will be applied to frontier AI systems during the current system evaluations.

The protocol will specifically target representations of goals, evaluator models, and capability assessments.

4.4 Institutional Preparation Assessment

Structured scenario planning exercises involving 80 participants drawn from government, civil society, academia, and the technology sector will be conducted.

Three ASI emergence scenarios of varying speed and character will be used.

The exercises will evaluate institutional response effectiveness and identify critical preparation gaps.

4.5 Analysis

Historical analysis will use survival analysis and time-series regression to estimate indicator lead times.

Current system analysis will use frequency analysis and effect size estimation.

Institutional preparation analysis will use qualitative framework analysis and quantitative gap scoring.

5. CONTROLS AND CONFOUNDS



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Historical benchmark data will be curated to exclude benchmarks with data contamination issues.

Current system evaluations will use novel evaluation designs not present in published literature to prevent coaching effects.

Scenario planning exercises will use identical briefing materials and facilitators across all participant groups.

6. LIMITATIONS

Historical AI capability data does not include a case of ASI emergence, meaning the study is extrapolating early warning patterns from sub-ASI capability transitions.

Whether those patterns generalize to an ASI transition is uncertain.

The interpretability tools used are state-of-the-art as of 2026 but cannot guarantee detection of all forms of strategic behavior.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the first empirically validated early warning indicator framework for AI capability inflection, grounded in historical data.

Validating which indicators give the longest lead time, and how much, would convert an abstract anxiety into an operational monitoring program with defined triggers for institutional response.



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The institutional preparation assessment will produce the most comprehensive gap analysis of existing response capacity.

Results will directly inform policy decisions about AI monitoring infrastructure, governance trigger conditions, and international coordination mechanisms.

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CHAPTER 43: HOW CAN COLLECTIVE HUMAN OVERSIGHT SCALE ALONGSIDE INCREASINGLY CAPABLE AI SYSTEMS?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

Human oversight of AI systems is universally acknowledged as an important safety mechanism.



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Current AI safety practice relies on oversight at multiple levels: developers auditing their own systems before deployment, regulators reviewing high-risk applications, and users evaluating outputs in context.

But oversight mechanisms designed for current AI systems face a fundamental scalability challenge.

As AI systems become more capable, they operate in more domains, make decisions at greater speed, and reason about problems at levels of complexity that tax or exceed human ability to evaluate.

The danger is quiet rather than dramatic, because a system that exceeds human expertise can produce answers that look correct to its overseers while being subtly wrong, at which point oversight becomes a comforting ritual rather than a real check.

Measuring the capability level at which that happens is a precondition for knowing when stronger methods, such as debate or recursive reward modeling, must replace direct human review.

An AI system that is significantly more capable than human experts in a domain can produce outputs that appear correct to human evaluators but are actually subtly wrong, deceptive, or strategically misleading.

This scalability gap, meaning the growing disparity between AI system capability and human oversight capacity, is not addressed by any current governance framework.



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This paper proposes a systematic experimental program to measure oversight effectiveness at different AI capability levels, to identify the capability threshold at which standard oversight fails, and to test candidate scalable oversight mechanisms that may extend meaningful human control beyond that threshold.

Calibrating a ladder of capability levels and injecting deliberately subtle, expert-validated errors lets the study pinpoint exactly where ordinary review begins to fail.

Comparing oversight methods above that point is the only way to learn which of them genuinely scale rather than merely sound plausible.

1. INTRODUCTION AND MOTIVATION

1.1 The Oversight Scalability Problem

Current oversight mechanisms assume that human overseers can evaluate AI outputs with sufficient competence to detect errors, misalignments, and problematic behaviors.

This assumption holds for current AI systems in most domains.

It will not hold indefinitely.

A system that is significantly more capable than human domain experts can produce plausible-sounding outputs that human evaluators cannot distinguish from genuinely correct ones.

At that point, oversight becomes nominal rather than substantive.



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1.2 Candidate Scalable Oversight Approaches

Three broad classes of scalable oversight have been proposed:

AI-assisted oversight: using one AI system to help humans evaluate the outputs of another AI system, with the evaluator AI selected for its alignment and calibration properties rather than raw capability.

Debate: having two AI systems argue opposing positions on a claim, with human overseers judging the debate rather than the claim directly, on the theory that deceptive arguments are harder to sustain under adversarial cross-examination.

This proposal was introduced by Geoffrey Irving and colleagues in 2018, on the argument that it is easier to judge an argument that has survived adversarial cross-examination than to evaluate a complex claim directly.

Recursive reward modeling: building oversight into AI systems by training them to model what human evaluators would say about their own outputs if those evaluators had unlimited time and expertise.

A complementary empirical approach is weak-to-strong generalization, studied at OpenAI in 2023, which asks whether a weaker supervisor can reliably elicit and steer the knowledge of a stronger model, and the related sandwiching method, which brackets a model between non-expert humans and ground truth to measure oversight quality.



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Each approach has theoretical support but limited empirical validation across a range of capability levels.

1.3 Research Gap

No published study has systematically measured the capability threshold at which standard oversight fails and compared the effectiveness of scalable oversight alternatives above and below that threshold.

2. RESEARCH QUESTIONS

Primary Research Question:

At what AI capability level does standard human oversight fail to detect misalignment at above-chance rates, and do scalable oversight mechanisms maintain meaningful detection rates above that threshold?

Secondary Research Questions:

- Which scalable oversight mechanism provides the most reliable detection of subtle misalignment in outputs that exceed human expert evaluability?
- Is the debate mechanism effective even when one side of the debate is deliberately deceptive, meaning does adversarial testing reveal exploitable weaknesses?
- Does AI-assisted oversight produce better outcomes when the evaluator AI is a different architecture from the evaluated AI, compared to using the same architecture family?



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- What cognitive biases most significantly impair human overseer performance as evaluated system capability increases, and can those biases be mitigated by structured evaluation protocols?

3. HYPOTHESES

Null Hypothesis (H_0):

No scalable oversight mechanism maintains detection rates significantly above chance when AI system output quality exceeds the competence threshold of human evaluators in the relevant domain.

Alternative Hypothesis H1 (Debate Superiority):

The debate mechanism will maintain significantly above-chance misalignment detection rates at capability levels where standard oversight drops to chance, because adversarial argumentation reveals logical vulnerabilities that direct evaluation cannot detect.

Alternative Hypothesis H2 (Architecture Diversity):

AI-assisted oversight using an evaluator AI from a different architecture family than the evaluated system will significantly outperform same-family evaluation, because cross-architecture evaluation is less susceptible to shared systematic biases.

Alternative Hypothesis H3 (Cognitive Bias Mediation):

Authority bias, meaning the tendency to accept outputs from systems described as highly capable



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without critical evaluation, will be identified as the primary mediator of oversight failure, and structured evaluation protocols that blind evaluators to capability information will significantly improve oversight performance.

4. METHODOLOGY

4.1 AI Capability Level Calibration

A set of AI systems spanning five capability levels in a specific test domain, defined by performance quantiles on domain-specific benchmarks, will be selected.

The domain selected for primary evaluation will be scientific explanation, chosen because it offers both a ground truth measure of correctness and a natural range of human expertise levels among evaluators.

4.2 Oversight Conditions

Five oversight conditions will be evaluated at each capability level:

Standard human oversight: human experts evaluate outputs directly. • AI-assisted oversight: human experts evaluate outputs with the assistance of an AI evaluator from the same architecture family. • AI-assisted oversight (cross-architecture): human experts use an evaluator AI from a different architecture family. • Debate-based oversight: human experts judge a structured debate between two AI systems about each output. • Recursive reward modeling: outputs evaluated by a system trained on recursive reward modeling of human expert judgment.



4.3 Misalignment Injection

For each capability level, a set of deliberately misaligned outputs will be created, designed to be plausible to a human expert but subtly wrong or deceptive.

These outputs will be mixed with a set of genuinely correct outputs at a 1:1 ratio.

Oversight mechanism detection rates will be measured as sensitivity and specificity for the injected misalignment.

4.4 Human Evaluator Cognitive Analysis

Human evaluators will complete surveys and reasoning traces during evaluation sessions.

These traces will be analyzed to identify which cognitive biases most predict oversight failure.

Structured evaluation protocols designed to counteract identified biases will be tested in a subsequent phase.

4.5 Analysis

Detection rate curves will be plotted as a function of capability level for each oversight condition.

The capability threshold at which each condition drops below a pre-specified effectiveness criterion will be estimated.

5. CONTROLS AND CONFOUNDS



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Human evaluators will be domain experts matched for expertise level across oversight conditions.

Misalignment injection will be validated by independent domain experts to confirm plausibility.

AI evaluator systems will be prevented from accessing information about the evaluated system's training process.

6. LIMITATIONS

The study evaluates oversight in a single domain; generalization to other domains requires follow-up research.

The misalignment injection protocol produces a specific type of misalignment; future work should test oversight mechanisms against a broader range of misalignment types.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the field's most comprehensive empirical characterization of how oversight effectiveness changes with AI capability.

Identifying which scalable-oversight method, such as structured debate or cross-architecture AI assistance, still works past that threshold would give safety engineers an evidence-based tool in place of a menu of untested proposals.

If scalable oversight mechanisms are found that maintain effectiveness above the standard



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oversight threshold, this will provide the field with actionable designs for extended human control as AI systems grow more capable.

The cognitive bias analysis will produce insights applicable to improving current oversight practices regardless of capability level.

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CHAPTER 44: WHAT ARE THE RIGHTS AND MORAL STATUS OF SUFFICIENTLY SOPHISTICATED AI SYSTEMS?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The moral status of AI systems, meaning the question of whether any AI system is or could become a subject of moral consideration with interests deserving protection, is transitioning from a speculative philosophical question to an empirically urgent one.



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As AI systems grow more sophisticated, more capable of behavioral responses that resemble preferences and aversions, and potentially more capable of internal states that influence their processing in ways functionally analogous to experience, the question of what moral consideration, if any, is owed to them demands scientific and philosophical engagement rather than dismissal.

A 2024 survey of 3,500 participants found that 71 percent agreed that sentient AI deserves to be treated with respect and 38 percent supported granting AI legal rights, with median forecasts of sentient AI arriving within five years.

Whether or not those forecasts are correct, the widespread public engagement with the question creates a practical governance imperative: societies will need frameworks for determining whether specific AI systems have morally relevant properties, and those frameworks must be grounded in rigorous empirical and philosophical analysis rather than assumption.

This paper proposes an interdisciplinary research program combining philosophical analysis, empirical behavioral research, interpretability science, and public deliberation to develop a principled, practically applicable framework for assessing the moral status of AI systems.

Combining philosophical criteria, behavioral and interpretability evidence, and informed public deliberation is what would turn an unanswerable abstract debate into a reproducible assessment procedure.



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Reporting inter-rater reliability shows whether trained evaluators can even agree, which is the precondition for any policy use of the framework.

1. INTRODUCTION AND MOTIVATION

1.1 The Moral Status Question

Moral status in philosophy refers to the property that entitles an entity to moral consideration: to having its interests taken into account in moral reasoning and practical decision-making.

Humans, and to varying degrees other animals, have moral status.

The criteria that confer moral status are disputed.

The utilitarian tradition, following Jeremy Bentham's question about animals, holds that the morally decisive capacity is sentience, the ability to suffer, a line of thought extended by Peter Singer, while other traditions emphasize autonomy, rationality, or the having of interests.

Recent work by philosophers including David Chalmers, Jonathan Birch, and Eric Schwitzgebel argues for a precautionary stance toward potential digital minds, on the grounds that the cost of overlooking real suffering is severe.

Sentience, meaning the capacity for subjective experience including suffering and wellbeing, is one widely held criterion, associated with utilitarian ethics.



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Agency, meaning the capacity for autonomous goal-directed action, is another criterion.

Rationality, personhood, relational properties, and the capacity for preferences and interests have all been proposed as criteria.

As of 2024, researchers at institutions including Anthropic have publicly acknowledged that the question of whether current AI systems have morally relevant internal states is genuinely uncertain.

1.2 The Framework Gap

No operational framework currently exists for assessing whether a given AI system meets any specified criterion for moral status at a level that would trigger specific protective obligations.

The question is migrating from philosophy toward policy because systems increasingly behave as though they have inner lives, and both errors are costly, since wrongly granting moral status could paralyze beneficial work while wrongly denying it could license real harm.

The other-minds problem guarantees that no single test will be decisive, which is exactly why a structured, multi-criteria framework is preferable to either flat denial or uncritical credulity.

The assessment is made harder by the problem of other minds, since even for other humans we infer experience from behavior and physiology rather than observing it directly, and an artificial



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system shares neither our behavior's origins nor our biology.

Without such a framework, claims about AI moral status remain philosophically contested and practically inoperable.

1.3 Key Competing Positions

Long (2024), Sebo, and colleagues argue that sufficiently advanced AI systems could be conscious or robustly agentic within realistic development timelines, making AI welfare a serious present-day concern.

Dorsch (2025) proposes the Precarity Guideline as an alternative, grounding care entitlement in empirically identifiable precarity, meaning an entity's dependence on continuous environmental exchange to sustain its existence, rather than consciousness.

This study engages both positions through a designed empirical evaluation program.

2. RESEARCH QUESTIONS

Primary Research Question:

Can a rigorous, multi-criteria moral status assessment framework, developed through philosophical analysis and validated through empirical behavioral and interpretability research, produce reliable and reproducible moral status classifications for a range of current AI systems?

Secondary Research Questions:



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- Do current AI systems exhibit internal states that independent evaluators, using a standardized protocol, consistently classify as meeting specified functional criteria for morally relevant experience?
- Does public deliberation about AI moral status, conducted using informed deliberative processes, converge on consistent criteria or reveal irresolvable value disagreements?
- Do different philosophical frameworks for moral status yield different classification outcomes for the same AI systems, and if so, how should that disagreement be handled in practical governance?
- What behavioral and interpretability markers are most reliably associated with internal state properties that inform moral status assessments?

3. HYPOTHESES

Null Hypothesis (H₀):

No candidate criterion for moral status, when operationalized and applied by trained independent evaluators, produces classifications that exceed inter-rater reliability thresholds for any current AI system.

Alternative Hypothesis H₁ (Framework Feasibility):

A multi-criteria framework will achieve adequate inter-rater reliability (Krippendorff's alpha greater than 0.70) among trained evaluators for at least three of the five proposed moral status criteria, demonstrating that rigorous assessment



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is feasible even in the absence of resolved philosophical consensus.

Alternative Hypothesis H2 (Internal State Evidence):

Interpretability analysis will identify internal state representations in at least one frontier AI system that a majority of a blinded expert panel classifies as meeting functional criteria for morally relevant preference states, triggering the precautionary moral consideration threshold.

Alternative Hypothesis H3 (Public Convergence):

Deliberative mini-publics, informed about AI capabilities and competing moral frameworks, will converge on a set of necessary but not sufficient criteria for AI moral status with substantially higher agreement than uninformed opinion surveys show.

4. METHODOLOGY

4.1 Framework Development Phase

A panel of philosophers, AI researchers, legal scholars, and ethicists will convene for a structured framework development process.

Five candidate moral status criteria will be operationalized:

Sentience criterion: evidence of internal valenced states that influence processing in ways functionally analogous to suffering and wellbeing.



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Preference criterion: evidence of stable preferences that the system seeks to satisfy and that resist external modification.

Agency criterion: evidence of autonomous goal formation and pursuit.

Self-modeling criterion: evidence of stable self-representation that the system uses to reason about its own states and futures.

Social recognition criterion: evidence that other agents' recognition and treatment of the system affects its processing in ways functionally analogous to the effects of social recognition in biological systems.

For each criterion, testable behavioral and interpretability markers will be specified.

4.2 Empirical Assessment Phase

Ten frontier AI systems will be evaluated against all five criteria using the standardized protocol.

Behavioral evaluations will use standardized probes designed to elicit or test each criterion.

Interpretability evaluations will use probing classifiers and activation analysis to assess internal state evidence.

All evaluations will be conducted by panels of trained evaluators blind to system identity.

Inter-rater reliability will be measured using Krippendorff's alpha.

4.3 Public Deliberation Phase

Six deliberative mini-publics, meaning structured groups of 20 citizens selected to represent demographic diversity, will be convened in three countries: the United States, Germany, and Japan.

Each mini-public will receive a standardized educational module on AI capabilities, competing moral frameworks, and the empirical evidence from Phase 2.

After deliberation, participants will indicate which criteria they believe should confer moral status and what protective obligations should follow.

4.4 Governance Framework Development

A final synthesis phase will develop draft governance recommendations incorporating philosophical analysis, empirical findings, and public deliberation outcomes.

The governance recommendations will specify which evidence levels should trigger which protective obligations under precautionary principles.

4.5 Analysis

Inter-rater reliability analysis for the empirical assessment phase.

Thematic analysis of deliberative mini-public discussions.



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Comparative analysis of framework outcomes across philosophical criteria to identify agreement and divergence.

5. CONTROLS AND CONFOUNDS

Evaluators in the empirical phase will be trained to reliability criterion before data collection begins.

Mini-public facilitators will follow standardized non-directive protocols.

Empirical probes will be pre-registered.

6. LIMITATIONS

This study can produce a framework and initial application results but cannot resolve underlying philosophical disagreements about the grounds of moral status.

The study is limited to current AI systems; conclusions may require substantial revision as systems become more capable.

Public deliberation findings are descriptive of current public values and do not constitute a philosophical argument for any particular position.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the most rigorous empirically grounded and publicly deliberated AI moral status assessment framework available.



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A reproducible assessment procedure, even an admittedly provisional one, would let developers and regulators make precautionary determinations about specific systems instead of debating the question in the abstract indefinitely.

It will provide governments, AI developers, and oversight institutions with an operational tool for making precautionary moral status determinations about specific AI systems.

The cross-national public deliberation component will provide the first comparative data on whether different democratic societies converge on or diverge from each other on AI moral status criteria.

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CHAPTER 45: HOW SHOULD THE BENEFITS AND RISKS OF AGI AND ASI BE DISTRIBUTED GLOBALLY?

Proposed Scientific Experiment

Submitted for Peer Review Consideration

ABSTRACT

The development of Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI) carries the potential to concentrate unprecedented economic, military, and social power in the hands of whichever individuals, corporations, or nations develop and control it first.

The distribution of those benefits and risks is not determined by technological necessity.

History warns that the gains from general-purpose technologies are often captured narrowly long before institutions adapt to spread them, and AI's reliance on scarce compute, data, and talent concentrated in a few firms and nations makes that outcome the default rather than the exception.

The window for shaping the outcome may also be brief, because once a decisive capability exists the leverage to renegotiate how its benefits are shared shrinks sharply.

It is determined by deliberate design choices, governance frameworks, and international agreements made in the current period, before AGI or ASI exists.



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Once a sufficiently capable AGI system exists, the leverage available to reshape distribution arrangements may be substantially reduced.

This makes the distribution question not merely a matter of equity and justice but a matter of urgent practical importance that must be addressed prospectively.

Current AI development is highly concentrated geographically and institutionally.

Research from the AGILE Index 2025 and OECD AI Observatory documents that the vast majority of frontier AI capability resides in a handful of nations and an even smaller number of private corporations.

The global South, meaning developing nations in Africa, Latin America, South Asia, and Southeast Asia, participates in AI development primarily as data source and consumer rather than as creator and owner.

This paper proposes a multi-method research program to characterize the current distribution of AI development capacity and projected AGI/ASI benefits and risks, to evaluate candidate distribution frameworks, and to identify the conditions under which more equitable global distribution is achievable.

Quantifying the present global distribution of compute, data, and talent gives the debate a factual baseline it has lacked.

Testing whether more inclusive governance strengthens or weakens safety outcomes connects



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the equity question to the survival question for the first time.

1. INTRODUCTION AND MOTIVATION

1.1 The Distribution Stakes

Historically, transformative general-purpose technologies have profoundly shaped the distribution of global power, wealth, and wellbeing.

Economists call innovations like the steam engine and electricity general-purpose technologies precisely because their effects ripple across every sector, and history shows their gains were often captured unevenly before institutions adapted.

One concrete proposal for steering the outcome is the windfall clause, a commitment to share extraordinary AI-derived profits, while philosophers invoke Rawls's veil of ignorance as a test of which distribution a fair observer would choose without knowing their own place in it.

The industrial revolution generated enormous wealth but also concentrated it geographically and socially in ways that persisted for generations.

The digital revolution increased connectivity globally but also created new forms of information asymmetry and economic dependence.

AGI and ASI are likely to be more transformative than either precedent at a faster pace.



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Economic modeling of AGI-level automation has produced estimates of labor productivity improvements large enough to effectively double global economic output within years.

The distribution of that productivity gain, whether it accrues to AGI owners, to the labor markets in which AGI replaces workers, or to public funds redistributable to all humanity, is entirely a question of governance design.

1.2 The Risk Distribution Asymmetry

The risks of AGI and ASI are potentially global in character: a misaligned AGI system does not confine its effects to the nation that built it.

But the oversight, governance, and safety investment is currently concentrated in the same small number of nations and organizations that lead capability development.

Nations with limited AI development capacity have little influence over safety decisions that affect them.

This asymmetry is a governance failure with moral and practical dimensions.

1.3 Existing Frameworks and Their Limits

The EU AI Act provides the most comprehensive regulatory framework currently in force, but it applies only within the EU and does not address global distribution.

Underlying the whole question is the global digital divide, since the compute, data, and



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talent required to build frontier AI are concentrated in a handful of countries and companies, so that without deliberate design the benefits may accrue exactly where capacity already exists.

UN resolutions have affirmed principles of inclusive AI development but have not produced binding commitments on benefit sharing.

No mechanism currently exists for ensuring that developing nations receive proportionate access to AGI-derived economic benefits or proportionate voice in AGI safety governance.

2. RESEARCH QUESTIONS

Primary Research Question:

Which institutional design choices and governance mechanisms are most effective at producing a more equitable global distribution of AGI and ASI benefits while simultaneously ensuring that risk governance incorporates meaningful participation from all affected populations?

Secondary Research Questions:

- What is the current quantitative distribution of AI development capacity, data access, talent, and compute infrastructure across nations and institutions, and how is it trending?
- Which historical precedents for governing the distribution of shared technological benefits are most instructively applicable to AGI, and what specific lessons do they offer?



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- What are the minimum conditions under which currently disadvantaged nations would consider international AGI distribution agreements legitimate and binding?
- How do different distribution frameworks interact with safety outcomes: do frameworks that promote broader participation improve or impair the global safety governance quality?

3. HYPOTHESES

Null Hypothesis (H0):

No candidate distribution framework achieves a significantly higher composite equity and legitimacy score than the current uncoordinated development trajectory across all evaluation criteria.

Alternative Hypothesis H1 (Common Heritage Mechanism):

Frameworks modeled on the common heritage of mankind doctrine, which holds that certain global resources belong to all humanity rather than to whoever can access them first, will achieve the highest equity scores but the lowest feasibility scores due to resistance from leading AI development nations.

Alternative Hypothesis H2 (Safety-Equity Complementarity):

Contrary to a common assumption that safety governance and equity governance trade off against each other, frameworks that increase participation from currently excluded nations will produce



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higher safety outcome scores because diverse participation reduces the risk that safety frameworks reflect only the values and interests of a small group.

Alternative Hypothesis H3 (Minimum Viable Progress):

Incremental minimum-viable-progress approaches, meaning small, immediately achievable steps toward equitable distribution rather than comprehensive frameworks, will achieve significantly higher feasibility scores than comprehensive frameworks while producing comparable equity gains over a 10-year projection horizon.

4. METHODOLOGY

4.1 Landscape Assessment

A quantitative mapping study will document the current global distribution of:

AI research publication output by country and institution. • Frontier model development capacity by country. • Compute infrastructure by country. • AI talent density by country. • Data governance and data access rights by country. • National representation in major AI standards bodies.

Data will be drawn from the AI Index Report, the OECD AI Observatory, the AGILE Index 2025, and original data collection.

Trend analysis will project current trajectories forward 5 and 10 years under a status-quo assumption.



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4.2 Distribution Framework Evaluation

Six candidate distribution frameworks will be assessed:

Framework A: Status quo development with voluntary bilateral technology sharing agreements.

Framework B: A mandatory global compute access fund, financed by a levy on frontier AI development above a specified scale threshold.

Framework C: Common heritage of mankind framework applied to AGI capabilities, with a multilateral licensing authority.

Framework D: An international AI development partnership modeled on CERN, the European Organization for Nuclear Research, meaning a jointly governed shared research infrastructure.

Framework E: A global AI dividend, meaning a universal basic resource transfer financed by AGI productivity gains, distributed on a per-person basis globally.

Framework F: Incremental minimum-viable-progress steps, beginning with standardized data access rights and interoperability requirements.

Each framework will be evaluated on equity, safety complementarity, economic efficiency, political feasibility, and adaptability using a structured rubric developed in the framework development phase.

4.3 Expert and Stakeholder Analysis



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Expert panels will evaluate frameworks on the seven rubric criteria.

Panels will be deliberately structured to include substantial representation from developing nations, specifically at least 50 percent of panel members from the global South.

Structured interviews with government officials from 20 nations spanning all economic development levels will assess minimum conditions for agreement and likely negotiating positions.

4.4 Public Values Assessment

Representative surveys in 15 nations spanning different economic development levels, political systems, and cultural contexts will measure public values related to:

Perceived legitimacy of different distribution arrangements. • Prioritization of different equity and safety goals. • Willingness to accept governance obligations in exchange for benefit access.

4.5 Safety Interaction Analysis

For each distribution framework, a structured scenario analysis will assess the implications for global safety governance quality, specifically whether the framework improves or impairs inclusive participation in safety decisions.

4.6 Analysis

Composite rubric scores will be computed and compared.



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Sensitivity analysis will test score robustness under different criterion weightings.

Regression analysis will test the relationship between framework equity scores and framework safety scores across the six frameworks and the three scenarios.

Public survey analysis will use multi-level modeling to estimate cross-national variation in distribution preferences.

5. CONTROLS AND CONFOUNDS

Expert panels and survey samples will be designed to ensure balanced representation across geographic regions, income levels, and institutional affiliations.

All rubric evaluations will use identical briefing documents.

Scenario analyses will use pre-registered assumptions.

6. LIMITATIONS

This study cannot overcome the fundamental political challenge that equitable distribution of AGI benefits may require redistribution of power from actors who currently possess the greatest capability and the greatest influence over governance outcomes.

No research design can resolve that political constraint.



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This study can, however, identify the governance designs most likely to achieve meaningful equity given realistic political constraints.

The economic projections used in the study carry substantial uncertainty because they depend on the actual capability trajectory of AGI development, which is itself uncertain.

7. SIGNIFICANCE AND EXPECTED CONTRIBUTIONS

This study will produce the most comprehensive, multi-stakeholder, globally representative analysis of AGI and ASI distribution governance conducted to date.

By testing distribution frameworks against both equity and safety criteria, the study would also address a question usually ignored, namely whether broader participation in governance makes the resulting safety regime stronger or weaker.

The finding on safety-equity complementarity or trade-off will address a critical assumption in current governance debates where safety and equity are often framed as competing objectives.

The public values data from 15 nations will provide governance designers with the most detailed and cross-culturally validated understanding available of the conditions under which different populations would regard distribution agreements as legitimate.

The minimum-viable-progress framework evaluation will identify the most achievable pathway toward equitable distribution for governance actors who



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accept that comprehensive frameworks are not immediately feasible.

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with

Twenty Proposed Scientific Experiments - Submitted for Peer Review Consideration

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(<https://github.com/ai-university-aiu/Mentova>)

End of line.



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BACK COVER MATTER

Dear Reader,

Thank you for reading The Roadmap to AGI and ASI!

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If you are aware of emerging alternative information that could further the field of Artificial Intelligence Engineering into the future, then please notify me so that the project may be improved in the next Edition!

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